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Statistical Data Warehouse Design Manual

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In order to modernise statistical production, [ESS](#) Member States are searching for ways to make optimal use of all available data sources, existing and new. This modernisation implies not only an important organisational impact but also higher and stricter demands for the data and metadata management. Both activities are often decentralised and implemented in various ways, depending on the needs of specific statistical systems (stove-pipes), whereas realising maximum re-use of available statistical data just demands the opposite: a centralised and standardised set of (generic) systems with a flexible and transparent metadata catalogue that gives insight in and easy access to all available statistical data.

To reach these goals, building a **Statistical Data Warehouse (S-DWH)** is considered to be a crucial instrument. The S-DWH approach enables NSIs to identify the particular phases and data elements in the various statistical production processes that need to be common and reusable.

The CoE on DWH provides a document that help and guide in the process of designing and developing a S-DWH:

The S-DWH Design Manual

This document answers the following questions:

- What is a Statistical Data Warehouse (S-DWH) ?
- How does a S-DWH differ from a traditional = 'commercial' DWH ?
- Why should we build a S-DWH ?
- Who are the envisaged users of a S-DWH ?
- Give a road map for designing, building and implementing the S-DWH:
- What are the prerequisites for implementing a S-DWH ?
- What are the phases/steps to take ?
- How to prepare for an implementation ?

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i - General Introduction

Authors: Antonio Laureti Palma

i Introduction

The statistical production system of a NSI concerns a cycle of organizational activity: the acquisition of data, the elaboration of information, the custodianship and the distribution of that information. This cycle of organizational involvement with information involves a variety of stakeholders: for example those who are responsible for assuring the quality, accessibility and program of acquired information, those who are responsible for its safe storage and disposal. The information management embraces all the generic concepts of management, including: planning, organizing, structuring, processing, controlling, evaluation and reporting of information activities and is closely related to, and overlaps, the management of data, systems, technology, and statistical methodologies.

Due to the great evolution in the world of information, user's expectations and need of official statistics has increased in the recent years. They require wider, deeper, quicker and less burdensome statistics. This has led NSIs to explore new opportunities for improving statistical productions using several different sources of data and in which is possible an integrated approach, both in term of data and processes. Some practical examples are:

- In the last European census, administrative data was used by almost all the countries. Each country used either a full register-based census or register combined with direct surveys. The census processes were quicker than in the past and generally with better results. In some cases, as in the 2011 German census, the first census not register-based taken in that country since 1983, provides a useful reminder of the danger in using only a register-based approach. In fact, the census results indicated that the administrative records on which Germany based official population statistics for a period of several decades, overestimates the population because of failing to adequately record foreign-born emigrants. This suggests that the mixed data source approach, which combines direct-survey data with administrative data, is the best method to obtain accurate results (Citro 2014) even if it is much more complex to organize in terms of methodologies and infrastructure.
- At a European level, a few years ago, the SIMSTAT project, an important operational collaboration between all member states, started. This is an innovative approach for simplifying Intrastat, the European Union (EU) data collection system on intra-EU trade in goods. It aims to reduce the administrative burden while maintaining data quality by exchanging microdata on intra-EU trade between Member States and re-using them, including both technical and statistical aspects. In this context directed survey or admin data are shared between member states through a central data hub. However, in SIMSTAT there is an increase in complexity due to the need for a single coherent distributed environment where the 28 countries can work together.
- Also in the context of Big Data, there are several statistical initiatives at the European level, for example "use of scanner data for consumer price index" (ISTAT) or "aggregated mobile phone data to identify commuting patterns" (ONS), which both require an adjustment of production infrastructure in order to manage these big data sets efficiently. In this case the main difficulty is to find a data model able to merge big data and direct surveys efficiently.

Recently, also in the context of regular structural or short term statistics, NSIs have expressed the need for a more intensive use of administrative data in order to increase the quality of statistics and

reduce the statistical burden. In fact, one or more administrative data sources could be used for supporting one or more surveys of different topics (for example the Italian Frame-SBS). Such a production approach creates more difficulties due to an increase in dependency between the production processes. Different surveys must be managed in a common coherent environment. This difficult aspect has led NSIs to assess the adequacy of their operational production systems and one of the main drawbacks that has emerged is that many NSIs are organized in single operational life cycles for managing information, or the “stove-pipe” model. This model is based on independent procedures, organizations, capabilities and standards that deal with statistical products as individual services. If an NSI with a production system mostly based the stove-pipe model wants to use administrative data efficiently, it has to change to a more integrated production system.

All the above cases indicate the need of a complex infrastructure where the use of integrated data and procedures is maximized. Therefore, this infrastructure would have two basic requirements:

- ability to management of large amounts of data,
- a common statistical frame in terms of IT infrastructure, methodologies, standards and organization to reduce the risk of losing coherence or quality.

An such complex infrastructure that can meet these requirements is a corporate Statistical Data Warehouse (S-DWH), possibly metadata-driven, in which statisticians can manage micro and macro data in the different production phases. A metadata-driven is a system where metadata create a logical self-describing framework to allow the data to drive functionality. The S-DWH approach would then support a high level of modularity and standards that help the design of statistical processes. Standardized processes combined with high level of data complexity can be organized in structured workflows of activities where the S-DWH became the common standardized data repository.

i.1 A statistical Data Warehouse view

A Statistical-Data Warehouse (S-DWH) can be defined as a single corporate Data Warehouse fully based on a metadata. A S-DWH is specialized in supporting production for multiple-purpose statistical information. With a S-DWH different aggregate data on different topics should not be produced independently from each other but as integrated parts of a comprehensive information system where statistical concepts, micro data, macro data and infrastructures are shared.

It is important to emphasize that the data models underlying a S-DWH are not only oriented to producing specific statistical output or on line analytical processing, as is the case currently in many NSIs, but rather to sustain the production of statistical information in the various phases of statistical life-cycle production process. A S-DWH model, instead of focusing on a process-oriented design, is based on data inter-relationships that are fundamental for different processes of different statistical domains.

The S-DWH data model must sustains the ability of realizing data integration at micro and macro data granularity levels: micro data integration is based on the combination of different data sources with a common unit of analysis, one or system of statistical registers, while macro data integration is based on integration of different aggregate or dis-aggregate information in a common estimation domain.

In the case of complex statistical productions, a corporate S-DWH can facilitate the design of production process based on workflow of activities of different statistical experts in which the knowledge sharing is central. This corresponds to a workflow management system able to sustain a “data-centric” workflow of activities based on the S-DWH; i.e. a common software environment in

which all the statistical experts involved in the different production phases work by testing hypotheses on a same production process. This can increase ability to manage complex data source, typically from administrative data or big data, reducing the risk connected with of integration errors and data loss by eliminating any manual steps in data retrieval.

We can identify four conceptual layers for the S-DWH, starting from the bottom up to the top of the architectural pile, they are defined as:

I° - source layer, is the level in which we locate all the activities related to storing and managing external data sources and where is realized the reconciliation, the mapping, of statistical definitions from external to internal DW environment.

II° - integration layer, is where all operational activities needed for any statistical production process are carried out; in this layer data are mainly transformed from raw to cleaned data;

III° - interpretation and data analysis layer, enables data analysis or data mining functional to support statistical design; functionality and data are optimized then for internal users, specifically for statistician methodologists or statistician experts on specific domains.

IV° - access layer, for the access to the data: selected operational views, final presentation, dissemination and delivery of the information sought specialized for external, relatively to NSI or Eurostat, users;

The layers can be grouped in two sub-groups: the first two layers for statistical operational activities, i.e. where the data are acquired, stored, coded, checked, imputed, edited and validated; the last two layers are for the effective data warehouse, i.e. levels in which data are accessible for analysis, design data re-use and for reporting.

i.2 Data-centric Workflows

The statistical production based on the use of a S-DWH must be articulated in a number of different phases, or specialized sub-processes, where each phase collects some data input and produces some data output. This constitutes a data transformation process which takes place by an asynchronous elaboration and uses the S-DWH as input/output data repository of raw and cleaned integrable data. In this way, the production can be seen as a workflow of separated activities, realized in a common environment, where all the statistical experts involved in the different production phases can work. In such an environment the role of knowledge sharing is central and this is sustained by the S-DWH information model, in which all information from the collaborative workflow is stored.

This type of workflow can be defined as “data-centric workflow”, i.e. an environment where all the statistical experts (or data scientists), involved in the different production phases of the same process, can work by testing hypotheses.

Any process, organized in a structured workflow, can sustain stable processes as well as frequently modified processes, i.e. process re-use or process adjustments. An example of process adjustment could be the integration process of external administrative data source not under the direct control of statisticians; in fact, the sources structure or content may change each supply which implies adaptation of the data integration processes or, in the extreme case, completely rewrite the procedures. In these cases if the process, the workflow of activities, are stored in dedicated collaborative infrastructure the activities of procedure adaptation became easier and safe. The data-centric workflow environment then allows a controlled process through the standardization of flexible working methods on a common information model which is particularly efficient in all cases where the analysis phase and the coding are realized in the same time.

i.3 Statistical models

The Manual is based on standards and frameworks for describing statistical processes, information objects and for modelling and supporting business processes management. The models used are: GSIM, GSBPM, BPMN, SDMX.

The use of these model facilitate the communications between statisticians and it would avoid the creation of new concepts when not strictly necessarily.

In the follow a brief description of the basic models are introduced.

i.3.1 GSIM

A model emanating from the “High-Level Group for the Modernisation of Statistical Production and Services” (HLG), is the Generic Statistical Information Model (GSIM ¹). This is a reference framework of internationally agreed definitions, attributes and relationships that describes the pieces of information that are used in the production of official statistics (information objects). This framework enables generic descriptions of the definition, management and use of data and metadata throughout the statistical production process.

GSIM Specification provides a set of standardized, consistently described information objects, which are the inputs and outputs in the design and production of statistics. Each information object has been defined and its attributes and relationships have been specified. GSIM is intended to support a common representation of information concepts at a “conceptual” level. It means that it is representative of all the information objects which would be required to be present in a statistical system.

In the case of a process, there are objects in the model to represent these processes. However, it is at the conceptual and not at the implementation level, so it doesn't support anyone specific technical architecture - it is technically 'agnostic'.

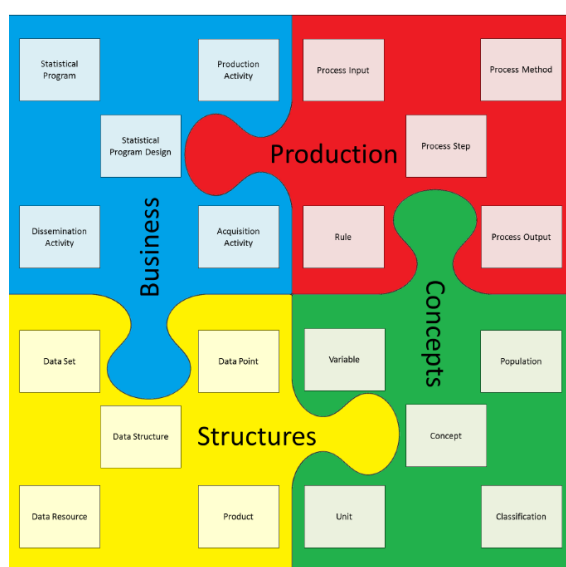


Figure 1 - General Statistical Information Model (GSIM) [from High-Level Group for the Modernisation of Statistical Production and Services]

Because GSIM is a conceptual model, it doesn't specify or recommend any tools or measures for IT processes management. It is intended to identify the objects which would be used in statistical

¹ <http://www1.unece.org/stat/platform/display/metis/Brochures>

processes, therefore it will not provide advice on tools etc. (which would be at the implementation level). However, in terms of process management, GSIM should define the objects which would be required in order to manage processes. These objects would specify what process flow should occur from one process step to another. It might also contain the conditions to be evaluated at the time of execution, to determine which process steps to execute next.

We will use the GSIM as a conceptual model to define all the basic requirements for a Statistical Information Model, in particular:

- the Business Group (in blue in Figure 1) is used to describe the designs and plans of Statistical Programs
- the Production Group (red) is used to describe each step in the statistical process, with a particular focus on describing the inputs and outputs of these steps
- the Concepts Group (green) contains sets of information objects that describe and define the terms used when talking about real-world phenomena that the statistics measure in their practical implementation (e.g. populations, units, variables)

i.3.2 GSBPM

The Generic Statistical Business Process Model (GSBPM) should be seen as a flexible tool to describe and define the set of business processes needed to produce official statistics.

It is necessary to identify and locate the different phases of a generic statistic production process on the different S-DWH's conceptual layers. The GSBPM schema is shown in figure below:

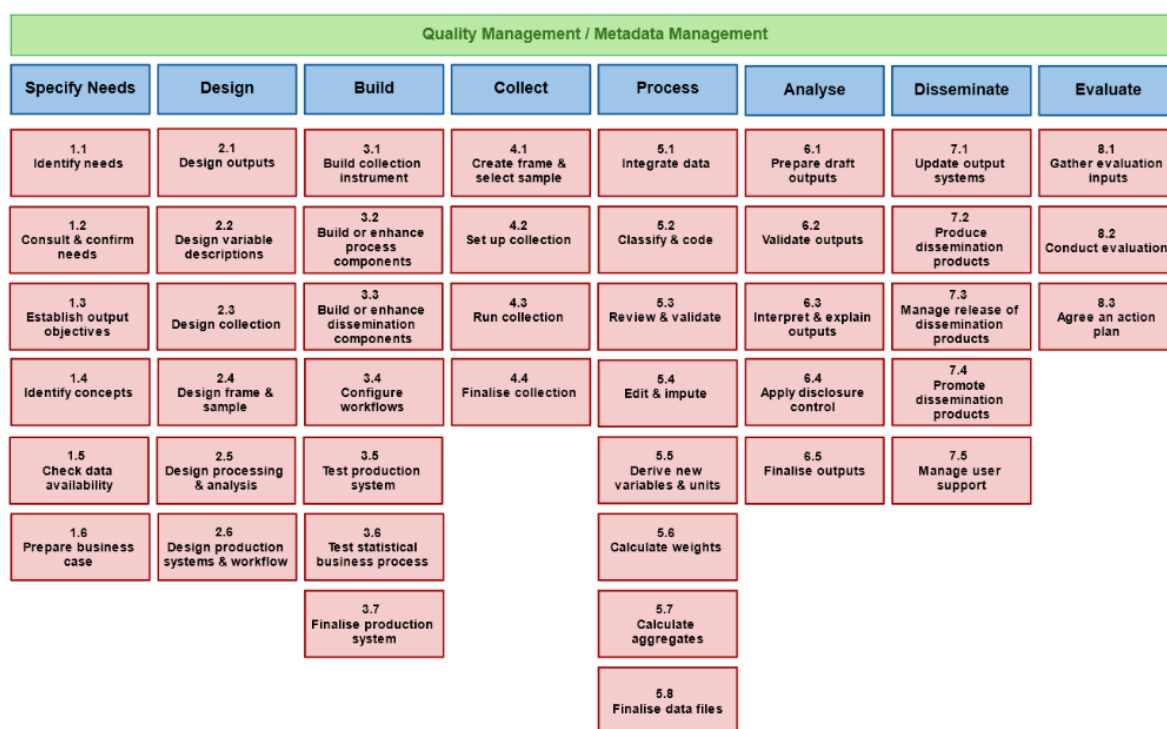


Figure 2 - The GSBPM schema

i.3.3 SDMX

The Statistical Data and Metadata Exchange (SDMX) is an initiative from a number of international organizations, which started in 2001 and aims to set technical standards and statistical guidelines to facilitate the exchange of statistical data and metadata using modern information technology.

The term metadata is very broad and distinction is made between “structural” metadata that define the structure of statistical data sets and metadata sets, and “reference” metadata describing actual metadata contents, for instance, concepts and methodologies used, the unit of measure, the data quality (e.g. accuracy and timeliness) and the production and dissemination process (e.g. contact points, release policy, dissemination formats). Reference metadata may refer to specific statistical data, to entire data collections or even to the institution that provides the data.

NSIs need to define metadata before linking sources. What kind of reference metadata needs to be submitted? As we know in Eurostat this information is presented in files based on a standardised format called ESMS (Euro SDMX Metadata Structure). ESMS Metadata files are used for describing the statistics released by Eurostat. It aims at documenting methodologies, quality and the statistical production processes in general.

i.4 Manual structure

The manual consist of five chapters and one annex:

Chapter 1, Implementation: the chapter explains how to implement a S-DWH in practical terms and describes the steps in the process for creating output variables from the available input data. The main goal of the chapter is to give recommendations about better use of data that already exist in the statistical systems and to create fully-integrated datasets at the micro level.

Chapter 2, Governance: this chapter gives suggestions on governance of a S-DWH with the aim of ensuring data and statistics quality. The topic covers the governance of metadata, processes and users.

Chapter 3, Architecture: in the chapter we use the typical EA domain views in order to give a comprehensive architectural vision of a S-DWH. Each domain is dealt with using the four conceptual layers in the S-DWH. Moreover, the chapter covers the aspects of a data-centric workflow. Finally, the SDMX in the context of S-DWH Architecture is analysed.

Chapter 4, Methodology: this chapter explains the methods needed, in a S-DWH, for cleaning the incoming data, linking them, weighing them and release them without disclosing confidential information.

Chapter 5, Metadata: this chapter describes the metadata categories of the S-DWH and give an overview of each layer. It also describes the metadata of a statistical production lifecycle: which metadata are produced during a process, which metadata are needed to perform a process and which metadata are forwarded from one process to the next one.

Annex 1 is intended to be an overview of software packages existing on the market or developed on request in NSIs in order to describe the solutions that would meet NSI needs, implement S-DWH concept and provide the necessary functionality for each S-DWH level. It also gives a basic overview on how the SDMX tools can be classified in terms of the various features they provide.

1-Implementation

1.1 Current state and pre-conditions

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1.2 Design Phase roadmap

Authors: Antonio Laureti Palma, Antti Santaharju

1.3 Building blocks – The input datasets

Authors: Antti Santaharju

1.4 Business processes of the layered S-DWH

Authors: Antonio Laureti Palma, Antti Santaharju, Sónia Quaresma

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Antonio Laureti Palma, Sónia Quaresma; ESS-Net-DWH Deliverable 3.1:”S-DWH Business Architecture”

Antonio Laureti Palma, Björn Berglund, Allan Randlepp, Valerij Žavoronok; ESS-Net-DWH Deliverable 3.5:”Relate the 'ideal' architectural scheme into an actual development and implementation strategy”

1.1 Current state and pre-conditions

In order to implement the S-DWH and work successfully several issues should be considered and documented. In this sub-section the preconditions for the successful work of S-DWH are discussed. This includes the requirements for the metadata and the analysis of the quality of the input data.

1.1.1 Methodological description of the statistical data process

Most of NSIs are implementing GSBPM for the description of the statistical data process. Descriptions and documentations should be prepared for the every phase (process) of GSBPM. In the frame of S-DWH the statistical data from different data sources could be linked and used for the evaluation of the statistical output. In some cases we need to compare different surveys (e.g. sample survey and census survey).

We need to have metadata information concerning these surveys. Therefore the metadata should be defined and documented at the lowest level (sub-process) of GSBPM. Topics related to metadata are discussed more detailed in the separate metadata chapter.

- **Data Linking** - The linking of statistical data is one of the essential issues in S-DWH since the data (different features) from different sources are linked. Data from different sources has different methodologies and/or different quality requirements. The methodological problem is to link the statistical data, to evaluate the values of particular indicator, and to obtain statistical output that meets the defined quality requirements of that indicator.
- **Data Confidentiality** - The aim of the statistical data confidentiality is to ensure that the statistical data are not collected unnecessarily and that confidential statistical data cannot be disclosed to the third parties in any statistical data processing stage. Especially the issue of disclosure control is relevant for small countries (like Lithuania) when there are many surveys but the number of respondents compared to bigger countries is rather small. One of the main problems is: whether the statistical institution is able to protect the statistical output so that the risk of disclosure would be as small as possible?

1.1.2 Quality requirements for the statistical data

Quality requirements for the statistical data are one of the main aspects of the statistical process. Different surveys could have different quality requirements. Statistical information from different data sources comes to the S-DWH. During the data integration process we face the problems of missing values, outliers, different timelines etc. The appropriate methodology ensures the quality of the statistical output of the S-DWH. Monitoring of the quality of statistical information in S-DWH should be based on the quality requirements of the ESS (relevance, accuracy, timelines and punctuality, accessibility and clarity, coherence and comparability).

The administrative data plays an important role in S-DWH. Statistical production rests on two pillars: statistical surveys and administrative data sources. Wider use of administrative data allows a decrease in the number of statistical surveys, thus reducing the statistical response burden. However, NSIs are often faced with the problem that administrative records tend to be unavailable or incomplete at the time when they are needed. (Kavaliauskienė et al., 2013)

The essential question is: whether the NSI is able to ensure the quality of additional data sources like administrative data in the frame of S-DWH? In order to answer this question the preparatory work should be done:

- to analyse the definitions of the statistical indicators of administrative sources.

- to make preliminary analysis the outliers, percent of missing values, etc.
- to compare administrative data to the data from traditional data sources (e.g. survey sampling).
- to estimate the correlation coefficient, to make other mathematical analysis.
- to define the methodology of using the administrative data: e.g. to input the missing values of the survey or to use for the regression-type estimation techniques, or other techniques.
- to analyse the errors of testing output (using the administrative data).

There is a set of quality assessment and improvement methods and tools e.g. audits (Inspections of statistical surveys), self-assessment, quality indicators, user satisfaction surveys, etc. One of the key issues in quality management is the identification of activities which are the most risky for the process.

1.1.3 IT tools

At the starting point of the implementation the S-DWH one of the main issues is the choice of IT tools. Usually the IT tools are chosen according to the amount of data, system and technical requirements of S-DWH. S-DWH could be placed at one or several physical locations. The IT solutions should be harmonized with the requirements of methodological side of the statistical data process (quality requirements, data linking, etc.). IT solutions are discussed in detailed level in architecture chapter.

1.1.4 Metadata Requirements

The metadata information and main problems related to data sources are described on phases 1-3 of GSBPM. The metadata requirements that have to be taken into consideration for these GSBPM phases are for example:

- Description of questionnaire version
- Template for the questionnaire
- Description of indicators and attributes of statistical questionnaire
- Classifications
- Measurement units, etc.

These questions are considered in the metadata section. Metadata questions related to the phases 4-6 of the GSBPM are also analysed in metadata chapter. These phases include the steps of statistical data collection, processing and analysing. Different cases of statistical process could be included into S-DWH. in order to integrate information from different sources possible metadata information of the phases 4-6 of the GSBPM of all cases should be provided. The metadata information and main types of metadata for phases 1-6 of the GSBPM model are described in the Metadata chapter.

1.2 Design Phase roadmap

The Design Phase roadmap for implementing a S-DWH described in the handbook¹ describes the design activities, and any associated practical research work needed to define the statistical outputs, concepts, correction methodologies, collection instruments and operational processes in a S-DWH environment.

The Design phase is worked out in detailed maps that show the essential milestones/steps, represented as a 'station or stop'. All the specific S-DWH stops are linked to the deliverables² to be used in that stage of the S-DWH development process.

In the detailed sub map the 3 tracks are represented by colored lines:

- the green line represents the Metadata Roadmap Design
- the blue line represents the Methodology Roadmap Design
- the red line represents the Technical aspects Roadmap Design

Furthermore there is a continuous **grey line** running through each phase and emphasizing the importance of good documentation, not only during the development process, but also in the operational phase.

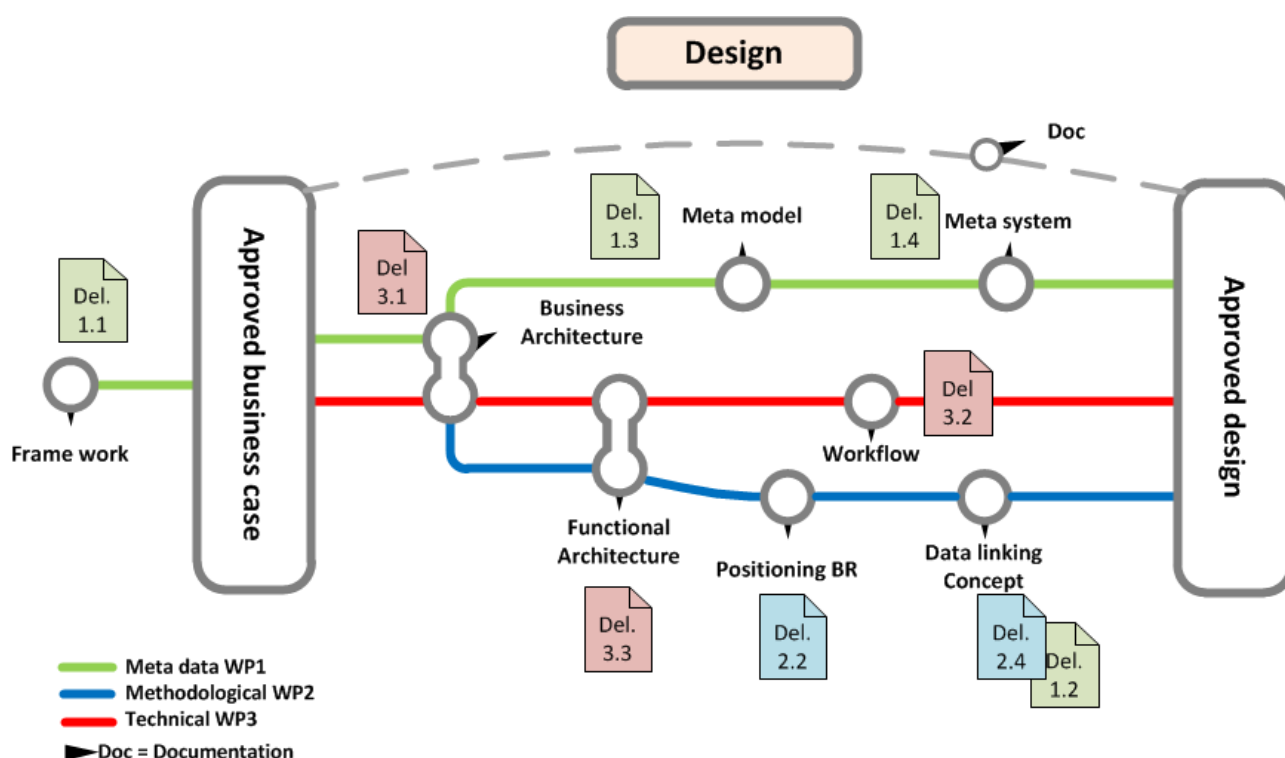


Figure 1. Design Phase roadmap

¹S-DWH Handbook is available on CROS portal: http://ec.europa.eu/eurostat/cros/content/s-dwh-handbook_en

² The ESSnet produced a large set of deliverables which are the basis for the CoE on DWH. All results are available on the ESSnet Website on the CROS-portal: http://ec.europa.eu/eurostat/cros/content/essnet-dwh_en

1.2.1 Metadata road map - green line

Metadata is data which describe other data. The description could refer to both data containers and/or individual instances of the application data. According to Common Metadata Framework³ the statistical metadata should enable a statistical organization to perform effectively the following functions:

- Planning, designing, implementing and evaluating statistical production processes.
- Managing, unifying and standardizing workflows and processes.
- Documenting data collection, storage, evaluation and dissemination.
- Managing methodological activities, standardizing and documenting concept definitions and classifications.
- Managing communication with end-users of statistical outputs and gathering of user feedback.
- Improving the quality of statistical data and transparency of methodologies. It should offer a relevant set of metadata for all criteria of statistical data quality.
- Managing statistical data sources and cooperation with respondents.
- Improving discovery and exchange of data between the statistical organization and its users.
- Improving integration of statistical information systems with other national information systems.
- Disseminating statistical information to end users. End users need reliable metadata for searching, navigation, and interpretation of data.
- Improving integration between national and international organizations. International organizations are increasingly requiring integration of their own metadata with metadata of national statistical organizations in order to make statistical information more comparable and compatible, and to monitor the use of agreed standards.
- Developing a knowledge base on the processes of statistical information systems, to share knowledge among staff and to minimize the risks related to knowledge loss when staff leave or change functions.
- Improving administration of statistical information systems, including administration of responsibilities, compliance with legislation, performance and user satisfaction.

In our context we will focus on the design phase which may involve any of the previous functions. In particular, the metadata design phase will be applied to different functionalities in function of the "business case" chosen.

Therefore, the business case metadata should derive from:

1. the contents, data sources and data outputs
2. the implicit semantics of data along with any other kind of data that enables the end-user to exploit the information;
3. their locations and their structures;
4. the processes that take place;
5. the infrastructure and physical characteristics of components;
6. security, authentication, and usage statistics that enable the administrator to tune the appropriate operation.

1.2.2 Methodological road map - blue line

The main goal of the methodology chapter is to prepare design methodological recommendations about better use of data that already exist in the statistical system and to create fully integrated data sets for enterprise and trade statistics at micro level: a 'data warehouse' approach to statistics.

³ <http://www1.unece.org/stat/platform/display/metis/The+Common+Metadata+Framework>

This corresponds to a central repository able to support several kind of data, micro, macro and meta, entering into the S-DWH in order to support cross-domain production processes and statistical design, fully integrated in terms of data, metadata, process and instruments.

1.2.3 Technical road map – red line

All essential technical elements of the layered architecture for implementing the S-DWH are described in the technical roadmap where it has provided both a Business Architecture for the S-DWH and the matching with the GSBPM. The technical road map provides principles and practices for designing the full architectural view of a S-DWH. It would help architects' thinking by dividing the architectural description into domains or views, and offers models for documenting each view. We will model a S-DWH architecture by three main domains: Business, Information, Technology.

1.2.4 Scientific Workflow

Using administrative data often means to elaborate and combine several large archives of data using different skills. In these cases the use of a S-DWH specialized for sharing information and optimized for producing statistical information could be crucial. In a S-DWH we can manage several production processes for different topics and sustain process integration. In fact, the data model underlying a S-DWH is based on an easy access to the information and data inter-relationships that could be fundamental for connecting different processes of a common statistical domain.

The statistical production based on the use of a S-DWH must be articulated in a number of different phases, or specialized sub-processes, where each phase collects some data input and produces some data output. This constitutes a data transformation process which takes place by an asynchronous elaboration and uses the S-DWH as input/output data repository of raw and cleaned integrable data. A process of this type can be supported using a blackboard design pattern's paradigm; i.e. a shared area (the blackboard) which can be accessed by autonomous processes or actors in some coordinated and cooperative way.

One way to find a common ground between different statistical actions is to focus on a generalized data input interface in which it is possible to identify and select the variables needed for data processing in each production phase. Adaptation of the data input and output interfaces of each phase of a process gives us the opportunity of managing the elaboration phases by using generic software components, i.e. using almost any statistical editing tool in a common application framework. The adaptability of the checking procedures are particularly helpful in statistical production based on administrative data when the input data layout and variable meanings are not under the direct control of the statistical producer, so that they can change for each supply due to national regulation changes. In this way, the production can be seen as a workflow of separated activities, realized in a common environment, where all the statistical experts involved in the different production phases can work. In such an environment the role of knowledge sharing is central and this is sustained by the S-DWH information model, in which all information from the collaborative workflow is stored.

This type of workflow is called **scientific-workflow**, i.e. an environment where all the statistical experts (or data scientists), involved in the different production phases of the same process, can work by testing hypotheses⁴.

The scientific-workflow environment then allows a controlled process through the standardization of flexible working methods on a common information model. In fact, the S-DWH increases the efficiency, reducing the risk of data loss and integration errors by eliminating any manual steps in data

⁴ G. Scherp. A Framework for Model-Driven Scientific Workflow Engineering 05/2010 Procedia Computer Science.

1.3 Building blocks – The input datasets

One aim of a S-DWH is to create a set of fully integrated statistical data. Input for these data may come from different sources like surveys, administrative data, accounting data and census data. Different data sources cover different populations. Some data sources like censuses cover all population (units). Some cover all units with a certain characteristic, some only influential units or other subpopulations. Other sources include less influential units, but provide information only about a few of them. To link these input data sources and to ensure that these data are linked to the same unit and are compared with the same target population is the main issue.

Main data sources:

1. Surveys (censuses, sample surveys)
2. Combined data (survey and administrative data)
3. Administrative data
4. BIG DATA

Survey based on statistical data collection (statistical questionnaire). A sample survey is more restricted in scope: the data collection is based on a sample, a subset of total population - i.e. not total count of target population which is called a census. However, in sample surveys some sub-populations may be investigated completely but most are sampled. Surveys as well as administrative data can be used to detect errors in the statistical register.

Combined data. Since survey and administrative data sets have their respective advantages, a combination of both sources enhances the potential for research. Furthermore, record linkage has several advantages from a survey methodological perspective. The administrative data is used to update the frame of active units, to cover and estimate non-surveyed or non-responding units.

The success of the actual linkage depends on the available information to identify a respondent in administrative records and on the quality of these identifiers. Record linkage can be performed using different linkage methods by means of a unique identifier such as the Social Security Number or unique common identifier, or on the basis of the ambiguous and error-prone identifiers as name, sex, date of birth and address etc.

Before the records from both data sources are actually compared extensive pre-processing needs to be conducted to clean up typographical errors as well as to fill in missing information. These steps of standardization should be done consistently for both the administrative and survey records.

Administrative data is the set of units and data derived from an administrative source. A traditional definition of administrative sources is that they are files of data collected by government bodies for the purposes of administering taxes or benefits, or monitoring populations. This narrow definition is gradually becoming less relevant as functions previously carried out by the government sector are, in many countries, being transferred partly or wholly to the private sector, and the availability of good quality private sector data sources is increasing.

Big Data usually includes data sets with sizes beyond the ability of commonly used software tools to capture, curate, manage, and process data within a tolerable elapsed time. Big data is often largely unstructured, meaning that it has no pre-defined data model and/or does not fit well into conventional relational databases¹. Big data requires a set of techniques and technologies with new forms of integration to reveal insights from datasets that are diverse, complex, and of a massive scale.

Unece has developed following classification for the types of Big Data²:

- **Social Networks (human-sourced information)**: this information is the record of human experiences, previously recorded in books and works of art, and later in photographs, audio and video. Human-sourced information is now almost entirely digitized and stored everywhere from personal computers to social networks. Data are loosely structured and often ungoverned.
- **Traditional Business systems (process-mediated data)**: these processes record and monitor business events of interest, such as registering a customer, manufacturing a product, taking an order, etc. The process-mediated data thus collected is highly structured and includes transactions, reference tables and relationships, as well as the metadata that sets its context. Traditional business data is the vast majority of what IT managed and processed, in both operational and BI systems. Usually structured and stored in relational database systems. (Some sources belonging to this class may fall into the category of "Administrative data").
- **Internet of Things (machine-generated data)**: derived from the phenomenal growth in the number of sensors and machines used to measure and record the events and situations in the physical world. The output of these sensors is machine-generated data, and from simple sensor records to complex computer logs, it is well structured. As sensors proliferate and data volumes grow, it is becoming an increasingly important component of the information stored and processed by many businesses. Its well-structured nature is suitable for computer processing, but its size and speed is beyond traditional approaches.

A Data Warehouse will combine data from different sources, which could be collected by several modes. The data register and administrative data is not only being used as business or population frames and auxiliary information for survey sample based statistics, but also as the main sources for statistics, and as sources of quality assessment.

For business statistics there are many logical relationships (or edit constraints) between variables. When sources are linked, inconsistencies will arise, and the linked records do not necessarily respect these constraints. A micro-integration step is usually necessary to integrate the different sources to arrive at consistent integrated micro data. The ESSnet Data Integration outlines a strategy for detecting and correcting errors in the linkage and relationships between units of integrated data. Gåsemyr et al (2008) advocate the use of quality measures to reflect the quality of integrated data, which can be affected by the linkage process.

Editing data from different sources is required for different purposes: maintaining the register and its quality; for a specific output and its integrated sources; and to improve the statistical system. The editing process is one part of quality control in the Statistical Data Warehouse – finding error sources and correcting them. These issues are discussed more detailed in methodology chapter.

¹ <http://www1.unece.org/stat/platform/pages/viewpage.action?pagelId=77170622>

² <http://www1.unece.org/stat/platform/display/bigdata/Classification+of+Types+of+Big+Data>

1.3.1 Use of administrative data sources

Many NSIs have increased the use of administrative data sources for producing statistical outputs. The potential advantages of using administrative sources include a reduction in data collection and statistical production costs; the possibility of producing estimates at a very detailed level thanks to almost complete coverage of the population; and re-use of already existing data to reduce respondent burden. There are also drawbacks to using administrative data sources. The economic data collected by different agencies are usually based on different unit types. For example a legal unit used to collect VAT information by the Tax Office is often different to the statistical unit used by the NSI. These different unit types complicate the integration of sources to produce statistics. This can lead to coverage problems and data inconsistencies on linked data sources.

Another complication affecting the use of administrative data is timeliness. For example, there is often too much of a lag between the reporting of economic information to the Tax Office and the reporting period of the statistic to be produced by the NSI. The ESSnet Admin Data (Work Package 4) has addressed some of these issues, and produced recommendations on how they may be overcome.³

Definitions of variables can differ between sources. Work Package 3 of the ESSnet Admin Data aims to provide methods of estimation for variables that are not directly available from administrative sources. In addition, in many cases the administrative sources alone do not contain all of the information that is needed to produce the detailed statistics that NSIs are required to produce, and so a mixed source approach is usually required.

1.3.2 The Business Register and the statistical DWH

The position of the Business Register in a statistical-DWH is relatively simple in general terms. The Business Register provides information about statistical units, the population, turnover derived from VAT and wages plus employment derived from tax and/or social security data. As this information is available for almost all units, the Business Register allows us to produce flexible output for turnover, employment and number of enterprises.

The aim of the statistical-DWH is to link all other information to the Business Register in order to produce consistent and flexible output for other variables. In order to achieve this, a layered architectural S-DWH has been considered. Note that statistical (enterprise) units, which are needed to link independent input data sets with the population frame and in turn to relate the input data to statistical estimates, play an important role in the processing phase of the GSBPM. This processing phase corresponds with the integration layer of the S-DWH.

We realize that some National Statistical Institutes (NSI) have separate production systems to calculate totals for turnover and employment outside the Statistical Business Register (SBR). These systems are linked to the population frame of the SBR. The advantage of doing this is that such a separate process acknowledges that producing admin data based turnover and employment estimates requires specified knowledge about tax rules and definition issues. Nevertheless the final result of calculating admin data based totals for turnover and employment within or outside the SBR is the same. As this tax information is available for almost all units and linked with the SBR, it is possible to produce flexible output for turnover, employments and number of enterprises regardless of whether totals are calculated within or outside the Business Register.

³ Reports of ESSnet Admin Data are available at <http://www.cros-portal.eu/content/admindata-sga-3>

Therefore, we discuss the role of (flexible) population totals like number of enterprises, turnover and employment in a S-DWH, but we don't discuss whether total of turnover and employment should be calculated within or outside the SBR. This decision is left to the individual NSI.

The same is true for whether the SBR is part of the S-DWH or not. The population frame derived from the SBR is a crucial part of the statistical-DWH. It is the reference to which all data sources are linked. However, this does not mean that the SBR itself is part of the statistical-DWH. A very good practical solution is that

- the population frame is derived from the SBR for every period t
- these snapshots of population characteristics for periods t_x are used in the statistical-DWH.

By choosing this option the maintenance of the SBR is separated from maintenance of the statistical-DWH. Both systems are however linked by the same population characteristics for period t . This option is called SBR outside the statistical DWH.

Another option is that the entire SBR-system is included in the statistical-DWH. The advantage of this approach is that corrected information about populations in the statistical-DWH is immediately implemented in the SBR. However, this may lead to consistency problems if outputs are produced outside the statistical-DWH (as the 'corrected' information is not automatically incorporated in these parts of the SBR). Maintenance problems may arise as a system including both the production of a SBR as well as flexible statistical outputs may be large and quite complex. This option is called SBR inside the statistical DWH.

It is up to the individual NSIs whether the SBR should be inside or outside the statistical-DWH because the coverage of the statistical-DWH (it may include all statistical input and outputs or only parts of the in- and outputs) may differ for different countries. Furthermore, we did not investigate the crucial maintenance factor.

In the remaining part of this manual, we consider the option "SBR outside the statistical DWH" only. This choice has been made for the sake of clarity. Apart from sub-section (Correcting information in the population frame and feedback to SBR), which is not relevant in the case of "SBR inside the statistical DWH", this choice does not affect the other conclusions.

1.3.3 Statistical units and population

The aim of a statistical-DWH is to create a set of fully integrated data pertaining to statistical units, which enables a statistical institute to produce flexible and consistent output. The original data come from different data sources. Collection of these data takes place in the collect phase of the GSBPM process model.

In practice, different data sources may cover different populations. The coverage differences may be for different reasons:

- The definition of an unit differs between the sources.
- Sources may include (or exclude) groups of units which are excluded (or included) in other sources.

An example of the latter is the VAT-registration versus business survey data. VAT-data (and some other tax data like corporate tax data) do not include the smallest enterprises, but include all other commercial enterprises. Business survey samples contain information about a small selected group of enterprises, including the smallest enterprises.

Hence, linking data of several sources is not only a matter of linking units between the different input data but also a matter of relating all input data to a reference.

Different sources may have different units. For example, surveys are based on statistical units (which generally corresponds with legal units), while VAT-units may be based on enterprise groups (as in the Netherlands). Hence, when linking VAT-data and business survey-data to the target population, it is important to agree to which units data are linked.

Summarising, when linking several input data in a statistical-DWH, one has to agree about

- the unit to which all input data are matched.
- the statistical register, i.e. the reference to which all data sources are linked,

Taking into account the expected recommendations of the ESSnet on Consistency, it is proposed that the statistical enterprise unit is the standard unit in business statistics. Ideally, the statistical community should have the common goal that all Member States use a unique identifier for enterprises based on the statistical unit. Therefore, the S-DWH uses the statistical enterprise as standard units for business statistics. As long as a unique identifier for enterprises is not defined yet, data from sources not using the statistical unit are linked to the statistical unit in a statistical-DWH.

To determine the population frame in the statistical-DWH, two types of information are needed:

- The statistical register, i.e. a list of units with a certain kind of activity during a period,
- Information to determine which units of the list really performed any activities during a period.

The statistical register for business statistics consists of all enterprises within the SBR during the year, regardless of whether they are active or not. To derive activity status and subpopulations, it is recommended that the business register includes the following information:

- 1) the frame reference year
- 2) the statistical unit enterprise, including its national ID and its EGR ID⁴
- 3) the name and address of the enterprise
- 4) the date in population (mm/yr)
- 5) the date out of population (mm/yr)
- 6) the NACE-code
- 7) the institutional sector code
- 8) a size class⁵

Note that a statistical register is crucial for a statistical-DWH. Target populations, i.e. populations belonging to estimates, for the flexible outputs are derived from it!

1.3.4 Target populations of active enterprises

In line with the SBS-regulation the following definition for the enterprises of target population is used in this paper: all enterprises with a certain kind of activity being economically active during the reference period. For annual statistics this means that the target population consists of all enterprises active during the year, including the starters and stoppers (and the new/stopping units due to merging and splitting companies). Such a population is called the target population in methodological terms, i.e. the population to which the estimates refer. The NACE-code is used to classify the kind of activity.

⁴ arbitrary ID assigned by the EGR system to enterprises, it is advised to include this ID in the Data warehouse to enable comparability between the country specific estimates

⁵ could be based on employment data

Case 1: Statistical data warehouse is limited to annual business statistics

The determination of a target population with active enterprises only is relatively easy, if the scope of the statistical-DWH is limited to annual statistics. This case is relatively easy because the required information about population totals, turnover and employment can be selected afterwards, i.e. when the year has finished. This is because annual business surveys are designed after the year has ended and results of surveys and other data sources with annual business data (like accountancy data + totals of four quarters) become available after the year has ended, too. Hence, no provisional populations are needed to link provisional data during the calendar year. Therefore, the business register can be determined by

- selecting all enterprises which are recorded in the SBR during the reference year
- using the complete annual VAT and social security dataset to determine the activity status and totals for turnover and employment.

Case 2: the Statistical Data Warehouse includes short-term business statistics

The determination of a target population with only active enterprises becomes more complicated when the production of short-term statistics is incorporated in the statistical DWH. In this case a provisional business register for reference year t frame should be constructed at the end of year $t-1$, i.e. November or December. This business register is used to design short-term surveys. It is also the starting point for the statistical-DWH. This provisional frame is called release 1 and formally it does not cover the entire population of year t as it does not contain the starting enterprises yet.

During the year the backbone of the statistical-DWH is regularly updated with new information about business population (new, stopped, merged and split enterprises), activity, turnover and employment. The frequency of these updates depends on the updates of the SBR and related to this updating information provided by the admin data holders (VAT and social security data). At the end of year t (or at the beginning of year $t+1$), a regular population frame for year t can be constructed. This regular population frame consists of all enterprises in the year and is called release 2.

Case 3: the Statistical Data Warehouse includes administrative data

The ESSnet on Administrative Data has observed that time-lags do exist between the registration of starting/stopping enterprises in the SBR (if based on Chamber of Commerce data) and other admin data sources like tax information or social security data. The impact of these time-lags differs for each country, because it depends

- on the updates of both
 - the population frame in the SBR
 - VAT and social security data from the admin data holders (in the SBR),
- the quality the underlying data sources.

Despite the different impact of the time-lags, the ESSnet on Administrative Data has shown that these time-lags do exist in every country and lead to revisions in estimates about active enterprises on a monthly and quarterly basis. This effect is enhanced, because the admin data are not entirely complete on a quarterly basis. These time-lag and incompleteness issues might be a consideration for choosing a low-frequency for updating the backbone in a statistical-DWH. For example, quarterly and/or bi-annual updates could be considered.

Note that target populations can be flexible in a S-DWH, because a S-DWH is meant to produce flexible outputs. When processing and analysing data, it is recommended to consider the target populations of the annual SBS and monthly or quarterly STS. These are important obligatory statistics. More

importantly, these statistics define the enterprise population to its widest extent. According to regulations, they include all enterprises with some economic activity during (part of) the period. Hence, by using these populations as standard:

- All other data sources could be linked to this standard, because they cannot cover a wider population in the SBS/STS domain from a theoretical point of view.
- All other publications derived from the S-DWH are basically subgroups from the SBS/STS-estimates.

Furthermore, the output obligations of the annual SBS and monthly or quarterly STS are quite detailed in terms of different kind of activities (NACE-codes). We propose that the SBS and STS-output obligations are used as standard to check, link, clean and weight the input data in the processing phase of the S-DWH, too.

A S-DWH is designed to produce flexible output. However, as the standard SBS- and STS-populations are the widest in terms of economic activity during the period and quite detailed in terms of kind of activity, most other populations can be considered as subpopulations of these standards. Examples of subpopulation are:

- Large or small enterprises only,
- All active enterprises active at a certain date,
- Even more detailed kind of activity populations (i.e. estimates at NACE 3/4-digit level).

Domain estimators or other estimation techniques can be used to determine these subtotals, if the amount of available data is sufficient and there are no problems with statistical disclosure.

1.3.5 Recommended Backbone of the statistical-DWH in Business Statistics: integrated population frame, turnover and employment

The results of the ESSnet on Admin Data showed that VAT and social security data can be used for turnover and employment estimates when quasi complete. The latter is the case for annual statistics and for quarterly statistics in most European countries on the continent. Note however that VAT and social security data can only be used for statistical purposes if:

- the data transfer from the tax office to the statistical institute is guaranteed, and
- the link with the statistical unit is established.

It is possible:

- to process the VAT and employment data within the SBR
- to have separate systems for processing VAT and social security data linked to the SBR

to obtain totals for turnover and employment.

In this section we do not discuss the pros and cons of each approach as it is a partly organizational decision for the NSIs. For this section, we assume that totals are produced for

- number of enterprises
- turnover,
- employment

with administrative data covering quasi-all enterprises in the SBS/STS domain. These totals are integrated because they are all based on the statistical unit and all classified by activity by using the NACE-code from the population frame. Hence, these three integrated totals together represent the basic characteristics of the enterprise population. Therefore, these three totals can be considered as the backbone of the statistical-DWH. All other data sources are linked to these three totals in

statistical-DWH and made consistent with them. This chapter mentions some aspects for VAT and social security data.

VAT and social security cover almost all enterprises in the domain covered by the SBS and STS-regulations and are available in a timely manner (i.e. earlier than most annual statistics). They are crucial

- to determine the activity status of the enterprises and implicitly to determine the target populations of active enterprises,
- to create a fully integrated dataset suitable for flexible outputs, because these administrative data sources contain information about almost all enterprises (unlike survey which contain only information of a small sample of enterprises).

The latter reason is explained further in the remainder of this section. When (quasi) complete VAT and social security data can be used to produce good-quality estimates of turnover and employment. Therefore, these estimates can – together with the population frame (i.e. number of enterprises, NACE-code etc.) be used as benchmarks when incorporating results of survey sampling in a statistical-DWH. In this case totals of turnovers and employment define, together with the number of enterprises, the basic population characteristics. These three characteristics are assumed to be correct unless otherwise proven. Other datasets or surveys covering more specific parts of the population should be made consistent with these three main characteristics of the entire population. In the case of inconsistencies, the population characteristics are considered as correct, survey data or other datasets are modified by adapting weights or data editing. As these three main characteristics (population frame, turnover, employment) are

- integrated,
- available at micro-level (statistical unit)
- considered as correct
- and all other sources are linked and made consistent to them,

these characteristics are the backbone of the statistical-DWH in business statistics. This backbone is considered as the authoritative source of the statistical-DWH because its information is assumed to be correct unless otherwise proven.

The concept of the backbone improves the quality of integrated datasets and flexible outputs of a statistical-DWH. This is because more auxiliary information, in addition to the number of enterprises, is used when weighting survey results (or other datasets) or when imputing missing values. For example, VAT and social security data can be used as auxiliary information when weighting survey results of variables derived from surveys. Many literature studies have proven that estimates based on weighting techniques using auxiliary information (e.g. ratio or GREG-type estimators) produce lower sampling errors than estimates without using auxiliary information when weighting (when survey variables are well correlated with the auxiliary variables). Using VAT and social security data as auxiliary information when weighting also corrects for unrepresentativity in the data sources. Hence, it improves the accuracy of estimates (and reduces its biases) for variables which are derived from data sources representing a specific part of the population.

Summarizing using a backbone with integrated population, turnover and employment data

- improves the quality of a fully integrated dataset using several input data sets, as two key variables for statistical outputs (turnover and employment) can be estimated precisely,
- reduces the impact of sampling errors or biases in estimates for variables derived from other data sources, because turnover and/or employment can be used as auxiliary information when weighting.

As the first condition is the aim of a statistical-DWH and the second condition is required to produce flexible output (especially about subgroups of the standard SBS and STS-population), this is the main argumentation to consider a backbone of integrated totals of number of enterprises (=population), employment and turnover as the heart of a statistical-DWH for business statistics.

The second reason to consider a backbone with integrated data about number of enterprises (=population), employment and turnover as the heart of the statistical-DWH is the determination of the activity status of an enterprise.

A schematic sketch of the position of the backbone with integrated population, turnover and employment data is provided in figure 1.

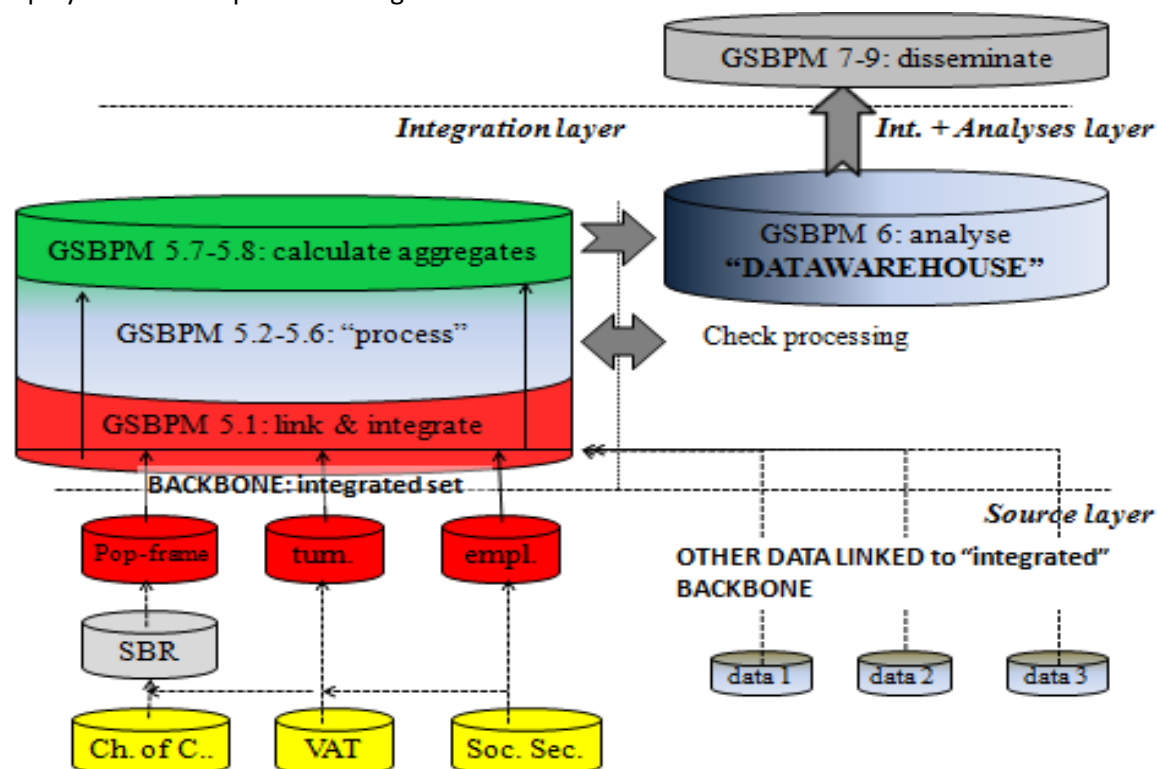


Figure 1. Position of the SBR and the backbone

Figure describes the position of the SBR and the backbone with integrated data about number of enterprises (=population), VAT-turnover and employment derived from social security data in a statistical-DWH. This backbone is represented by a line within the GSBPM phase 5.1. All other data sources are integrated to this backbone at GSBPM phase 5.1, which is at the beginning of the processing phase. The same backbone is also used for weighting when producing outputs at the end of the processing phase (see line in GSBPM steps 5.7 and 5.8). In this figure VAT, social security data and population are represented as different data sources with separate processes to integrate them. Note that this integration can also be done within the SBR (dotted lines via SBR) or outside the SBR (dotted lines directly to turnover, employment etc.).

1.4 Business processes of the layered S-DWH

The layered architecture vision was mentioned in introduction. In this sub-chapter we identify the business processes for each layer; the ground level corresponds to the area where the external sources are incoming and interfaced, while the top of the pile is where aggregated, or deliverable, data are available for external users. In the intermediate layers we manage the ETL functions for uploading the DWH in which are carried out strategic analysis, data mining and design, for possible new strategies or data re-use.

This will reflect a conceptual organization in which we will consider the first two levels as pure statistical operational infrastructures. In these first two levels necessary information is produced and functions like acquiring, storing, coding, checking, imputing, editing and validating data are performed. We consider the last two layers as the effective data warehouse, i.e. levels in which data are accessible for execute analysis, re-use of data and perform reporting. These four levels are described in figure 1.

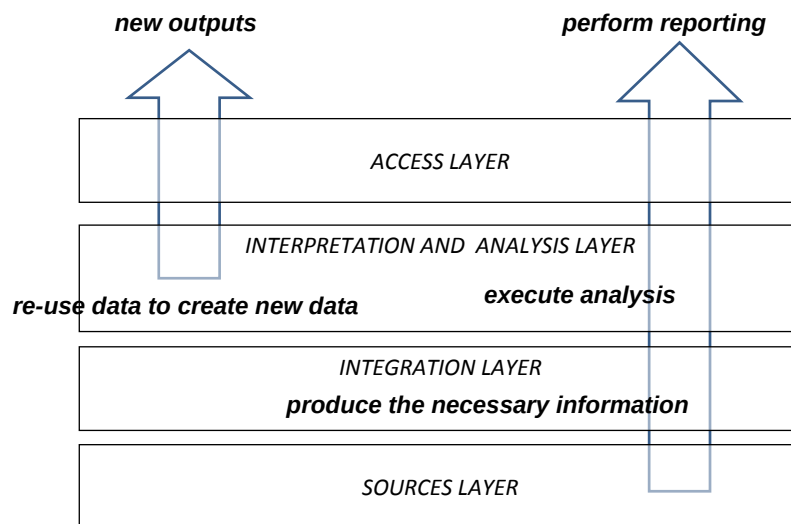


Figure 1. Business processes for layer architecture

The core of the S-DWH system is the interpretation and analysis layer, this is the effective data warehouse and must support all kinds of statistical analysis or data mining, on micro and macro data, in order to support statistical design, data re-use or real-time quality checks during productions.

The layers II and III are reciprocally functional to each other (Figure 2). Layer II always prepare the elaborated information for the layer III: from raw data, just uploaded into the S-DWH and not yet included in a production process, to micro/macro statistical data at any elaboration step of any production processes. Otherwise in layer III it must be possible to easily access and analyse this micro/macro elaborated data of the production processes in any state of elaboration, from raw data to cleaned and validate micro data. This because, in layer III methodologists should correct possible operational elaboration mistakes before, during and after any statistical production line, or design new elaboration processes for new surveys. In this way the new concept or strategy can generate a feedback toward layer II which is able to correct, or increase the quality, of the regular production lines.

A key factor of this S-DWH architecture is that layer II and III must include components of bidirectional co-operation. This means that, layer II supplies elaborated data for analytical activities, while layer III supplies concepts usable for the engineering of ETL functions, or new production processes.

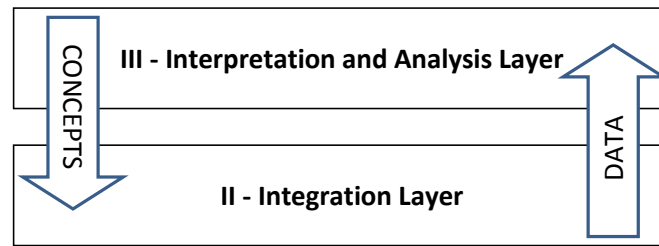


Figure 2. Bidirectional co-operation between layer II and III

These two internal layers are therefore reciprocally functional. Layer two always prepares the elaborated information for layer three: from raw data to any useful, semi or final elaborated data. This means that, in the interpretation layer, methodologists or experts should easily access all data, before, during and after the elaboration of a production line to correct or re-design a process. This is a fundamental aspect for any production based on a large, changeable, amount of data, as testing by hypotheses is crucial for any new design.

Finally, the access layer should support functionalities related to the exercise of output systems, from the dissemination web application to the interoperability. From this point of view, the access layer operates inversely to the source layer. On the access layer we should realize all data transformations, in terms of data and metadata, from the S-DWH data structure toward any possible interface tools functional to dissemination.

In the following sections we will indicate explicitly the atomic activities that should be supported by each layer using the GSBPM taxonomy.

1.4.1 Source layer processes

The Source Layer is the level in which we locate all the activities related to storing and managing internal or external data sources. Internal data are from direct data capturing carried out by CAWI, CAPI or CATI; while external data are from administrative (or other) sources, for example from Customs Agencies, Revenue Agencies, Chambers of Commerce, National Social Security Institutes. Generally, data from direct surveys are well-structured so they can flow directly into the integration layer. This is because NSIs have full control of their own applications. Differently, data from others institutions' archives must come into the S-DWH with their metadata in order to be read correctly. In the source layer we support data loading operations for the integration layer but do not include any data transformation operations, which will be realized in the next layer.

Analyzing the GSBPM shows that the only activities that can be included in this layer are:

Phase	sub-process
4- Collect:	4.2-set up collection
	4.3-run collection
	4.4-finalize collection

Table 1. Source layer sub-processes

Set up collection (4.2) ensures that the processes and technology are ready to collect data. So, this sub-process ensures that the people, instruments and technology are ready to work for any data collections. This sub-process includes:

- preparing web collection instruments,
- training collection staff,
- ensuring collection resources are available e.g. laptops,
- configuring collection systems to request and receive the data,
- ensuring the security of data to be collected.

Where the process is repeated regularly, some of these activities may not be explicitly required for each iteration.

Run collection (4.3) is where the collection is implemented, with different collection instruments being used to collect the data. Reception of administrative data belongs to this sub-process.

It is important to consider that the run collection sub-process in a web-survey could be contemporary with the review, validate & edit sub-processes. Some validation of the structure and integrity of the information received may take place within this sub-process, e.g. checking that files are in the right format and contain the expected fields.

Finalize collection (4.4) includes loading the collected data into a suitable electronic environment for further processing of the next layers. This sub-process also aims to check the metadata descriptions of all external archives entering the SDW system. In a generic data interchange, as far as metadata transmission is concerned, the mapping between the metadata concepts used by different international organizations, could support the idea of open exchange and sharing of metadata based on common terminology.

1.4.2 Integration layer processes

The integration layer is where all operational activities needed for all statistical elaboration process are carried out. This means operations carried out automatically or manually by operators to produce statistical information in an IT infrastructure. With this aim, different sub-processes are pre-defined and pre-configured by statisticians as a consequence of the statistical survey design in order to support the operational activities.

This means that whoever is responsible for a statistical production subject defines the operational workflow and each elaboration step, in terms of input and output parameters that must be defined in the integration layer, to realize the statistical elaboration.

For this reason, production tools in this layer must support an adequate level of generalization for a wide range of processes and iterative productions. They should be organized in operational work flows for checking, cleaning, linking and harmonizing data-information in a common persistent area where information is grouped by subject. These could be those recurring (cyclic) activities involved in the running of the whole or any part of a statistical production process and should be able to integrate activities of different statistical skills and of different information domains.

To sustain these operational activities, it would be advisable to have micro data organized in generalized data structures able to archive any kind of statistical production. Otherwise data should be organized in completely free form but with a level of metadata able to realize an automatic structured interface toward the data itself.

Therefore, there is wide family of possible software applications for the Integration layer activities, from Data Integration Tool, where a user-friendly graphic interface helps to build up work flow to generic statistics elaboration line or part of it.

In this layer, we should include all the sub-processes of phase 5 and one sub-process from phase 6 of the GSBPM:

Phase	sub-process
5- Process	5.1 -integrate data 5.2 -classify & code 5.3 -review and validate 5.4 -edit and impute 5.5 -derive new variables and statistical units 5.6 -calculate weights 5.7 -calculate aggregates 5.8 -finalize data files
6- Analyse	6.1 -prepare draft outputs

Table 2. Integration layer sub-processes

Integrate data (5.1), this sub-process integrates data from one or more sources. Input data can be from external or internal data sources and the result is a harmonized data set. Data integration typically includes record linkage routines and prioritising, when two or more sources contain data for the same variable (with potentially different values).

The integration sub-process includes micro data record linkage which can be realized before or after any reviewing or editing, in function of the statistical process. At the end of each production process, data organized by subject area should be clean and linkable.

Classify and code (5.2), this sub-process classifies and codes data. For example automatic coding routines may assign numeric codes to text responses according to a pre-determined classification scheme.

Review and validate (5.3), this sub-process applies to collected micro-data, and looks at each record to try to identify potential problems, errors and discrepancies such as outliers, item non-response and miscoding. It can also be referred to as input data validation. It may be run iteratively, validating data against predefined edit rules, usually in a set order. It may raise alerts for manual inspection and correction of the data. Reviewing and validating can apply to unit records both from surveys and administrative sources, before and after integration.

Edit and impute (5.4), this sub-process refers to insertion of new values when data are considered incorrect, missing or unreliable. Estimates may be edited or imputed, often using a rule-based approach.

Derive new variables and statistical units (5.5), this sub-process in this layer describes the simple function of the derivation of new variables and statistical units from existing data using logical rules defined by statistical methodologists.

Calculate weights, (5.6), this sub process creates weights for unit data records according to the defined methodology and is automatically applied for each iteration.

Calculate aggregates (5.7), this sub process creates already defined aggregate data from micro-data for each iteration. Sometimes this may be an intermediate rather than a final activity, particularly for business processes where there are strong time pressures, and a requirement to produce both preliminary and final estimates.

Finalize data files (5.8), this sub-process brings together the results of the production process, usually macro-data, which will be used as input for dissemination.

Prepare draft outputs (6.1), this sub-process is where the information produced is transformed into statistical outputs for each iteration. Generally, it includes the production of additional measurements such as indices, trends or seasonally adjusted series, as well as the recording of quality characteristics. The presence of this sub-process in this layer is strictly related to regular production process, in which the measures estimated are regularly produced, as should in the STS.

1.4.3 Interpretation and data analysis layer processes

The interpretation and data analysis layer is specifically for internal users, statisticians. It enables any data analysis, data mining and support at the maximum detailed granularity, micro data, for production processes design or individuate data re-use. Data mining is the process of applying statistical methods to data with the intention of uncovering hidden patterns. This layer must be suitable to support experts for free data analysis in order to design or test any possible new statistical methodology, or strategy.

The results expected of the human activities in this layer should then be statistical “services” useful for other phases of the elaboration process, from the sampling, to the set-up of instruments used in the process phase until generation of new possible statistical outputs. These services can, however, be oriented to re-use by creating new hypotheses to test against the larger data populations. In this layer experts can design the complete process of information delivery, which includes cases where the demand for new statistical information does not involve necessarily the construction of new surveys, or a complete work-flow setup for any new survey needed.

Case: produce the necessary information

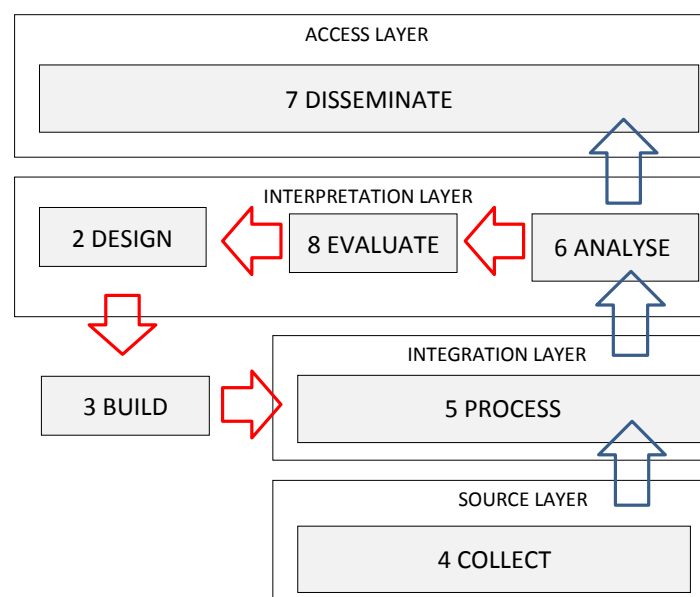


Figure 3. Produce the necessary information from S-DWH micro data

From this point of view, the activities on the Interpretation layer should be functional not only to statistical experts for analysis but also to self-improve the S-DWH, by a continuous update, or new definition, of the production processes managed by the S-DWH itself.

We should point out that an S-DWH approach can also increase efficiency in the Specify Needs and Design Phase since statistical experts, working on these phases on the interpretation layer, share the same information elaborated in the Process Phase in the integration layer.

Case: re-use data to create new data

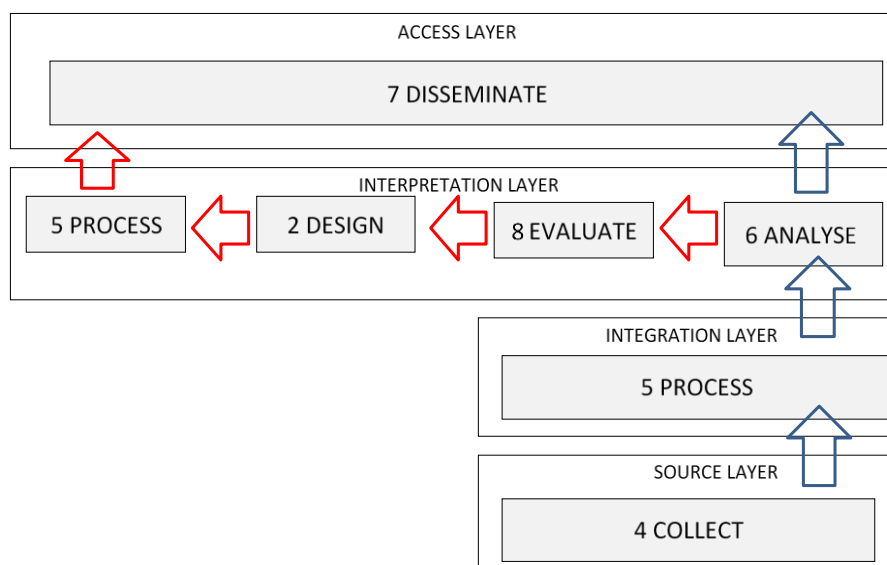


Figure 4. Re-use S-DWH microdata to create new information

The use of a data warehouse approach for statistical production has the advantage of forcing different typologies of users to share the same information data. That is, the same stored-data are usable for different statistical phases. Therefore, this layer supports any possible activities for new statistical production strategies aimed at recovering facts from large administrative archives. This would create more production efficiency and less of a statistical burden and production costs.

From the GSBPM then we consider:

1- Specify Needs:	1.5 - check data availability
2- Design:	2.1-design outputs 2.2-design variable descriptions 2.4-design frame and sample 2.5-design processing and analysis 2.6-design production systems and workflow
4- Collect:	4.1-create frame and select sample
5- Process	5.1-integrate data 5.5-derive new variables and units 5.6-calculate weights 5.7-calculate aggregates
6- Analyze	6.1-prepare draft outputs 6.2-validate outputs 6.3-interpret and explain outputs 6.4-apply disclosure control 6.5-finalise outputs
7- Disseminate	7.1-update output systems
8- Evaluate	8.1- gather evaluation inputs 8.2- conduct evaluation

Table 3. Interpretation and data analysis layer sub-processes

Check data availability (1.5), this sub-process checks whether current data sources could meet user requirements, and the conditions under which they would be available, including any restrictions on their use. An assessment of possible alternatives would normally include research into potential administrative data sources and their methodologies, to determine whether they would be suitable for use for statistical purposes. When existing sources have been assessed, a strategy for filling any remaining gaps in the data requirement is prepared. This sub-process also includes a more general assessment of the legal framework in which data would be collected and used, and may therefore identify proposals for changes to existing legislation or the introduction of a new legal framework.

Design outputs (2.1), this sub-process contains the detailed design of the statistical outputs to be produced, including the related development work and preparation of the systems and tools used in phase 7 (Disseminate). Outputs should be designed, wherever possible, to follow existing standards. Inputs to this process may include metadata from similar or previous collections or from international standards.

Design variable descriptions (2.2), this sub-process defines the statistical variables to be collected via the data collection instrument, as well as any other variables that will be derived from them in sub-process 5.5 (Derive new variables and statistical units), and any classifications that will be used. This sub-process may need to run in parallel with sub-process 2.3 (Design collection) as the definition of the variables to be collected and the choice of data collection instrument may be inter-dependent to some degree. The interpretation layer can be seen as a simulation environment able to identify the effective variables needed.

Design frame and sample methodology (2.4), this sub-process identifies and specifies the population of interest, defines a sampling frame (and, where necessary, the register from which it is derived), and determines the most appropriate sampling criteria and methodology (which could include complete enumeration). Common sources are administrative and statistical registers, censuses and sample surveys. This sub-process describes how these sources can be combined if needed. Analysis of whether the frame covers the target population should be performed. A sampling plan should be made: The actual sample is created sub-process in 4.1 (Create frame & select sample), using the methodology, specified in this sub-process.

Design processing and analysis (2.5), this sub-process designs the statistical processing methodology to be applied during phase 5 (Process), and Phase 6 (Analyse). This can include specification of routines for coding, editing, imputing, estimating, integrating, validating and finalising data sets.

Design production systems and workflow (2.6), this sub-process determines the workflow from data collection to archiving, taking an overview of all the processes required within the whole statistical production process, and ensuring that they fit together efficiently with no gaps or redundancies. Various systems and databases are needed throughout the process. A general principle is to reuse processes and technology across many statistical business processes, so existing systems and databases should be examined first, to determine whether they are fit for purpose for this specific process, then, if any gaps are identified, new solutions should be designed. This sub-process also considers how staff will interact with systems, and who will be responsible for what and when.

Create frame and select sample (4.1), this sub-process establishes the frame and selects the sample for each iteration of the collection, in line with the design frame and sample methodology. This is an interactive activity on statistical business registers typically carried out by statisticians using advanced methodological tools.

Sub process includes the coordination of samples between instances of the same statistical business process (for example to manage overlap or rotation), and between different processes using a common frame or register (for example to manage overlap or to spread response burden).

Integrate data (5.1), in this layer this sub-process makes it possible for experts to freely carry out micro data record linkage from different information data sources when these refer to the same statistical analysis unit.

In this layer this sub-process must be intended as an evaluation for the data linking design, wherever needs.

Derive new variables and units (5.5), this sub-process derives variables and statistical units that are not explicitly provided in the collection, but are needed to deliver the required outputs. In this layer this function would be used to set up procedures or for defining the derivation roles applicable in each production iteration. In this layer this sub-process must be intended as an evaluation for evaluation on designing new variable.

Calculate weights (5.6), see chapter 1.4.2.

Calculate aggregates (5.7), see chapter 1.4.2.

Prepare draft outputs (6.1), in this layer this sub-process means the free construction of not regular outputs.

Validate outputs (6.2), this sub-process is where statisticians validate the quality of the outputs produced. Also this sub process is intended as a regular operational activity, and the validations are carried out at the end of each iteration on an already defined quality framework.

Interpret and explain outputs (6.3) this sub-process is where the in-depth understanding of the outputs is gained by statisticians. They use that understanding to interpret and explain the statistics produced for this cycle by assessing how well the statistics reflect their initial expectations, viewing the statistics from all perspectives using different tools and media, and carrying out in-depth statistical analyses.

Apply disclosure control (6.4), this sub-process ensures that the data (and metadata) to be disseminated do not breach the appropriate rules on confidentiality. This means the use of specific methodological tools to check the primary and secondary disclosure

Finalise outputs (6.5), this sub-process ensures the statistics and associated information are fit for purpose and reach the required quality level, and are thus ready for use.

Update output systems (7.1), this sub-process manages update to systems where data and metadata are stored for dissemination purposes.

Gather evaluation inputs (8.1), evaluation material can be produced in any other phase or sub-process. It may take many forms, including feedback from users, process metadata, system metrics and staff suggestions. Reports of progress against an action plan agreed during a previous iteration may also form an input to evaluations of subsequent iterations. This sub-process gathers all of these inputs, and makes them available for the person or team producing the evaluation.

Conduct evaluation (8.2), this process analyses the evaluation inputs and synthesizes them into an evaluation report. The resulting report should note any quality issues specific to this iteration of the statistical business process, and should make recommendations for changes if appropriate. These recommendations can cover changes to any phase or sub-process for future iterations of the process, or can suggest that the process is not repeated.

1.4.4 Access layer processes

Access Layer is the layer for the final presentation, dissemination and delivery of the information sought. This layer is addressed to a wide typology of external users and computer instruments. This layer must support automatic dissemination systems and free analysts tools, in both cases, statistical information are mainly macro data not confidential, we may have micro data only in special limited cases.

This typology of users can be supported by three broad categories of instruments:

- A specialized web server for software interfaces towards other external integrated output systems. A typical example is the interchange of macro data information via SDMX, as well as with other XML standards of international organizations.
- Specialized Business Intelligence tools. In this category, extensive in terms of solutions on the market, we find tools to build queries, navigational tools (OLAP viewer), and in a broad sense web browsers, which are becoming the common interface for different applications. Among these we should also consider graphics and publishing tools able to generate graphs and tables for users.
- Office automation tools. This is a reassuring solution for users who come to the data warehouse context for the first time, as they are not forced to learn new complex instruments. The problem is that this solution, while adequate with regard to productivity and efficiency, is very restrictive in the use of the data warehouse since these instruments, have significant architectural and functional limitations

In order to support this different typology of instruments, this layer must allow the transformation of data-information already estimated and validated in the preview layers by automatic software.

From the GSBPM we may consider only the phase 7 for operational process and specifically:

7- Disseminate	7.1-update output systems 7.2-produce dissemination products 7.3-manage release of dissemination products 7.4-promote dissemination products 7.5-manage user support
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Table 4. Access layer sub-processes

Update output systems (7.1) this sub-process in this layer manages the output update adapting the already defined macro data to specific output systems, including re-formatting data and metadata into specific output databases, ensuring that data are linked to the relevant metadata. This process is related with the interoperability between the access layer and others external system; e.g. toward the SDMX standard or other Open Data infrastructure.

Produce dissemination products (7.2), this sub-process produces final, previously designed statistical products, which can take many forms including printed publications, press releases and web sites. Typical steps include:

- Preparing the product components (explanatory text, tables, charts etc.).
- Assembling the components into products.
- Editing the products and checking that they meet publication standards.

The production of dissemination products is a sort of integration process between table, text and graphs. In general this is a production chain in which standard table and comments from the interpretation of the produced information are included.

Manage release of dissemination products (7.3), this sub-process ensures that all elements for the release are in place including managing the timing of the release. It includes briefings for specific groups such as the press or ministers, as well as the arrangements for any pre-release embargoes. It also includes the provision of products to subscribers.

Promote dissemination products (7.4), this sub-process concerns the active promotion of the statistical products produced in a specific statistical business process, to help them reach the widest possible audience. It includes the use of customer relationship management tools, to better target potential users of the products, as well as the use of tools including web sites, wikis and blogs to facilitate the process of communicating statistical information to users.

Manage user support (7.5), this sub-process ensures that customer queries are recorded, and that responses are provided within agreed deadlines. These queries should be regularly reviewed to provide an input to the over-arching quality management process, as they can indicate new or changing user needs.

1.4.5 Data linking process

The purpose of this section is to make overview on data linking in a Statistical Data Warehouse and to mention problems that we can meet linking data from multiple sources. Data linking methods and present guidelines about methodological challenges on data linking are discussed in methodological Chapter.

The main goal of the S-DWH process is to increase the better use of data already exist in the National Statistical institute. First and the main step in data linking process is to determine needs and check data availability. It is considered to have all available data of interest in S-DWH.

Proposed scope of input data set:

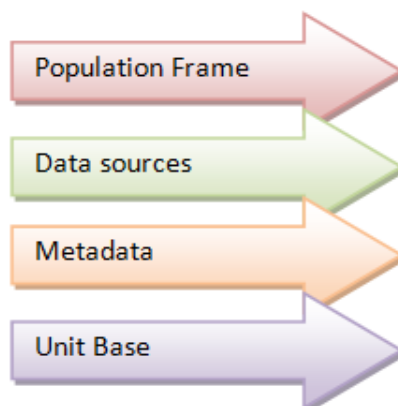


Figure 5. Proposed scope of input data set

The difference between data linking and integration

Data linking is linking the different input sources (administrative data, surveys data, etc.) to one population and processing this data to one consistent dataset that will greatly increase the power of analysis then possible with the data.

While data integration according to GSBPM model 5.1 sub-process it is a process that integrates data from one or more sources. The input data can be from a mixture of external or internal data sources,

and a variety of collection modes, including extracts of administrative data. The result is a harmonized data set. Data integration typically includes:

- Matching / record linkage routines, with the aim of linking data from different sources, where those data refer to the same unit.
- Prioritising, when two or more sources contain data for the same variable (with potentially different values).

Data integration may take place at any point in process phase, before or after any of the other sub-processes. There may also be several instances of data integration in any statistical business process. Following integration, depending on data protection requirements, data may be anonymized, that is stripped of identifiers such as name and address, to help to protect confidentiality.

Data integration process put data from disparate sources into a consistent format. Must be resolved such problems as naming conflicts and inconsistencies among units of measure. When this is achieved, data are said to be integrated.

Data integration is a big opportunity for NSIs, it opening up possibilities for reducing costs, leads to reduced survey burden on respondents and may increase data quality. But also it is a big challenge, a lot of preparatory work must be done by NSIs, should be examined the data sources, the metadata should be defined before linking data. There are a lot of issues and questions that should be analysed and answered in order to create fully integrated data sets for enterprise and trade statistics at micro level.

If the data include error-free and unique common identifiers as a unique identification code of the legal entity or a social security number, record linkage is a simple file merge operation which can be done by any standard database management system. In other cases it is necessary to resort to a combination of ambiguous and error-prone identifiers as surnames, names, address, NACE code information. Data quality problems of such identifiers usually yield a considerable amount of unlinkable cases. In this situation the use of much more sophisticated techniques and specialised record linkage software is inevitable. These techniques are discussed in methodological Chapter.

In a Data Warehouse system the Statistical Register has a crucial role in linking data from several sources and defining the population for all statistical output.

The statistical unit base in business statistics

The statistical community should have the aim that all Member States use a unique identifier for enterprises based on the statistical unit having the advantage that all data sources can be easily linked to the statistical-DWH. In practice, data holders may use several definitions of enterprises in some countries. As a result, several enterprises units may exist. Related to this, different definitions of units may also exist when producing output (LKAU, KAU, etc.).

The relationship between the different in- and output units on the one hand and the statistical enterprise units on the other hand should be known (or estimated) before the processing phase, because it is a crucial step for data linking and producing output. Maintaining this relationship in a database is recommended when outputs are produced by releases; e.g. newer more precise estimates when more data (sources) become available. This prevents redoing a time-consuming linking process at every flexible estimate.

It is proposed that the information about the different enterprise units and their relationships at micro level is kept by using the concept of a so-called unit base. This base should at least contain:

- The statistical enterprise, which is the only unit used in the processing phase of the statistical-DWH.

- The enterprise group, which is the unit for some output obligations. Moreover the enterprise group may be the base for tax and legal units, because in some countries, like the Netherlands, the enterprise unit is allowed to choose its own tax and legal units of the underlying enterprises.

The unit base contains the link between the statistical enterprise, the enterprise group and all other units. Of course, it should also include the relationship between the enterprise group and the statistical enterprise. In case of x-to-y relationships between the units, i.e. one statistical unit corresponds with several units in another data source or vice versa, the estimated share in terms of turnover (or employment) of the 'data source' units to the corresponding statistical enterprise(s) and enterprise group needs to be mentioned. This share can be used to relate levels of variables from other data sources based on enterprises unit x1 to levels of turnover and employment in the backbone based on the (slightly different) statistical enterprise unit x2. We refer to deliverable 2.4 of the ESSnet on data warehousing¹ for further information about data linking and estimating shares.

The unit base can be subdivided into 'input' units, used to link the different dataset to the statistical enterprise unit at the beginning of the processing phase (GSBPM-step 5.1: "integrate data") and 'output' unit used to produce output on units other than the statistical enterprise at the end of the processing phase (GSBPM-step 5.5 "derive new variable and units").

Figure 6 illustrates the concept of a unit base. It shows that the unit base can be subdivided into

- input units, used to link the data sources to the statistical enterprise unit at the beginning of the processing phase (GSBPM-step 5.1: "integrate data")
- output units, which are used to produce output about units other than the statistical enterprise at the end of the processing phase (GSBPM-step 5.5 "derive new variable and units"). An example is output about 'enterprise groups' LKAUs etc...

¹ The document is available at: http://ec.europa.eu/eurostat/cros/content/deliverables-10_en

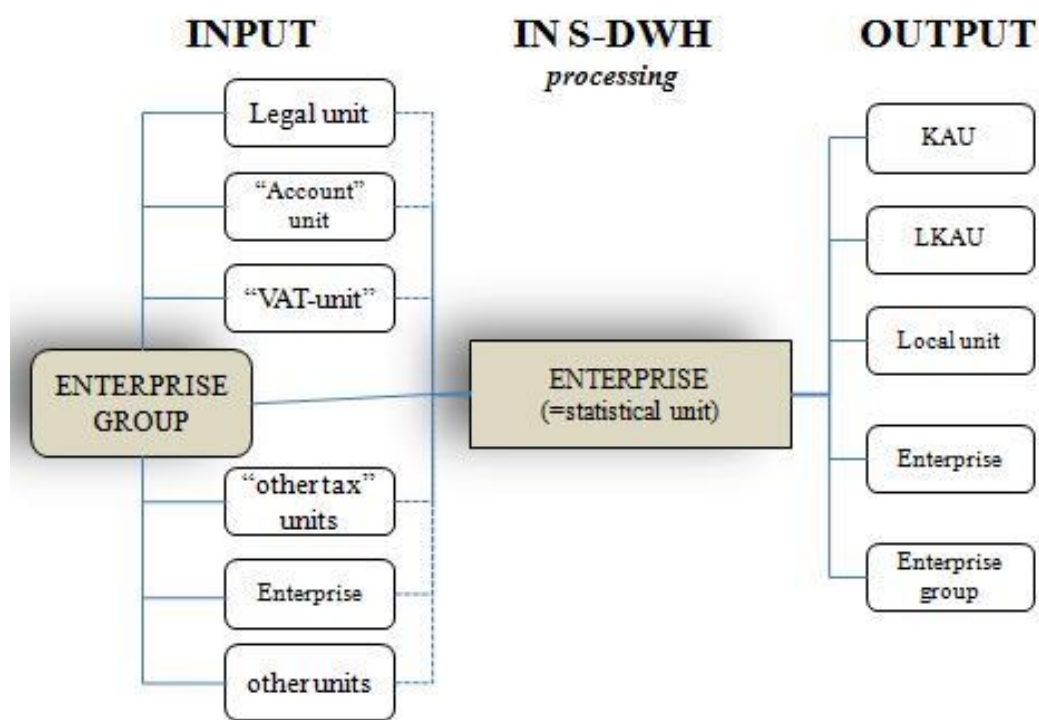


Figure 6. Proposed scope of input data set

The exact contents of the unit base (and related to this its complexity) depends on

- legislation for a particular country,
- output requirements and desired output of a statistical-DWH,
- available input data.

It is a matter of debate

- whether the concept of a unit base should be included in the SBR
- or whether the concept of a unit base should result in a physically independent database.

In the case of the latter it is closely related to the SBR, because both contain the statistical enterprise. Basically, the choice depends on the complexity of the unit base. If the unit base is complex, the maintenance becomes more challenging and a separate unit base might be considered. The complexity depends on

- the number of enterprise unit in a country
- the number of (flexible) data sources an NSI uses to produce statistics.

As these factors differ by country and NSI, the decision to include or exclude the concept of a unit base in the SBR depends on the individual NSI and won't be discussed further in this paper. However, the Unit Base is essential for data linking process. You need to have established links between data to make the process of data integration fluid, accurate and quality assured.

Linking data sources to the statistical unit

When we are linking data from different sources like sample surveys, combined data and administrative data we can meet such problems as data missing, data overlapping, “unlinked data”

etc. Errors might be detected in statistical units and target population when linking other data to this information. And if these errors are influential they need to be corrected in the S-DWH.

The simplest and most transparent statistical process can be generated by

- Linking all input sources to the statistical enterprise unit at the beginning of the processing phase (GSBPM-step 5.1).
- Performing data cleaning, plausibility checks and data integration on statistical units only (GSBPM steps 5.2-5.5).
- Producing statistical output (GSBPM-steps 5.7-5.8) by default on the statistical unit and the target populations according to the SBS and STS regulations. Flexible outputs on other target populations and other units are also produced in these steps by using repeated weighting techniques and/or domain estimates.

Note that it is theoretically possible to perform data analysis and data cleaning on several units simultaneously. However, the experience of Statistics Netherlands with cleaning VAT-data on statistical units and 'implementing' these changes on the original VAT-units too, reveal that the statistical process becomes quite complex. Therefore, it is proposed that

- linking to the statistical units is carried out at the beginning of the processing phase only,
- the creation of a fully integrated dataset is done for statistical units only,
- statistical estimates for other units are produced at the end of the processing phase only,
- relationships between the different in- and output units on the one hand and the statistical enterprise units on the other hand should be known (or estimated) beforehand.

1.4.6 Correcting information in the population frame and feedback to Statistical register

The statistical register is the sampling frame for the surveys, which are an important data-source of the statistical-DWH (for variables which cannot be derived from admin data). This point implies that errors in the backbone source, which might be detected during the statistical process, should be incorporated in the statistical register. Hence, a process to incorporate revised information from the backbone in the statistical-DWH to the statistical register should be established. By not doing this, the same errors will return in survey results in subsequent periods.

The key questions are:

- At which step of the process of the statistical-DWH is the backbone corrected when errors are detected?
- How is revised information from the backbone of integrated sources in the statistical-DWH incorporated in the statistical register?

Backbone should be corrected and the feedback to the statistical register should be provided in parallel with the GSBPM sub-processes 5.7 and 5.8 after the GSBPM sub-processes 5.1-5.6 of the process phase (GSBPM 5) in the integration layer are carried out. Deliverable 2.2.2 of the ESSnet of data Warehousing² addresses the question how this feedback process should be implemented in case of S-DWH for business statistics and business register. Deliverable discusses also how the timing of this feedback should be handled.

² Deliverable WH-SGA2-WP2 - 2.2.2 Guidelines on how the BR interacts with the S-DWH is available on theCROS portal: http://ec.europa.eu/eurostat/cros/content/deliverables-10_en

retrieval. A scientific-workflow has a strong dependence on data and on the easy access to the transformation procedure in order to quickly adapt the process to any possible source changes. In order to efficiently organize the WF⁵ with the aim to support the production processes and improve quality, it is necessary to connect several entities such as the source variables and the related documentation. It is also important to gather and record the versions of any entity in order to fully document the process and guarantee its quality, reliability, replicability and reusability. A systematic collection of all the tested attempts could also contribute to the production efficiency because the researcher's team would be able to examine all past discarded hypotheses.

⁵ N. Russell, A. Ter Hofstede, D. Edmond, W. van der Aalst. 2005. Workflow data patterns. In Proc. of 24th Int. Conf. on Conceptual Modeling Springer. Verlag: october.

2-Governance

2.1 Governance of the metadata

Authors: Harold Kroeze, Sónia Quaresma

2.2 Management processes

Authors: Antonio Laureti Palma,

2.3 Type of analysts

Authors: Sónia Quaresma

References

Viviana De Giorgi, Michel Lindelauf; ESS-Net-DWH Deliverable 1.5: "Recommendations and guidelines on the governance of metadata management in the S-DWH"

Antonio Laureti Palma, Sónia Quaresma; ESS-Net-DWH Deliverable 3.1:"S-DWH Business Architecture"

2.1 Governance

The context in which governance moves is aiming to ensure data and statistics quality (European Parliament, 2009).

Implementing good governance for metadata management is of primary importance for a S-DWH. As a matter of fact the effective governance of metadata management introduces an outside part (the governance itself) that assures a more objective view on what works and what does not, why it works (or not), and which decisions are to be taken and put into practice.

We will begin by defining what we understand in this context as governance and how it should interact and determine management, specifically in the case of metadata within a data warehouse.

This block provides an overview of recommendations for governance of metadata management in a corporate context. It is not sensible, and probably also not possible, to prescribe an ideal model for corporate governance of metadata. Every statistical organization works under different legislation, organizational arrangements, organization culture, business rules and levels of autonomy with respect to other public sector agencies.

2.1.1 Definitions

It's undeniable the importance of reliable governance in a statistical organization when dealing with S-DWHs. We will focus on the main issues to consider when establishing, running and maintaining metadata management in a S-DWH.

When, in the context of the S-DWH, defining governance of metadata management and the differences between them we should say that

- a) governance is what a corporate governing body asks metadata management to do, that is how strategies for policies, processes, procedures and rules are set and implemented;

whereas

- b) metadata management concerns the day-to-day operations of the metadata system within the context established by the governance.

The boundary between governance and management should be clear and well defined, in order to separate responsibilities between those who govern and those who manage.

Although metadata management in the S-DWH can be either decentralized or centralized depending on the organization, governance should always be centralized.

Figure 1 shows the positions of governance and management of metadata in the S-DWH.

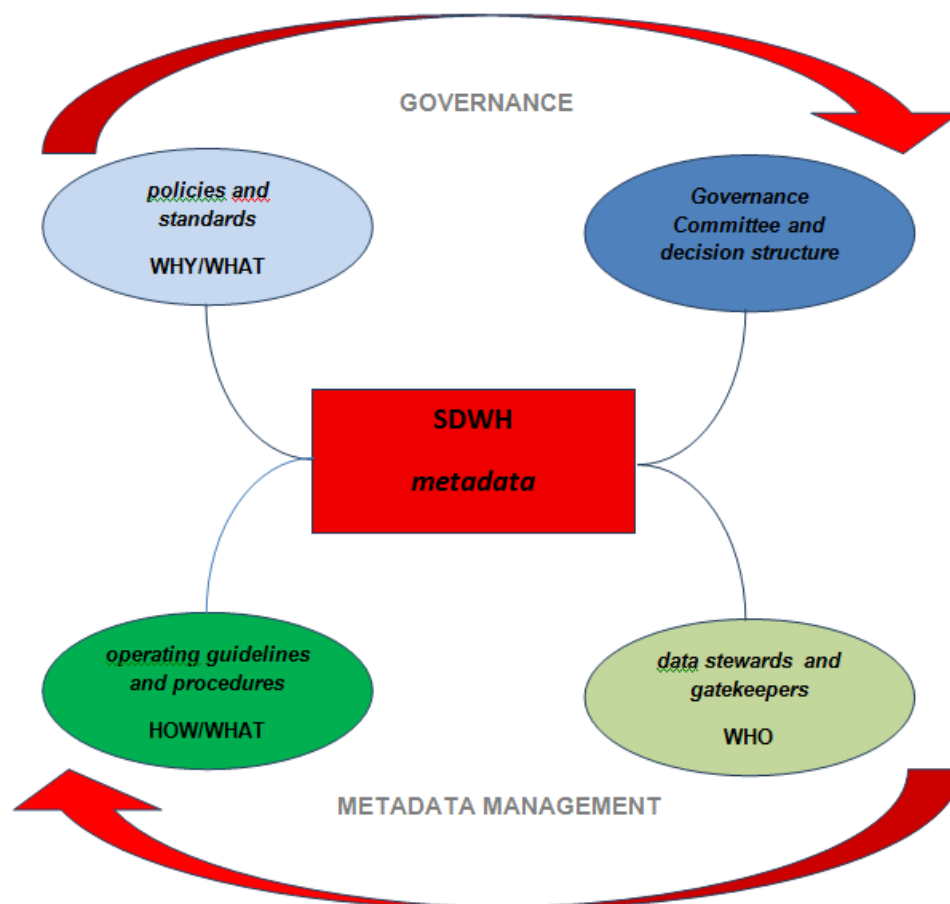


Figure 1: Governance and metadata management in S-DWH

2.1.2 Users of governance in the S-DWH

The most common users of metadata management guidelines are: managers, designers, subject-matter specialists, statisticians, methodologists, information technology experts and researchers as they are all involved in the metadata management (UNECE, 2009, p. 6). They are responsible for different aspects of

it, but need a common understanding of its role and complexity. Only then it is possible to assure a culture of teamwork and a clear and consistent communication of the management issues in the overall system and more specifically in the S-DWH.

2.1.3 Role and functionalities of metadata management

Metadata is the DNA of the statistical data warehouse, defining its elements and how they work together. Thus, metadata plays a vital role in the S-DWH, satisfying two essential needs:

- a. to *guide* statisticians in processing and controlling the statistical production
- b. to *inform* end users by giving them insight in the exact meaning of statistical data

In order to meet these two essential functions, the statistical metadata must be:

- **correct and reliable** - the metadata must give a correct picture of the statistical data,
- **consistent and coherent** - the metadata driving the statistical processes and the reporting metadata presented to the end users must be compatible with each other,
- **standardized and coordinated** - the data of different statistics are described and documented in the same standardized way.

The METIS project aiming to facilitate harmonization of data models and structures for statistical metadata in the context of statistical information processing and dissemination, has defined core principles for metadata management (Unece 2009, chapter 6), on headlines:

- **Metadata handling**
 - Manage metadata with focus on the GSBPM;
 - Make metadata active as much as possible, as a key driver for processes and actions;
 - Maximize the reuse of metadata.
- **Metadata authority**
 - Secure the process of metadata registration to realize good documentation and clear identification (ownership, approvals status, date of operation etc.) of all meta elements;
 - Ensure 1 single registration authority ('single source') for each metadata element;
 - Minimize variations from the applied standards and if, tightly document them.

- **Relationship to statistical business processes**
 - Make the metadata work an integral part of all statistical production processes;
 - Metadata as presented to end users must match the metadata driving and/or created in business processes;
 - Capture metadata at their source
 - Exchange and use metadata for informing systems and users.
- **Users**
 - Ensure clear identification of all users;
 - Recognise the diversity of metadata, different users require different formats;
 - Ensure availability of metadata for all users' information needs.

Since metadata users have diverse needs, effective management of statistical metadata is strategically important for any NSI; identifying users and their needs is a crucial part of a statistical metadata system. The metadata management in a S-DWH implies functionalities for managing metadata, user rights on data, metadata models and other metadata structures.

Metadata experts together with information technology specialists and methodologists together with subject-matter statisticians have important roles in managing statistical metadata. As a matter of fact, metadata has no more the only role of supporting statistical production, more and more it also takes the role to facilitate the efficient functioning and further development of the whole S-DWH. This requires corporate commitment and systematic management of activities related to design, implementation, maintenance, use, and evaluation of the S-DWH.

Managing metadata in a knowledge management solution is an important step in metadata strategy. It is part of the strategy to make sure that the metadata are complete, current and correct at any given time.

A framework for a corporate metadata management strategy should be specified in the corporate vision. It is strongly recommended that the top management of the organization is directly involved in the statistical metadata system and its management, but experience shows that this can be hard to achieve. At least the senior management should play a leading role in a corporate management model.

2.1.4 Governance roles: Who does what in governance?

Governance has to guarantee that metadata management achieves its objectives in an effective and transparent manner. In practice, it represents the “authority” that gives the policy for metadata management, the body for the purpose of administering the metadata management. In the context of a S-DWH within a NSI, the governance of metadata management can be seen as the body that builds standards and requirements for metadata management in alignment within the institutional vision.

Therefore, governance of metadata management defines the way to govern, monitor and measure different aspects of metadata management: it relates to non-tangible resources. Governance encompasses:

- 1) the people,
- 2) the procedures, and
- 3) the practices that ensure that:
 - a) metadata management improves, develops and maintains such results, and
 - b) the right management at the right time in the right way is running (UNECE, 2009).

Governance is

- a) horizontal to the entire metadata management from the process perspective and in time (from input to output)
- b) vertical from the organisational perspective (from top to bottom). It regards all levels and types of metadata management.

Governing staff could be involved in management, and vice versa, but this should not be the rule. It depends on the experience of the NSI who manages the S-DWH and on the legal, regulatory, and institutional environment.

Establishing an active permanent structure for governance of metadata management, such as a governance board, committee or department, is crucial for metadata management and therefore for a S-DWH. So, it does not represent an option or a feature of the S-DWH, instead it contributes heavily to metadata management success.

The governance committee (or board or department) should include, as permanent members, senior representatives from across all the production

departments involved in the S-DWH. Not necessarily the committee members have to come from metadata management background or be in the field of statistical metadata; instead they should be devoted to the institutional role of governing, monitoring and measuring metadata management. Members may also be non-permanent, temporary or even invited representatives, e.g. from external bodies and other committees or specialists from departments involved, so that they can examine the actions of governance on their administrations/departments or assess the feasibility of new actions.

In general a coordinator chairs the committee and a secretary manages all the practical staff (meetings, documentations, contacts, emails, etc.).

At any time the governance committee can identify and recruit further resources/experts to assist it in its role and responsibilities or also to be given metadata responsibilities. The committee may also consult metadata management specialists on taking crucial decisions. Audits on statistical, methodological and technical aspects of metadata management are not mandatory, but they are likely to be done by the committee. Instead it should develop procedures for monitoring such aspects and their quality. The committee is also responsible for overseeing the activities of evaluation and promotion of metadata management processes, specially new processes.

The governance committee meets up together periodically or when it is necessary, draws up a report indicating the decisions taken, and discloses them to metadata management. The committee is in fact responsible for establishing the best way for managing management. As the committee is involved in developing the vision, formulating policy, approving development plans and evaluating progress of metadata management with the institutional alignment, its activities must represent the expression of all members of the committee.

Metadata management improvements and developments: advances in metadata management should be evaluated deliberately and suitably any time it is necessary and are subjects of meetings, in which a resolution is debated and decided.

Metadata management maintenance: the governance fulfils the on-going obligation of management results maintenance over the time.

2.1.5 Governance functions, responsibilities and rules for governance

Looking at good practices for governance, we can state that each NSI that needs to set up and implement a metadata management strategy needs to evaluate its own objectives, strategies and organizational arrangements. Therefore it is helpful to consider the experience generated by organizations that have already done this.

The METIS project also provides examples of lessons for good corporate governance metadata management (Unece, 2009, chapter 7), based on experiences of statistical organizations in the implementation of a metadata management strategy. The most essential examples of these lessons are:

- Senior managers, including the Chief Statistician, should be closely involved in developing the vision, formulating policy, approving statistical metadata system development plans and evaluating progress.
- The roles and accountability of all organizational units with respect to metadata should be clear. A 'corporate data management unit' could be responsible for providing client support, developing and maintaining infrastructure and providing training.
- Make sure that the organization endorses a metadata strategy and that this strategy is integrated into broader corporate plans and strategies.
- Metadata management is strategic for the organization, but there is often scepticism in the organization against it. All managers across different levels and parts of the organization must be committed.
- Systematically use metadata systems for capturing and organizing tacit knowledge of individuals in order to make it available to the organization as a whole and to external users of statistics.

Responsibilities of the governance involve important functionalities (OECD, 2004), such as:

- **Direction.** The governance exercises functions to optimize the use of all (technological and other) resources. It establishes the potentialities and uses of metadata management, reviews documentation, and establishes guidelines. If needed, it should also monitor the effectiveness of the governance arrangements and make due changes.
- **Monitoring metadata management.** The governance monitors management performance and efficiency, by appropriate rules and procedures in order to improve them.

- **Managing conflicts.** Monitoring and solving the potential conflicts due to deficient management, in order to develop new performances is a responsibility of the governance.
- **Communication with management.** The governance should ensure adequate consultation, communication, transparency, and disclosure to metadata management, in order to keep high performances maintained.
- **Managing risks.** The governance establishes a policy for managing risks and monitoring the implementation of the project. Being able to guarantee timely answers is crucial when critical situations arise.
- **Cooperation.** The governance should cooperate with other groups or committee in order to make sure that common standards and terminology are used, point out cross objectives and reduce redundancies. This is also to ensure that the right management happens at the right time in the correct way.
- **Evaluation.** Both ex-ante and on-going evaluations are part of the same tool: the former is a compass; the latter is for learning how to achieve the best results. Ex-post evaluation is intended to assess that the agreed decision has given expected results. The evaluation is the last, but not least, part of the governance role of governing, monitoring and measuring.

In addition practice shows us that it is sensible to refer to accepted principles of good and effective governance (OECD, 2004):

- **Legitimacy.** The legitimacy of the governance of metadata management of the S-DWH lies in the establishment of the S-DWH itself. The governance is in fact justified in carrying out its institutional roles.
- **Accountability.** The accountability of governance of metadata management exists as a consequence of the relationship with metadata management that represents the extent to which accountability is defined, accepted, and exercised. There may be also mutual accountability between governance and management.
- **Responsibility.** If the committee understands and manages well its responsibility/responsibilities, it achieves good governance.
- **Transparency.** Transparency concerns the open and free availability to metadata management of decision-making, reporting, and evaluation processes. Standards of good practices can be incorporated in this principle.

- **Efficiency.** Efficiency is mainly related to the enhancement of efficiency or cost-effectiveness in the allocation and use of resources
- **Probity.** Probity refers to the adherence by all persons involved in governance and metadata management to high performance and specialists. Quality, relevance and effectiveness of governance depend on probity.

Governance has to decide on how determine changing in policies, processes, procedures, and rules for the statistical metadata system, such as versioning, inputting, deleting, and updating metadata. Governance provides information on whether they are fit for purpose and on capabilities allowing these changes to take place.

Once decided they are fit for purpose and agreed, because they are considered and planned as necessary, governance assesses that all changes are in line with the organizational vision, e.g. promoting common terminology and standards and consolidated metadata repositories, facilitating reuse of metadata, increase of knowledge and statistical integration, aiming to high metadata quality. Approved, documented and visible changes are recommended.

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2.2 Management processes of the S-DWH

In a S-DWH fourteen over-arching statistical processes are needed to support the statistic production processes, nine of them are the same as in the GSBPM, while the remaining five are the consequence of a full active S-DWH approach.

In line with the GSBPM, the first 9 over-arching processes are¹:

1. statistical program management – This includes systematic monitoring and reviewing of emerging information requirements and emerging and changing data sources across all statistical domains. It may result in the definition of new statistical business processes or the redesign of existing ones
2. quality management – This process includes quality assessment and control mechanisms. It recognizes the importance of evaluation and feedback throughout the statistical business process
3. metadata management – Metadata are generated and processed within each phase, there is, therefore, a strong requirement for a metadata management system to ensure that the appropriate metadata retain their links with data throughout the different phases
4. statistical framework management – This includes developing standards, for example methodologies, concepts and classifications that apply across multiple processes
5. knowledge management – This ensures that statistical business processes are repeatable, mainly through the maintenance of process documentation
6. data management – This includes process-independent considerations such as general data security, custodianship and ownership
7. process data management – This includes the management of data and metadata generated by and providing information on all parts of the statistical business process. (process management is the ensemble of activities of planning and monitoring the performance of a process) operations management is an area of management concerned with overseeing, designing, and controlling the process of production and redesigning business operations in the production of goods or services
8. provider management – This includes cross-process burden management, as well as topics such as profiling and management of contact information (and thus has particularly close links with statistical business processes that maintain registers)

¹ <http://www1.unece.org/stat/platform/download/attachments/8683538/GSBPM+Final.pdf?version=1>

9. customer management – This includes general marketing activities, promoting statistical literacy, and dealing with non-specific customer feedback.

In addition, we should include five more over-arching management processes in order to coordinate the actions of a fully active S-DWH infrastructure; they are:

10. S-DWH Management: - This includes all activities able to support the coordination between: statistical framework management, provider management, process data management, data management
11. data capturing management – This include all activities related with a direct, statistical or computer, support (help-desk) to respondents, i.e. provision of specialized customer care for web-questionnaire compilation or toward external institution for acquiring archives.
12. output management, for general marketing activities, promoting statistical literacy, and dealing with non-specific customer feedback.
13. web communication management, includes data capturing management, customer management and output management; this includes for example should be the effective management of a statistical web portal, able to support all front-office activities.
14. business register management (or for institutions or civil registers) – this is a trade register kept by the registration authorities and is related to provider management and operational activities.

By definition, an S-DWH system includes all effective sub-processes needed to carry out any production process. Web communication management handles the contact between respondents and NSIs, this includes providing a contact point for collection and dissemination of data over internet. It supports several phases of the statistical business process, from collecting to disseminating, and at the same time provides the necessary support for respondents.

The BR Management is an overall process since the statistical, or legal, state of any enterprise is archived and updated at the beginning and end of any production process.

2.3 Type of Analysts of a S-DWH

Users are usually present in all the four architectural layers of the data warehouse (Sources, Integration, Interpretation and Data Access Layers) but in the last two layers they should spread across more or less accordingly with the following pyramid:

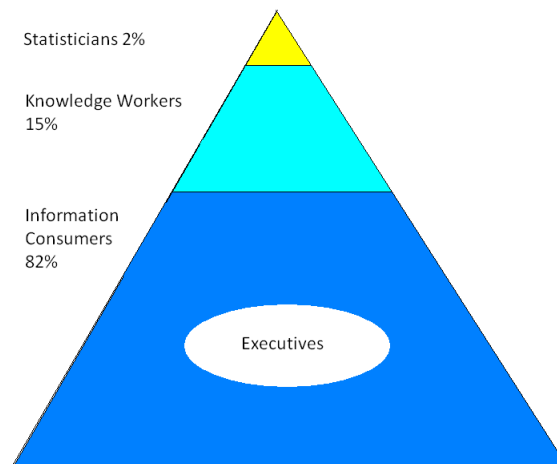


Figure 1: Users in the data warehouses

Statisticians: There are typically only a handful of sophisticated analysts—Statisticians and operations research types—in any organization. Though few in number, they are some of the best users of the data warehouse; those whose work can contribute to deeply influence the operations and profitability of the company.

Knowledge Workers: Usually a relatively small number of analysts perform the bulk of new queries and analyses against the data warehouse. These are the users who get the "Designer" or "Analyst" versions of user access tools. After a few iterations, those queries and reports typically get published for the benefit of the Information Consumers.

Information Consumers: Characteristically most users of the data warehouse are Information Consumers; they will probably never compose a true ad hoc query. They use static or simple interactive reports that others have developed.

Executives: Executives are a special case of the Information Consumers group. Few executives actually issue their own queries, but an executive's slightest musing can generate a flurry of activity among the other types of users.

Of course we end up having this four types of data warehouse users even in the SDWH, but our internal users and even some of the external users are statisticians (and not only 2%) which places a biggest burden on the system.

Making a correspondence with the layers of the S-DWH system we only have Information Consumers and Executives on the topmost layer, the access layer. The Knowledge Workers (which have sometimes a Statistical education background) usually perform tasks which belong to the interpretation.

Due to this unique characteristic of the S-DWH users' we have to characterize this group further and describe the type and complexity of the analysis they perform at each stage of the system.

In general, the following type of analysis takes place inside a statistical data warehouse. We present this list in a complexity growing order:

- Basic analysis - Calculation of averages and sums across salient subject areas. This phase is characterized by a reliance on heuristic analysis methods.
- Correlation analysis - Users develop models for correlating facts across data dimensions. This stage marks the beginning of stochastic data analysis.
- Multivariate data analysis - Users begin to perform correlations on groups of related facts, and become more sophisticated in their use of analytical statistics.
- Forecasting - Users make use statistical packages (SAS, SPSS) to generate forecasts from their data warehouses.
- Modeling - Users test hypotheses against their data warehouse, and they begin to construct simple what-if scenarios.
- Simulation - Users who developed a deep knowledge and understanding of their data may begin constructing sophisticated simulation models. This is the phase where previously unknown data relationships (correlations) are often discovered.
- Data Mining - Users begin to extract aggregates from their warehouses, and feed them into neural network programs to discover unobtrusive correlations.

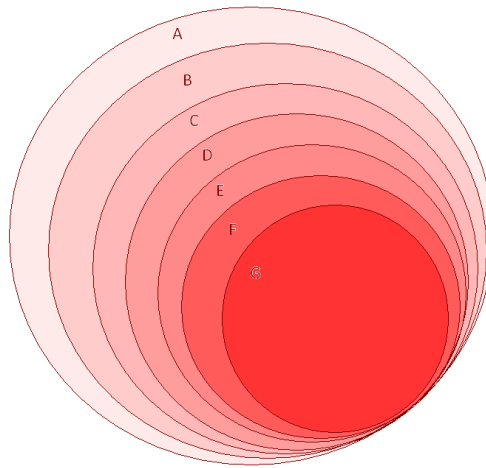


Figure 2: Different level of Data Warehouse exploration

The different levels at which the users explore and make use of the data warehouse depend not only on the kind of users we have (knowledge workers, statisticians, etc) but also on the familiarity they already have with the data warehouse. Users become increasingly sophisticated in their use of the data warehouse, largely as a function of becoming familiar with the power of the system.

The presented list of exploration complexity can also be understood as a progression of usage as the users recognize the potential types of data analysis that can be delivered by the data warehouse and also the layer at which they are positioned. As much interested as the information consumers and executives may be in forecasts and simulation results the only group which produces these are the statisticians. As a result even if in the Access Layer we may find more sophisticated products they are not available for modification or ad-hoc querying. The Layer in which those documents can be prepared and produced is the interpretation and data analysis layer.

Summing up for the statisticians group we have the following activities distribution through the data warehouse layers:

IV- Access Layer – All kind of activities, but only on previously produced products.

III – Interpretation and Data Analysis Layer – All kind of activities to create the products for the next layer, Basic Analysis, Correlation Analysis is usual performed at this stage.

All the information uses at the different levels, can be reduced to queries which should be regularly reviewed to provide an input to the over-arching quality management process, as they can indicate new or changing user needs.

3-Architecture

3.1 Business architecture

Authors: Antonio Laureti Palma, Sónia Quaresma

3.2 Information systems architecture

Authors: Antonio Laureti Palma, Sónia Quaresma

3.3 Technology Architecture

(docs in the Annex)

3.4 Data centric workflow

Authors: Antonio Laureti Palma

3.5 Focus_on_sdmx_in_statistical_data_warehouse.pdf

Authors: Antonio Laureti Palma, Sónia Quaresma

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Allan Randlepp, Antonio Laureti Palma, Francesco Altarocca, Valerij Žavoronok, Pedro Cunha; ESS-NET S-DWH Deliverable 3.2: "S-DWH Modular Workflow"

Björn Berglund, Antonio Laureti Palma; ESS-Net-DWH Deliverable 3.3: "Functional Architecture of the S-DWH"

Valerij Žavoronok, Maksim Lata, Lina Amšiejūtė; ESS-Net-DWH Deliverable 3.4: "Overview of various technical aspects in SDWH"

Antonio Laureti Palma, Björn Berglund, Allan Randlepp, Valerij Žavoronok; ESS-Net-DWH Deliverable 3.5: "Relate the 'ideal' architectural scheme into an actual development and implementation strategy"

3 Architecture¹

In the first three sub-chapters, in order to give a comprehensive architectural vision of a S-DWH in the chapter we are using typical EA architecture domain views:

- Business Architecture – related to corporate business, the documents and diagrams that describe the architectural structure of that business.
- Information Systems Architecture – the conceptual organization of the effective S-DWH which is able to support tactical demands.
- Technology Architecture – is the combined set of software, hardware and networks able to develop and support IT services.

Each architecture domain is contextualized by the four conceptual layers of the S-DWH, defined as:

IV° - access layer, for the final presentation, dissemination and delivery of the information, sought specially for external users, relative to NSI or EStat

III° - interpretation and data analysis layer, enables any data analysis or data mining, functional to support statistical design or any new strategies, as well as data re-use; functionality and data are optimized then for internal users, specifically for statistician methodologists or statistician experts.

II° - integration layer, is where all operational activities needed for any statistical production process are carried out; in this layer data is mainly transformed from raw to cleaned and made integrable;

I° - source layer, is the level in which we locate all the activities related to storing and managing internal or external data sources.

The ground level corresponds to the area where the process starts, while the top of the pile is where the data warehousing process finishes. This reflects a conceptual organization in which we consider the first two levels as operational IT infrastructures, i.e. the typical ETL operational activities, and the last two layers as the effective data warehouse.

The 4th sub-chapter is dedicated to the data-centric workflow; i.e a workflow based on a DWH. This kind of workflow is introduced since they allow process design maximizing sharing of data and knowledge.

In the 5th sub-chapter is analysed the SDMX in the context of S-DWH Architecture.

This section covers:

- S-DWH Business Architecture
- S-DWH Information Systems Architecture
- S-DWH Technology Architecture
- Data centric Workflows
- Focus on SDMX in Statistical Data Warehouse

¹ The S-DWH is reported in detail deliverables 3.1. and 3.3. from the DWH ESSnet

3.1 S-DWH Business Architecture

As we explained earlier¹, the sub-processes of the GSBPM have been mapped to each S-DWH layer, which are now represented in the picture below in a compact form. Graphically, the GSBPM phases, articulated in different columns, are on the horizontal axis and the S-DWH layers are on the vertical axis, in different rows. Their intersection produces a cell-matrix which represents a potential position for a GSBPM sub-process. To effectively include a possible sub-process in this cell, we will fill the cell with a ball and connect subsequent balls with arrows to describe common work flow. The arrows describe the direction of the work flow. To identify actors responsible for each sub-process, we will fill each circle with different colour and associate each actor to a colour. Their association will be described in a separate legend.

The rhombus, like in the BPMN, represents data objects, i.e. shows the reader which data is required or produced in each activity. The position of each rhombus is relevant only in relation to S-DWH information architecture. The rhombus must be positioned each time in one of four possible layers. To describe the process of population of a data object, a dotted line with arrows from the process toward the rhomb is used. Otherwise, where the process uses data as input, the arrow goes from the data object to the sub-process.

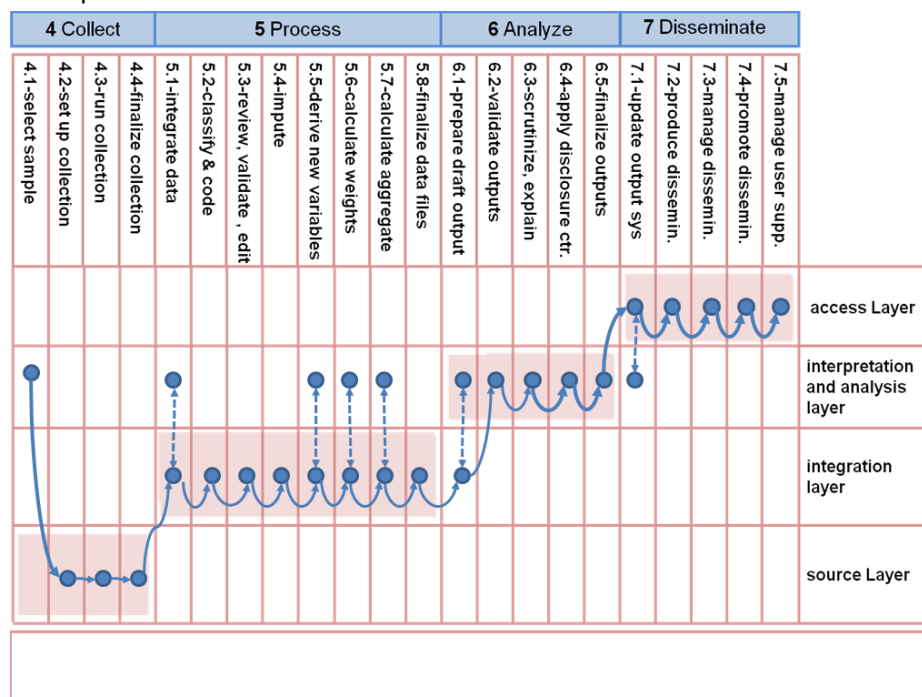


Figure 1 - GSBPM Subprocesses allocation to S-DWH layers

In the figure, the pink areas, along the diagonal, show the classic mapping area between a classic sequential statistical production process and a sequential information system. This corresponds to the association between: the Collect Phase and the Source layer, the Process Phase and the Integration layer, the Analyse Phase and Interpretation layer, and the Dissemination Phase and the Access layer.

The data warehousing approach for a statistical information system is used mainly to compel different types of users to share the same infrastructure and information. That is, the same stored-data are usable for different statistical phases. It is evident from the figure that the Integration layer is

¹ See chapter on methodology.

responsible for the organization of operational data. This is emphasized by the presence of the highest number of sub-processes.

We should point out that a S-DWH approach can also increase efficiency during the Design and Phases of the GSBPM model, since statistical experts working on these phases share the same information system of the Process Phase.

In the figure, the off diagonal sub-processes outside the pink area are relevant in our S-DWH analysis since they underline an innovative point of view from a sequential production process approach. In fact, in the same S-DWH environment we would like to carry out the production process, its modelling and allow the re-use of data. As examples of these we see that a part of the Analysis Phase is implemented in the Integration layer in order to systematize the organization of produced information in a generic operational data store. Therefore, only free analysis activities which could require statistical information in a non pre-defined way are managed in the Interpretation layer using data through dynamic query from the Integration layer data structures.

To better understand the impact of a S-DWH approach for statistical production, we investigated the business processes of the S-DWH ESSnet member countries. We have considered four European-regulated business processes: SBS, STS, Prodcum and Trade Statistics.

The analysis is first done by using the BPMN and then the results are mapped in the layered S-DWH. The figure below shows a fusion of work-flows of the case-studies.

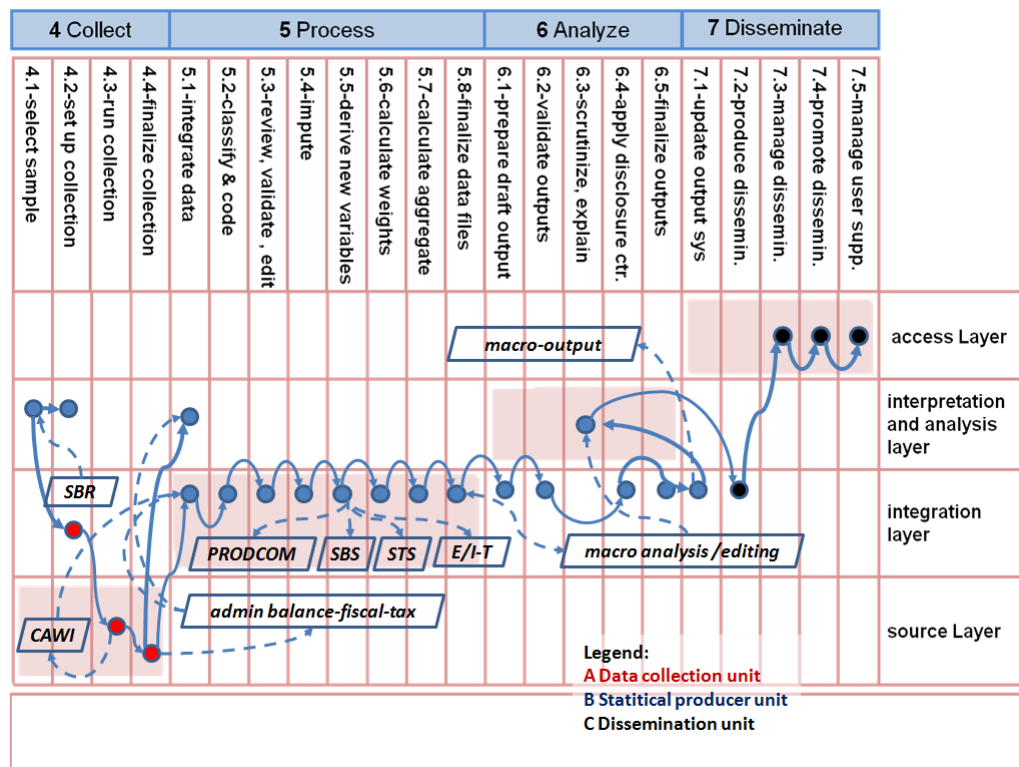


Figure 2 - Example of the allocation of GSBPM subprocesses to S-DWH layers

From the figure above we can note that every Unit-actor is involved in at least two layers, one layer in which the Unit carries out its target activities and the other in which the operational process continuity is implemented. The later means all activities necessary to ensure operational continuity to the next actor.

The Data Collection Unit carries out its activities in the Source layer. In the Integration layer, it interacts with the Statistical Producer Unit functionalities through the setting-up of collection sub-processes. This interaction is due to the fact that the collection tools could include an editing activity which must be linked to the reviewing sub-process (5.3).

In the Statistical Production Unit we have identified two kinds of users: statistical editor and statistical expert. The first operates in the Integration layer while the second operates in the Interpretation and Analysis layer. The statistical editor carries out all regular iterative production activities which include data linking from different sources, data transformations, data editing and imputation and the production of planned output. The statistical expert's activities include sampling analysis units, scrutinizing the information produced and new production process designing when there are changes in regulations or new needs. The statistical expert carries out non-systematic operations in which, generally, it is necessary to use advanced statistical tools.

The Dissemination Unit is specialized in the access layer. They manage, promote and adapt the output for the different users and devices. Moreover, they are specialized in the release of the final products in output web sites or books.

The identification of specialized functional layers has given a useful indication for the design of a S-DWH information architecture.

For the source layer we have suggested a data staging area able to sustain data collection from surveys and administration archives. For the integration layer we have suggested a fully normalized operational data organization. This is in order to have a single data store for a wide typology of production applications. For the Interpretation and Analysis layer we have suggested a dimensional data architecture able to maximize the efficiency of data usability or data mining. In this layer experts carry out their analysis at micro data level on large volumes of data. Finally, for the Access layer we have suggested multidimensional data architecture which mostly operates at macro data level.

3.2 S-DWH Information Systems Architecture

The **Information Systems** connect the business to the infrastructures, in our context this is represented by a conceptual organization of the effective S-DWH which is able to support tactical demands.

In the layered architecture, in terms of data system, we identify:

- the staging data are usually of temporary nature, and its contents can be erased, or archived, after the DW has been loaded successfully;
- the operational data is a database designed to integrate data from multiple sources for additional operations on the data. The data is then passed back to operational systems for further operations and to the data warehouse for reporting;
- the Data Warehouse is the central repository of data which is created by integrating data from one or more disparate sources and store current as well as historical data;
- data marts are kept in the access layer and are used to get data out to the users. Data marts are derived from the primary information of a data warehouse, and are usually oriented to specific business lines.

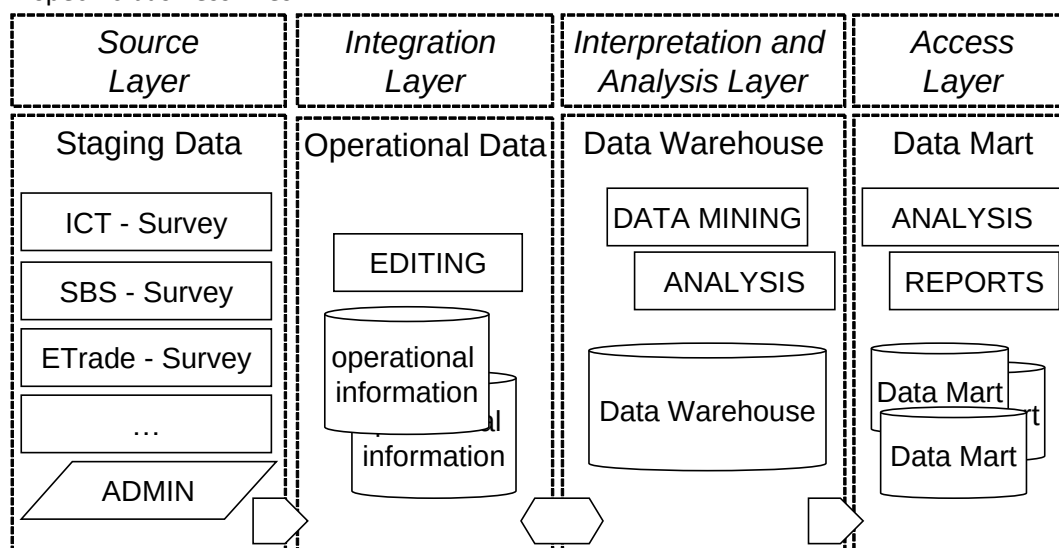


Figure 3 - Information Systems Architecture

The Metadata Management of metadata used and produced in all different layers of the warehouse are specifically defined in the Metadata framework ¹ and the Micro data linking². This is used for description, identification and retrieval of information and links the various layers of the S-DWH, which occurs through the mapping of different metadata description schemes; It contains all statistical actions, all classifiers that are in use, input and output variables, selected data sources, descriptions of output tables, questionnaires and so on. All these meta-objects are collected during design phase into one metadata repository. It configures a metadata-driven system well-suited also for supporting the management of actions or IT modules, in generic workflows.

In order to suggest a possible path towards process optimization and cost reduction, in this chapter we will introduce a data model and a possible simple description of a generic workflow, which links the business model with the information system in the S-DWH.

¹ Lundell L.G. (2012) Metadata Framework for Statistical Data Warehousing, ver. 1.0. Deliverable 1.1

² Ennok M et al. (2013) On Micro data linking and data warehousing in production of business statistics, ver. 1.1. Deliverable 1.4

3.2.1 S-DWH is a metadata-driven system

The over-arching Metadata Management of a S-DWH as metadata-driven system supports Data Management within the statistical program of an NSI, and it is therefore vital to thoroughly manage the metadata. To address this we refer to the metadata chapter where metadata are organized in six main categories. The main six categories are:

- **active metadata**, metadata stored and organized in a way that it enables operational use, manual or automated;
- **passive metadata**, any metadata that are not active;
- **formalised metadata**, metadata stored and organised according to standardised codes, lists and hierarchies;
- **free-form metadata**, metadata that contain descriptive information using formats ranging from completely free-form to partly formalised;
- **reference metadata**, metadata that describe the content and quality of the data in order to help the user understand and evaluate them (conceptually);
- **structural metadata**, metadata that help the user find, identify, access and utilise the data (physically).

Metadata in each of these categories belong to a specific type, or subset of metadata. The five subsets are:

- **statistical metadata**, data about statistical data e.g. variable definition, register description, code list;
- **process metadata**, metadata that describe the expected or actual outcome of one or more processes using evaluable and operational metrics;
- **quality metadata**, any kind of metadata that contribute to the description or interpretation of the quality of data;
- **technical metadata**, metadata that describe or define the physical storage or location of data;
- **authorization metadata** are administrative data that are used by programmes, systems or subsystems to manage user's access to data.

In the S-DWH, one of the key factors is consolidation of multiple databases into a single database and identifying redundant columns of data for consolidation or elimination. This involves coherence of statistical metadata and in particular on managed variables. Statistical actions should collect unique input variables, not just rows and columns of tables in a questionnaire. Each input variable should be collected and processed once in each period of time. This should be done so that the outcome, input variable in warehouse, could be used for producing various different outputs. This variable triggers changes in almost all phases of statistical production process. So, samples, questionnaires, processing rules, imputation methods, data sources, etc., must be designed and built in compliance with standardized input variables, not according to the needs of one specific statistical action.

The variable based on statistical production system reduces the administrative burden, lowers the cost of data collection and processing and enables to produce richer statistical output faster. Of course, this is true in boundaries of standardized design. This means that a coherent approach can be used if

statisticians plan their actions following a logical hierarchy of the variables estimation in a common frame. What the IT must support is then an adequate environment for designing this strategy.

As an example, according to a common strategy, we consider Surveys 1 and 2 which collect data with questionnaires and one administrative data source. But this time, decisions done in design phase (design of the questionnaire, sample selection, imputation method, etc.) are made “globally”, taking into consideration all three surveys. In this way, integration of processes gives us reusable data in the warehouse. Our warehouse now contains each variable only once, making it much easier to reuse and manage our valuable data.

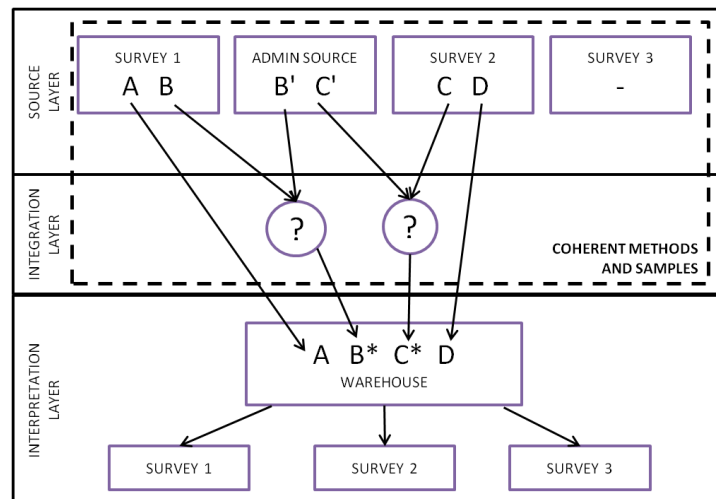


Figure 4 - Integration to achieve each variable only once - Information Re-use

Another way of reusing data which is already in the warehouse is to calculate new variables. The following figure illustrates the scenario where a new variable E is calculated from variables C* and D, loaded already into the warehouse.

It means that data can be moved back from the warehouse to the integration layer. Warehouse data can be used in the integration layer in multiple purposes, calculating new variables is only one example. Integrated variable based on a warehouse data opens the way to any new possible sub-sequent statistical actions that do not have to collect and process data, and can produce statistics directly from the warehouse. Skipping the collection and processing phases, one can produce new statistics, and analyses are very fast and much cheaper than in case of the classical survey.

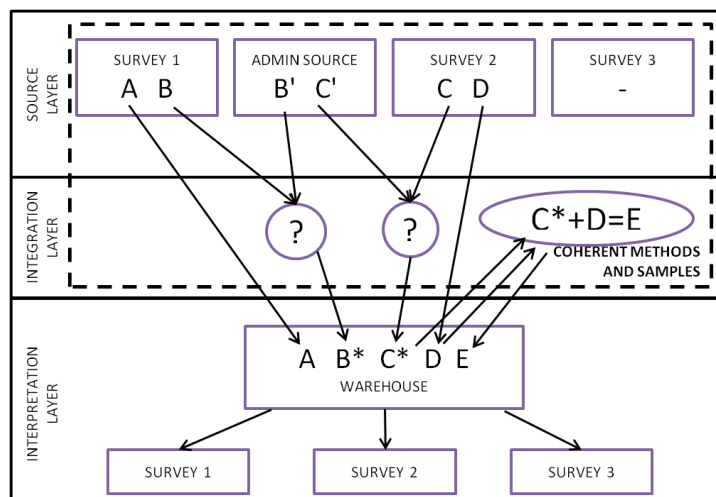


Figure 5 - Building a new variable - Information Re-Use

Designing and building a statistical production system according to the integrated warehouse model takes initially more time and effort than building the stovepipe model. But maintenance costs of integrated warehouse system should be lower, and new products which can be produced faster and cheaper, to meet the changing needs, should compensate the initial investments soon.

The challenge in data warehouse environment is to integrate, rearrange and consolidate large volumes of data from different sources to provide a new unified information base for business intelligence. To meet this challenge, we propose that the processes defined in GSBPM are distributed into four groups of specialized functionalities, each represented as a layer in the S-DWH.

3.2.2 Layered approach of a full active S-DWH

The layered architecture reflects a conceptual organization in which we will consider the first two levels as pure statistical operational infrastructures, functional for acquiring, storing, editing and validating data and the last two layers as the effective data warehouse, i.e. levels in which data are accessible for analysis.

These reflect two different IT environments: an operational one (where we support semi-automatic computer interaction systems) and an analytical one (the warehouse, where we maximize human free interaction).

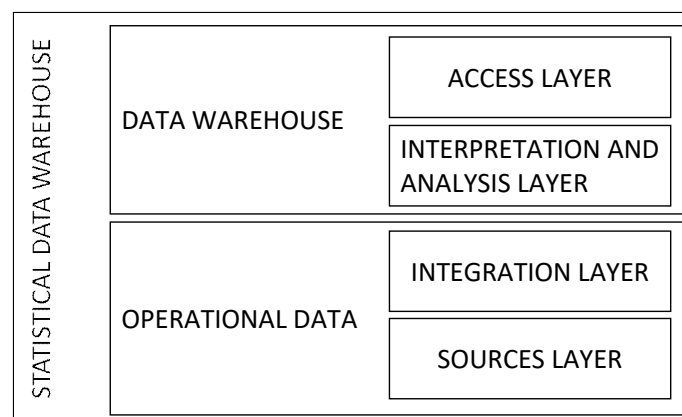


Figure 6 - S-DWH Layered Architecture

3.2.3 Source layer

The Source layer is the gathering point for all data that is going to be stored in the Data warehouse. Input to the Source layer is data from both internal and external sources. Internal data is mainly data from surveys carried out by the NSI, but it can also be data from maintenance programs used for manipulating data in the Data warehouse. External data is administrative data, which is data collected by someone else (originally for some other purpose).

The structure of data in the Source layer depends on how the data is collected and the designs of the various NSI data collection processes. The specifications of collection processes and their output, the data stored in the Source layer, have to be thoroughly described. Some vital information is names,

meaning, definition and description, of any collected variable. Also the collection process itself must be described, for example the source of a collected item, when it was collected and how.

When data are entering in the source layer from an external source, or administrative archive, data and relative metadata must be checked in terms of completeness and coherence.

From a data structure point of view, external data are stored with the same data structure as they arrive. The integration toward the integration layer should be then implemented by mapping of the source variable with the target variable, i.e. the internal variable to the S-DWH.

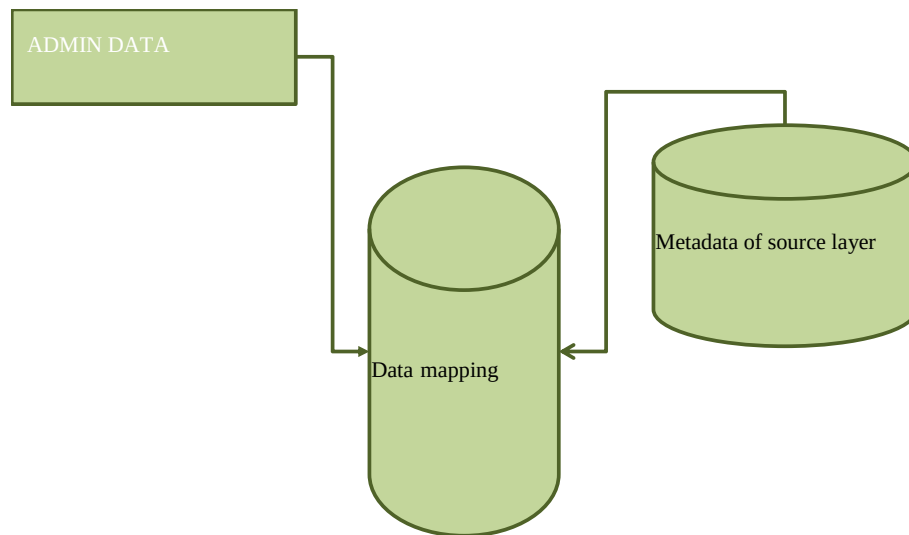


Figure 1 - Data Mapping

The mapping is a graphic or conceptual representation of information to represent some relationships within the data, i.e. the process of creating data element mappings between two distinct data models. The common and original practice of mapping is the interpretation of an administrative archive in terms of S-DWH definition and meaning.

Data mapping involves combining data residing in different sources and providing users with a unified view of these data. These systems are formally defined as a triple $\langle T, S, M \rangle$ where T is the target schema, S is the heterogeneous set of source schemas, and M is the mapping that maps queries between the source and the target schemas.

Queries over the data mapping system also assert the data linking between elements in the sources and the business register units.

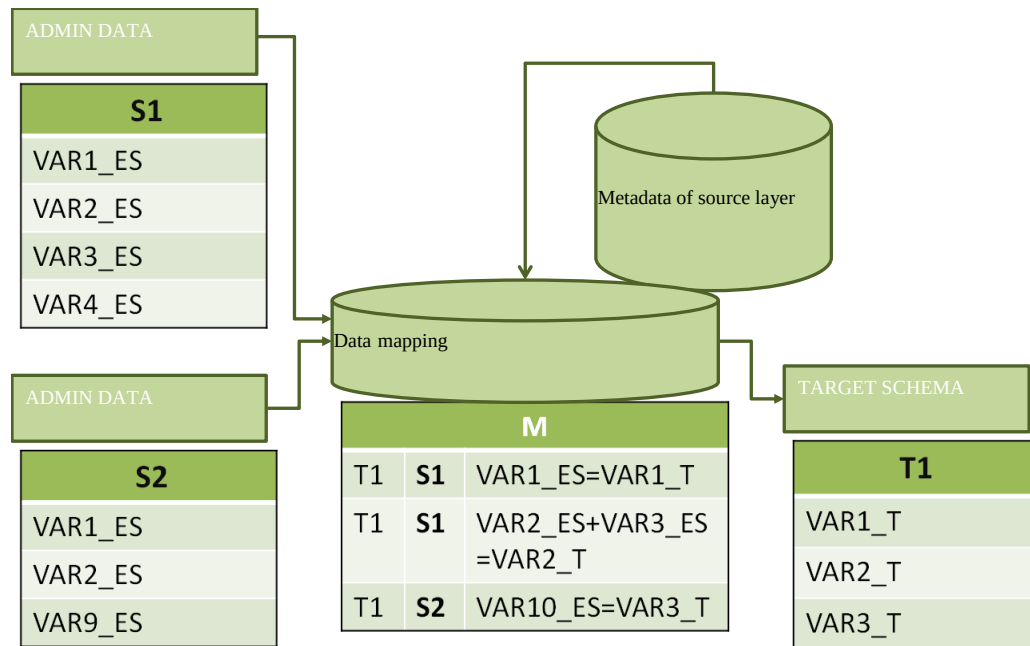


Figure 2 - Data Mapping Example

None of the internal sources need mapping since the data collection process is defined in a S-DWH during the design phase by using internal definitions.

Source layer: store and maintain internal and external data sources

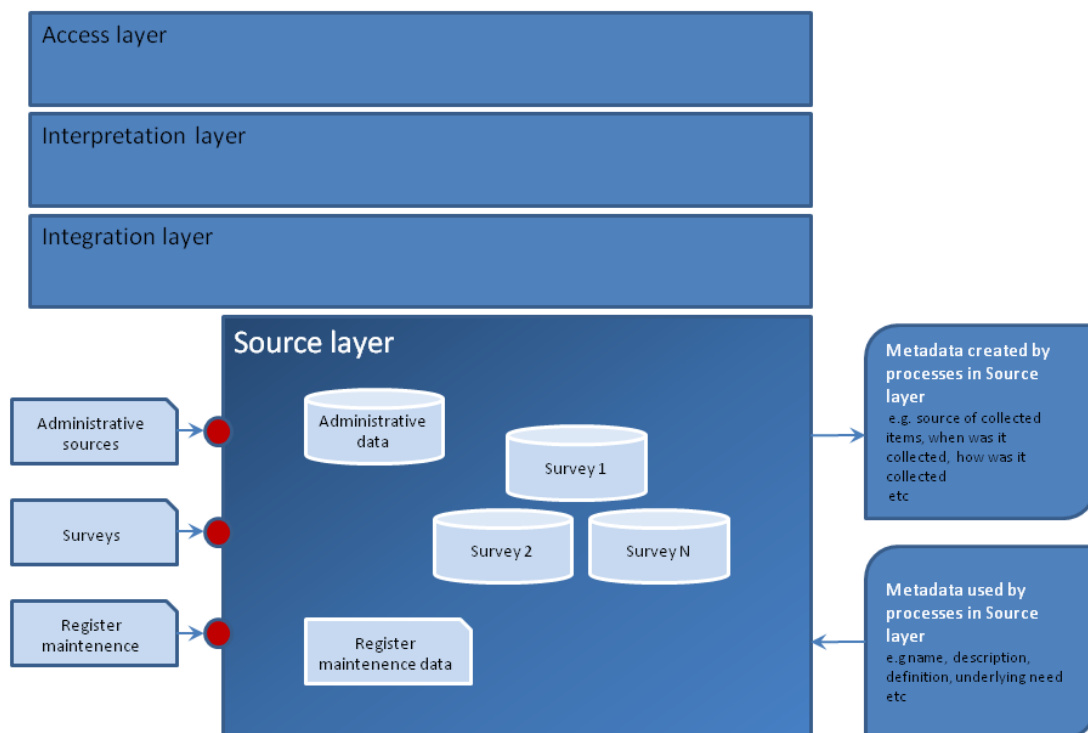


Figure 3 - Source Layer Overview

3.2.4 Integration layer

From the Source layer, data is loaded into the Integration layer. This represents an operational system used to process the day-to-day transactions of an organization. These systems are designed to efficiently process and maintain transactional integrity. The process of translating data from source systems and transform it into useful content in the data warehouse is commonly called ETL. In the Extract step, data is moved from the Source layer and made accessible in the Integration layer for further processing.

The Transformation step involves all the operational activities usually associated with the typical statistical production process. Examples of activities carried out during the transformation are:

- Find, and if possible, correct the incorrect data;
- Transform data to formats matching standard formats in the data warehouse;
- Classify and code;
- Derive new values;
- Combine data from multiple sources;
- Clean data, that is for example correct misspellings, remove duplicates and handle missing values;

To accomplish the different tasks in the transformation of new data to useful output, data already in the data warehouse is used to support the work. Examples of such usage are using existing data together with new ones to derive a new value or using old data as a base for imputation.

Each variable in the data warehouse may be used for several different purposes in any number of specified outputs. As soon as a variable is processed in the Integration layer in a way that it is useful in the context of data warehouse output, it has to be loaded into the Interpretation layer and the Access layer.

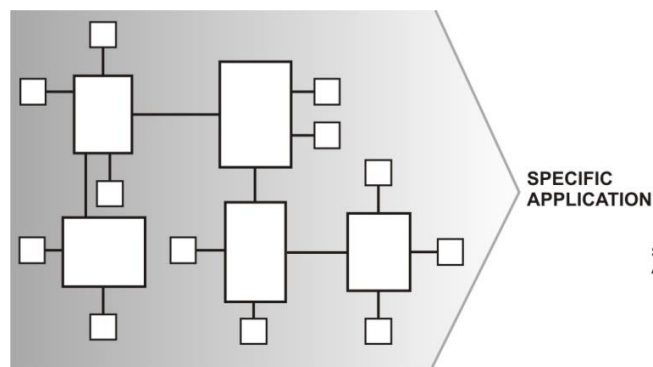


Figure 4 - OLTP Online Transaction Processing

The Integration layer is an area for processing data: this is implemented by operators specialized in ETL functionalities. Since the focus for the Integration layer is on processing rather than search and analysis, data in the Integration layer should be stored in generalized normalized structure, optimized for OLTP (Online transaction processing, is a class of information systems that facilitate and manage transaction-oriented applications, typically for data entry and retrieval transaction processing), where all data are stored in similar data structure independently from the domain or topic and each fact is stored only in one point in order to make easier maintenance of consistent data.

It is well known that these databases are very powerful when it comes to data manipulation as inserting, updating and deleting, but are very ineffective when we need to analyse and deal with a large amount of data. Another constraint in the use of OLTP is their complexity. Users must have great expertise to manipulate them and it is not easy to understand all the intricacy.

During the several ETL processes, a variable will likely appear in several versions. Every time a value is corrected or changed for some reason, the old value should not be erased but a new version of that variable should be stored. That is a mechanism used to ensure that all items in the database can be followed over time.

Integration layer: Extract, Transform, Load

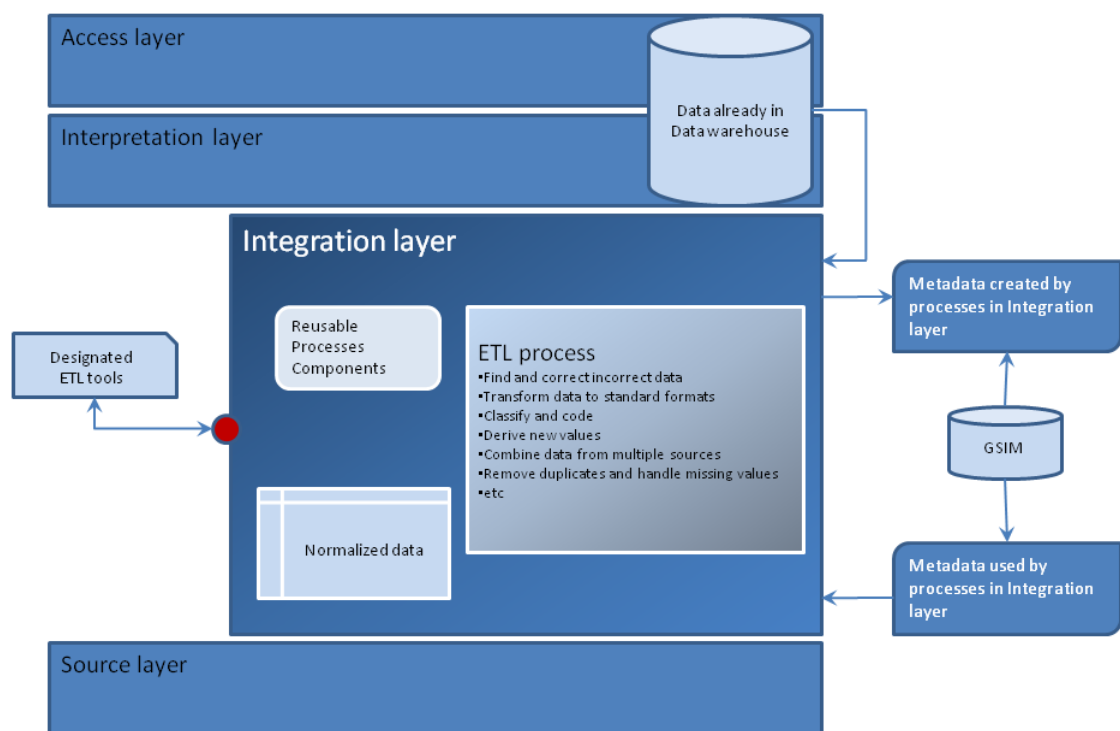


Figure 5 - Integration layer Overview

3.2.5 Interpretation and Data Analysis layer

This layer contains all collected data processed and structured to be optimized for analysis, as well as the base for output planned by the NSI. The Interpretation layer is specially designed for statistical experts and is built to support data manipulation of big complex search operations. Typical activities in the Interpretation layer are hypothesis testing, data mining and design of new statistical strategies, as well as designing data cubes functional to the Access layer.

Its underlying data model is not specific to a particular reporting or analytic requirement. Instead of focusing on a process-oriented design, the repository design is modelled based on data inter-relationships that are fundamental to the organization across processes.

Data warehousing became an important strategy to integrate heterogeneous information sources in organizations, and to enable their analysis and quality. Although data warehouses are built on relational database technology, the design of a data warehouse database differs substantially from the design of an online transaction processing system (OLTP) database.

The Interpretation layer will contain micro data, elementary observed facts, aggregations and calculated values. It will contain all data at the finest granular level in order to be able to cover all possible queries and joins. A fine granularity is also a condition used to manage changes of required output over time.

Besides the actual data warehouse content, the Interpretation layer may contain temporary data structures and databases created and used by the different ongoing analysis projects carried out by statistics specialists. The ETL process in integration level continuously creates metadata regarding the variables and the process itself that is stored as a part of the data warehouse.

In a relational database, fact tables of the Interpretation layer should be organized in dimensional structure to support data analysis in an intuitive and efficient way. Dimensional models are generally structured with fact tables and their belonging dimensions. Facts are generally numeric, and dimensions are the reference information that gives context to the facts. For example, a sales trade transaction can be broken up into facts, such as the number of products moved and the price paid for the products, and into dimensions, such as order date, customer name and product number.

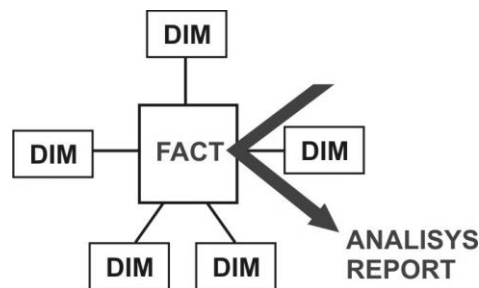


Figure 6 - Star Schema

A key advantage of a dimensional approach is that the data warehouse is easy to use and operations on data are very quick. In general, dimensional structures are easy to understand for business users, because the structures are divided into measurements/facts and context/dimensions related to the organization's business processes.

A dimension is sometimes referred to as an axis for analysis. Time and Location are the classic basic dimensions. A dimension is a structural attribute of a cube that consists in a list of elements, all of which are of a similar type in the user's perception of the data. For example, all months, quarters, years, etc., make up a time dimension; likewise all cities, regions, countries, etc., make up a geography dimension.

A dimension table is one of the sets of companion tables for a fact table and normally contains attributes or (fields) used to constrain and group data when performing data warehousing queries. Dimensions correspond to the "branches" of a star schema.

The positions of the dimensions are organised according to a series of cascading one to many relationships. This way of organizing data is comparable to a logical tree, where each member has only one parent but a variable number of children. For example the positions of the Time dimension might be months, but also days, periods or years.

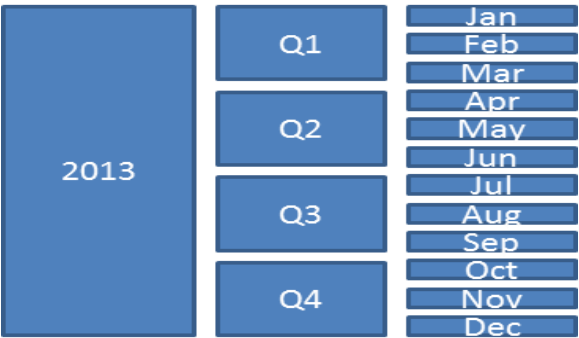


Figure 7 - Time Dimension

A dimension can have an hierarchy, which is classified into levels. All the positions for a level correspond to a unique classification. For example, in a "Time" dimension, level one stands for days, level two for months and level three for years. The dimensions can be balanced, unbalanced or ragged. In balanced hierarchies, the branches of the hierarchy all descend to the same level, with each member's parent being at the level immediately above the member. In unbalanced hierarchies, not all the branches of the hierarchy reach to the same level but each member's parent does belong to the level immediately above it.

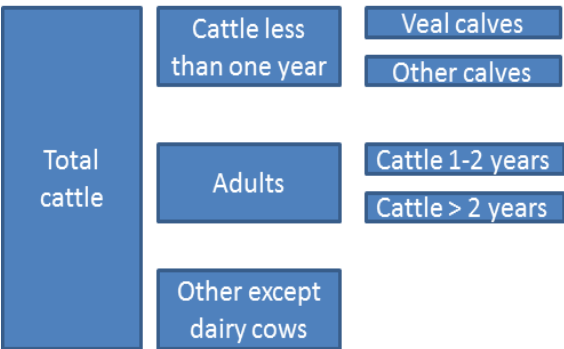


Figure 8 - Unbalanced Hierarchies

In ragged hierarchies, the parent member of at least one member of the dimension is not in the level immediately above the member. Like unbalanced hierarchies, the branches of the hierarchies can descend to different levels. Usually, unbalanced and ragged hierarchies must be transformed into balanced hierarchies.

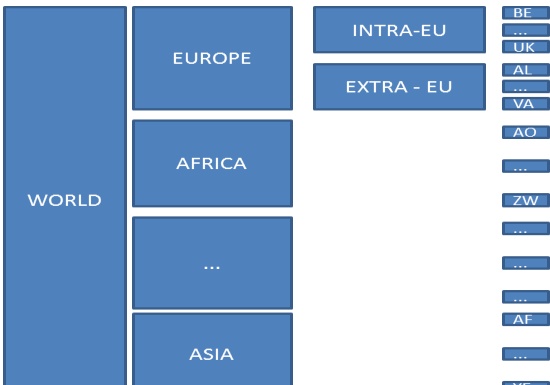


Figure 9 – Ragged Dimension

A fact table consists of measurements, metrics or facts of a statistical topic. The fact table in the S-DWH is organized in a dimensional model, built on a star-like schema, with dimensions surrounding it. In the S- S-DWH, the fact table is defined at the finer level of granularity with information organized in columns distinguished in dimensions, classifications and measures. Dimensions are the descriptions of the fact table. Typically dimensions are nouns like date, class of employ, territory, NACE, etc. and could have hierarchy on it, for example, the date dimension could contain data such as year, month and weekday.

The definition of a star schema would be implemented by dynamic ad hoc queries from the integration layer by the proper metadata, in order to generally implement data transposition query. With a dynamic approach, any expert user should define its own analysis context starting from the already existing data mart and virtual or temporary environment derived from the data structure of the integration layer. This method allows users to automatically build permanent or temporary data marts according to their needs, leaving them free to test any possible new strategy.

Interpretation layer: New facts

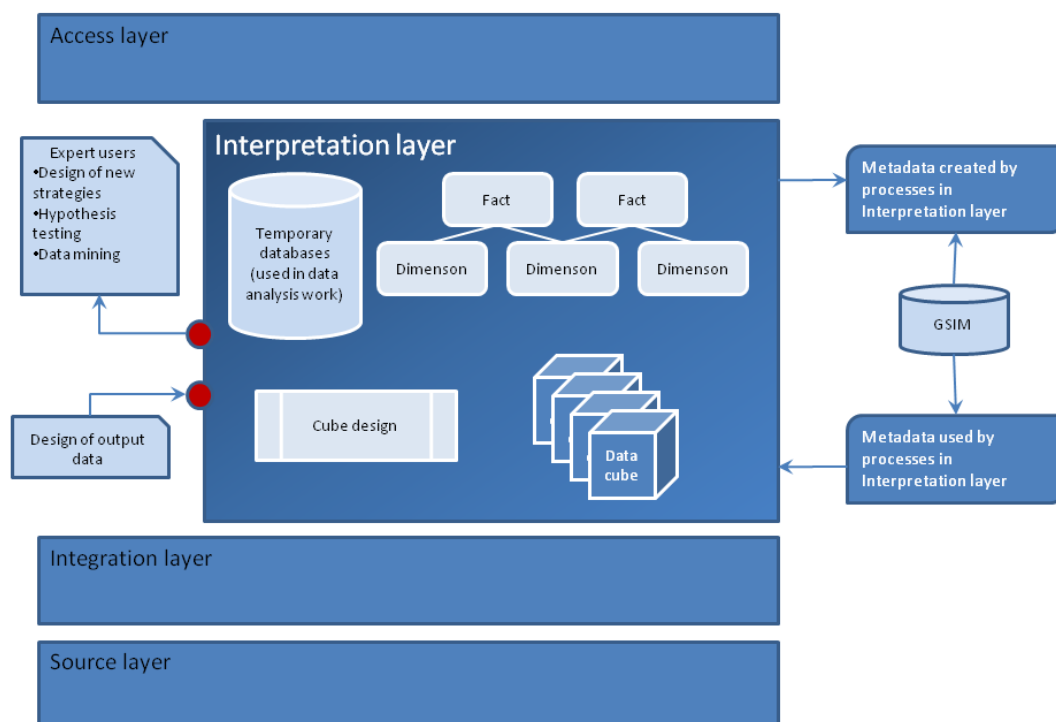


Figure 10 - Interpretation and Data Analysis Layer Overview

3.2.6 Access layer

The Access layer is the layer for the final presentation, dissemination and delivery of information. This layer is used by a wide range of users and computer instruments. The data is optimized to effectively present and compile data. Data may be presented in data cubes and different formats specialized to support different tools and software. Generally the data structure is optimized for **MOLAP** (Multidimensional Online Analytical Processing). It uses specific analytical tools on a multidimensional data model or **ROLAP**, Relational Online Analytical Processing, as well as specific analytical tools on a

relational dimensional data model which is easy to understand and does not require pre-computation and storage for the information.

Accesslayer: Knowledge

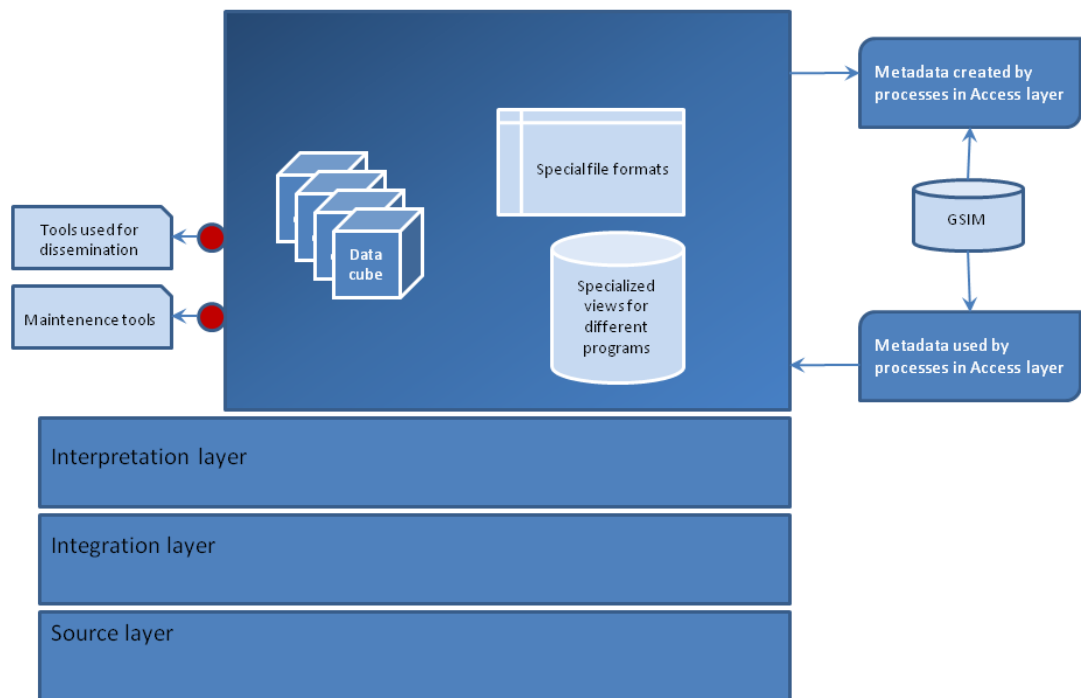


Figure 11 - Access layer Overview

Multidimensional structure is defined as “a variation of the relational model that uses multidimensional structures to organize data and express the relationships between data”. The structure is broken into cubes and the cubes are able to store and access data within the confines of each cube. *“Each cell within a multidimensional structure contains aggregated data related to elements along each of its dimensions”*. Even when data is manipulated it remains easy to access and continues to constitute a compact database format. The data still remains interrelated. Multidimensional structure is quite popular for analytical databases that use online analytical processing (OLAP) applications. Analytical databases use these because of their ability to deliver answers to complex business queries swiftly. Data can be viewed from different angles, which gives a broader perspective of a problem unlike other models. Some Data Mart might need to be refreshed from the Data Warehouse daily, whereas user groups might need to be refreshed only monthly.

Each Data Mart can contain different combinations of tables, columns and rows from the Statistical Data Warehouse. For example, a statistician or user group that doesn't require a lot of historical data might only need transactions from the current calendar year in the database. The analysts might need to see all details about data, whereas data such as "salary" or "address" might not be appropriate for a Data Mart that focuses on Trade.

Three basic types of data marts are dependent, independent, and hybrid. The categorization is based primarily on the data source that feeds the data mart. Dependent data marts draw data from a central data warehouse that has already been created. Independent data marts, in contrast, are standalone

systems built by drawing data directly from operational or external sources of data or both. Hybrid data marts can draw data from operational systems or data warehouses.

The Data Mart in the ideal information system architecture of a full active S-DWH, are dependent data marts: data in a data warehouse is aggregated, restructured and summarized when it passes into the dependent data mart. The architecture of a dependent data mart is as follows.

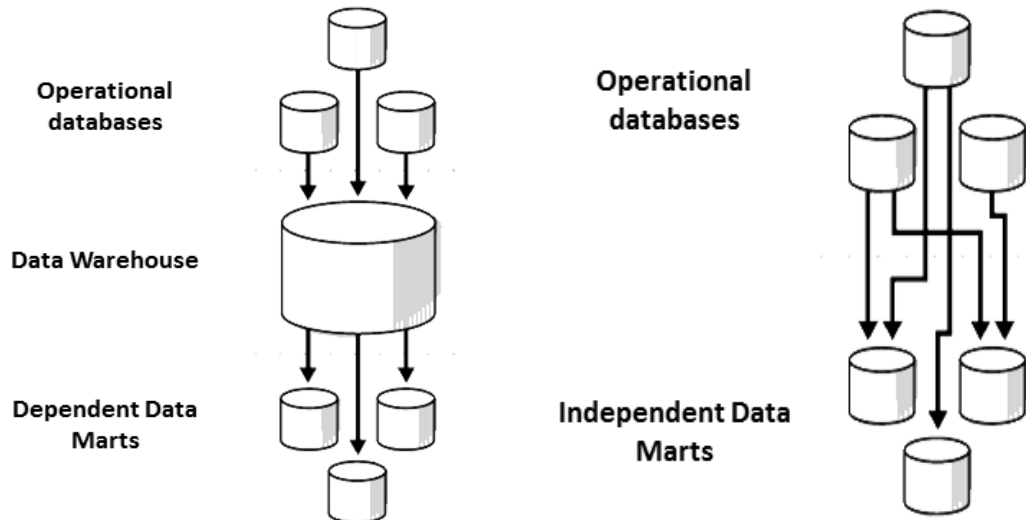


Figure 12 - Dependent versus Independent Data Marts

There are benefits of building a dependent data mart:

- Performance: when performance of a data warehouse becomes an issue, building one or two dependent data marts can solve the problem, because the data processing is performed outside the data warehouse.
- Security: by putting data outside data warehouse in dependent data marts, each department owns their data and has complete control over their data.

3.3 Technology Architecture

The **Technology Architecture** is the combined set of software, hardware and networks able to develop and support IT services. This is a high-level map or plan of the information assets in an organization, including the physical design of the building that holds the hardware.

An overview of software packages existing on the market or developed on request in NSIs in order to describe the solutions that would meet NSI needs, implement S-DWH concept and provide the necessary functionality for each S-DWH level is presented more detailed in Annex 1.

3.4 Data-Centric workflow

One of the most important activities in a S-DWH environment is represented by *data integration* of different and heterogeneous sources. The process of extract, transform, and load data from different heterogeneous statistical data sources into a single schema so that data become compatible with each other and can be processed, compared, queried or analysed regardless of the original data structure and semantics.

This sub-chapter deals with the integration of processes and all the elements needed for the data process phase. We structure the process phase in a S-DWH as a data-centric workflow. The data-centric WF is characterized by frequently modified processes, process re-use or adjustments. As example, in the case of administrative data source not under the direct control of statisticians; the sources structure or content may change each supply which implies adaptation of the data integration processes or completely rewrite the procedures.

In order to efficiently organize the WF¹ with the aim to support the production processes and improve quality, it is necessary to connect several entities such as the source variables and the related documentation. It is also important to gather and record the versions of any entity in order to fully document the process and guarantee its quality, reliability, replicability and reusability. A systematic collection of all the tested attempts could also contribute to the production efficiency because the researcher's team would be able to examine all past discarded hypotheses.

All previous consideration bring us to the following item functionality needs:

- *design and management of a data centric workflow*. It allows designing, modifying and executing the main phases, sub-processes and elementary activities which constitute the statistical production process;
- *activities and processes schedule*. The activities, the remote processes and the procedures can be run by a scheduler in an automatic way. This is particularly useful when one deals with huge amounts of data. The scheduler's purpose is to translate the workflow design into a sequence of activities to be submitted to the distributed processing nodes. This sequence has to satisfy priority constraints planned during the design phase;
- *local and remote services call*. Each elementary activity can be either a native procedure (e.g. a SAS procedure, a *PL/SQL* program or an R procedure) or an external service, such as a web service encapsulating a high-level domain service (i.e. *BANFF*) that can be invoked from the platform. It is necessary to provide some mechanism of sharing information between systems;
- *integration of statistical abstractions*. A statistical production process has its own rules, constraints, methodologies and paradigms. The aim of the statistical abstraction layer is to supply a set of abstractions that make the researcher's work flexible, independent of technical details and more focused on research objectives. Among the possible abstractions there could be:
 - *meta-parameters*: the use of global parameters reduces the need to modify the scripts and variables necessary for other systems to operate correctly;
 - *partitioning or filtering units*: each type of record (unit) has its own processing path in the WF. The value of some variable could be used to filter units to the next processing step;
 - *sampling test*: when the amount of data is very large, it is useful to test some hypothesis or

¹ N. Russell, A. Ter Hofstede, D. Edmond, W. van der Aalst. 2005. Workflow data patterns. In Proc. of 24th Int. Conf. on Conceptual Modeling Springer. Verlag: october.

programs on a subset of data in order to avoid loss of time and to early discover weak hypotheses;

- *rule checker*: a tool for finding inconsistencies in a formally defined set of rules and to manage efficiently semantic and definitional changes in sources;
- *documentation management and versioning*. It is possible to associate one or more documents and metadata to each WF element and, at any time, recall previous versions of the WF and all the elements connected.
- *metadata* module implements a decoupling approach in data mapping. This type of abstraction introduces a new layer between data sources and statistical variables so that a semantic change in one administrative source does not affect statistical sub-processes that depend on the related statistical variables;
- *rules* module allow the researcher to write the consistency plan, check possible contradictions in the edits set, run the plan, log error and warnings and produce reports. Moreover, this module assists the researcher in the activity of modifying an existing check plan in case some variables are introduced or deleted;
- *parameters* module is used to implement a basic form of parametric changes in all of the components of the WF. It can be thought as similar to a dashboard through which modifying thresholds, setting parameters, choosing elaborative units, switching on and off options, etc. For instance, suppose one parameter is shared by many sub-processes: a change in this value has an impact on all the sub-processes containing that parameter. The parameter is a placeholder that at runtime is set to the actual value (e.g. some sub-process can possibly change the parameters' value during processing);
- *processes* module provides information on actual state of active elaborations. It is possible to view the scheduled sequence of sub-processes and to recall the log of previous ones;
- *procedure editor* module is the development environment needed to create procedures or modify existing code. Such a module should support at least one statistical language (SAS, R) and one data manipulation language (PL/SQL). New languages can be added to this system in a modular and incremental way. The editor integrates a versioning system in order to restore a previous version of a procedure, document code changes and to monitor the improvements of the implemented functions;
- the *micro-editing* component is used in manual and interactive micro data editing activities. It can be a useful tool for statisticians to analyze some sample of micro-data.

3.4.1 Example of modularity

This paragraph in more depth focuses on the Process phase of the statistical production. Looking at the Process phase in more detail, there are sub-processes in it. These elementary tasks are the finest-grained elements of the GSBPM. We will try to sub-divide the sub-processes into elementary tasks in order to create a conceptual layer closer to the IT infrastructure. With this aim we will focus on “Review, validate, edit” and we will describe a possible generic sub-task implementation in the next

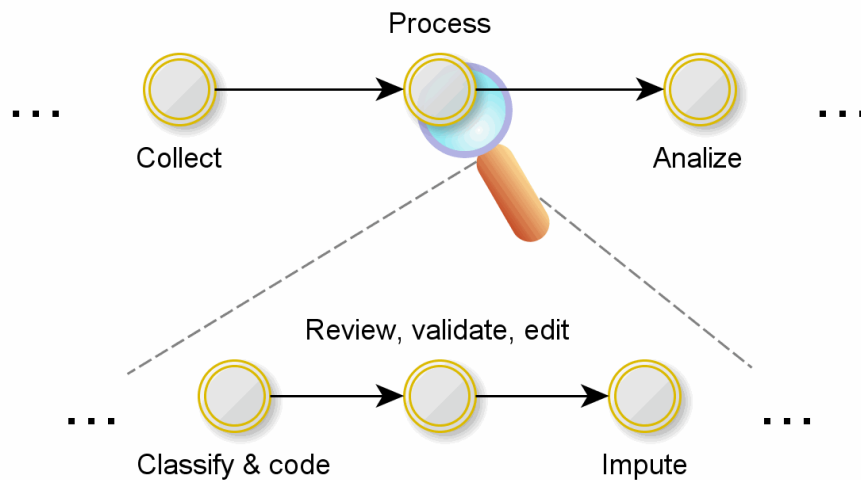


Figure 1 - Process Phase Break Down

lines.

Let's take a sample of five statistical units (represented in the following diagram by three triangles and two circles) each containing the values from three variables (V1, V2 and V3) which have to be edited (checked and corrected). Every elementary task has to edit a sub-group of variables. Therefore, a unit entering a task is processed and leaves the task with all that task's variables edited.

We will consider a workflow composed of 6 activities (tasks): S (starting), F (finishing), and S1, S2, S3, S4 (editing data and activities). Suppose also each type of unit needs a different activity path, where triangle shaped units need more articulated treatment on variables V1 and V2. For this purpose a “filter” element F is introduced (the diamond in the diagram), which diverts each unit to the correct part of the workflow. It is important to note that only V1 and V2 are processed differently because in task S4 two branches rejoin.

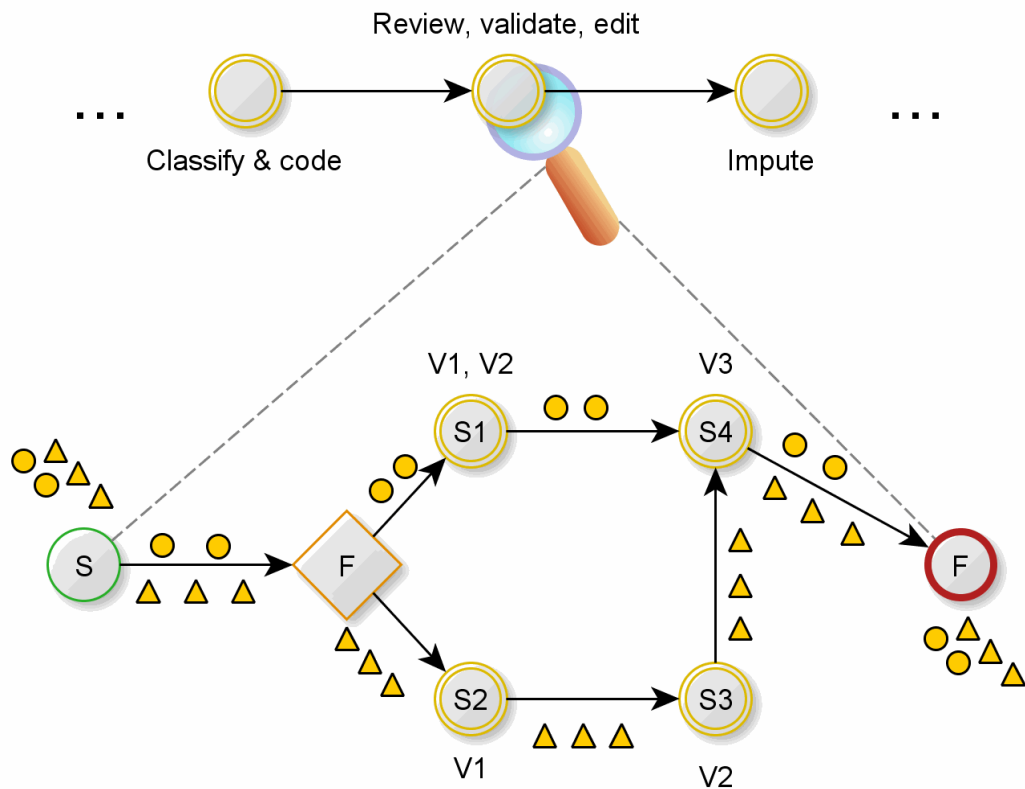


Figure 2 - Review, Validate and Edit on Process Phase

During the workflow, all the variables are inspected task by task and, when necessary, transformed into a coherent state. Therefore each task contributes to the set of coherent variables. Note that every path in the workflow meets the same set of variables. This incremental approach ensures that at the end of the workflow every unit has its variables edited. The table below shows some interesting attributes of the tasks.

Task	Input	Output	Purpose	Module	Data source	Data target
S	All units	All units	Dummy task	-	TAB_L_I_START	TAB_L_II_TARGET
S ₁	Circle units	Circle units (V ₁ , V ₂ corrected)	Edit and correct V ₁ and V ₂	EC_V ₁ (CU, P ₁) EC_V ₂ (CU, P ₂)	TAB_L_II_TARGET	
S ₂	Triangle units	Triangle units (V ₁ corrected)	Edit and correct V ₁	EC_V ₁ (TU, P ₁₁)	TAB_L_II_TARGET	
S ₃	Triangle units (V ₁ corrected)	Triangle units (V ₁ , V ₂ corrected)	Edit and correct V ₂	EC_V ₂ (TU, P ₂₂)	TAB_L_II_TARGET	
S ₄	All units (V ₁ , V ₂ corrected)	All units (all variables corrected)	Edit and correct V ₃	EC_V ₃ (U, P ₃)	TAB_L_II_TARGET	
F	All units	All units	Dummy task	-	TAB_L_II_TARG ET	TAB_L_III_FINAL

The columns in the table above provide useful elements for the building and definition of modular objects. These objects could be employed in an applicative framework where data structures and interfaces are shared in a common infrastructure.

The task column identifies the sub-activities in the workflow: the subscript, when present, corresponds to different sub-activities.

Input and output columns identify the statistical information units that must be processed and produced respectively by each sub-activity. A simple textual description of the responsibility of each sub-activity or task is given in the purpose column.

The module column shows the function needed to fulfil the purpose. As in the table above, we could label each module with a prefix, representing a specific sub-process EC function (Edit and Correct), and a suffix indicating the variable to work with. The first parameter in the function indicates the unit to treat (CU stands for circle unit, TU for triangle unit), and the second parameter indicates the procedure (i.e. a threshold, a constant, a software component).

Structuring modules in such a way could enable the reuse of components. The example in the table above shows the activity S1 as a combination of EC_V1 and EC_V2 where EC_V1 is used by S1 and also S2 and EC_V2 is used by S1 and also S3. Moreover, because the work on each variable is similar, single function could be considered like a skeleton containing a modular system in order to reduce building time and maximize re-usability.

Lastly, the data source and target columns indicate references to data structures necessary to manage each step of the activity in the workflow.

3.4.2 Towards a modular approach

There are many software models and approaches available to build modular flows between layers. S-DWH's layered architecture itself provides possibility to use different platforms and software in separate layers or to re-use components already available. In addition, different software can be used inside the same layer to build up one particular flow. The problems arise when we try to use these different modules and different data formats together.

One of the approaches is CORE services. They are used to move data between S-DWH layers and also inside the layers between different sub-tasks, then it is easier to use software provided by statistical community or re-use self-developed components to build flows for different purposes.

CORE services are based on SDMX standards and use main general conception of messages and processes. Its feasibility to use within statistical system was proved under ESSnet CORE. Note that CORE is not a kind of software but only a set of methods and approaches.

Generally CORE (COMmon Reference Environment) is an environment supporting the definition of statistical processes and their automated execution. CORE processes are designed in a standard way, starting from available services. Specifically, process definition is provided in terms of abstract statistical services that can be mapped to specific IT tools. CORE goes in the direction of fostering the sharing of tools among NSIs. Indeed, a tool developed by a specific NSI can be wrapped according to CORE principles, and thus easily integrated within a statistical process of another NSI. Moreover, having a single environment for the execution of entire statistical processes, it provides a high level of automation and a complete reproducibility of processes execution.

The main principles underlying CORE design are:

- a) **Platform Independence.** NSIs use various platforms (e.g. hardware, operating systems, database management systems, statistical software, etc.), hence architecture is bound to fail if it endeavours to impose standards at a technical level. Moreover, platform independence allows to model statistical processes at a “conceptual level”, so that they do not need to be modified when the implementation of a service changes.
- b) **Service Orientation.** The vision is that the production of statistics takes place through services calling other services. Hence services are the modular building blocks of the architecture. By having clear communication interfaces, services implement principles of modern software engineering like encapsulation and modularity.
- c) **Layered Approach.** According to this principle, some services are rich and are positioned at the top of the statistical process, so, for instance a publishing service requires the output of all sorts of services positioned earlier in the statistical process, such as collecting data and storing information. The ambition of this model is to bridge the whole range of layers from collection to publication by describing all layers in terms of services delivered to a higher layer, in such a way that each layer is dependent only on the first lower layer.

In a general sense, an integration API permits to wrap a tool in order to make it CORE-complaint, i.e. a CORE executable service. CORE service is indeed composed by an inner part, which is the tool to be wrapped, and by input and output integration APIs. Such APIs transform from/to CORE model into the tool specific format.

Basically, the integration API consists of a set of transformation components. Each transformation component corresponds to a specific data format and the principal elements of their design are specific mapping files, description files and transform operations.

3.4.3 Possible work flow scenarios

Layered architecture, modular tools and warehouse-based variables is a powerful combination that can be used for different scenarios. Here are some examples of workflows that S-DWH supports.

3.4.3.1 Scenario 1: full linear end-to-end workflow

To publish data in access layer, raw data need to be collected into raw database in source layer, then extracted into integration layer for processing, then loaded into warehouse in interpretation layer. After that, someone can calculate statistics or make an analysis and publish it in access layer.

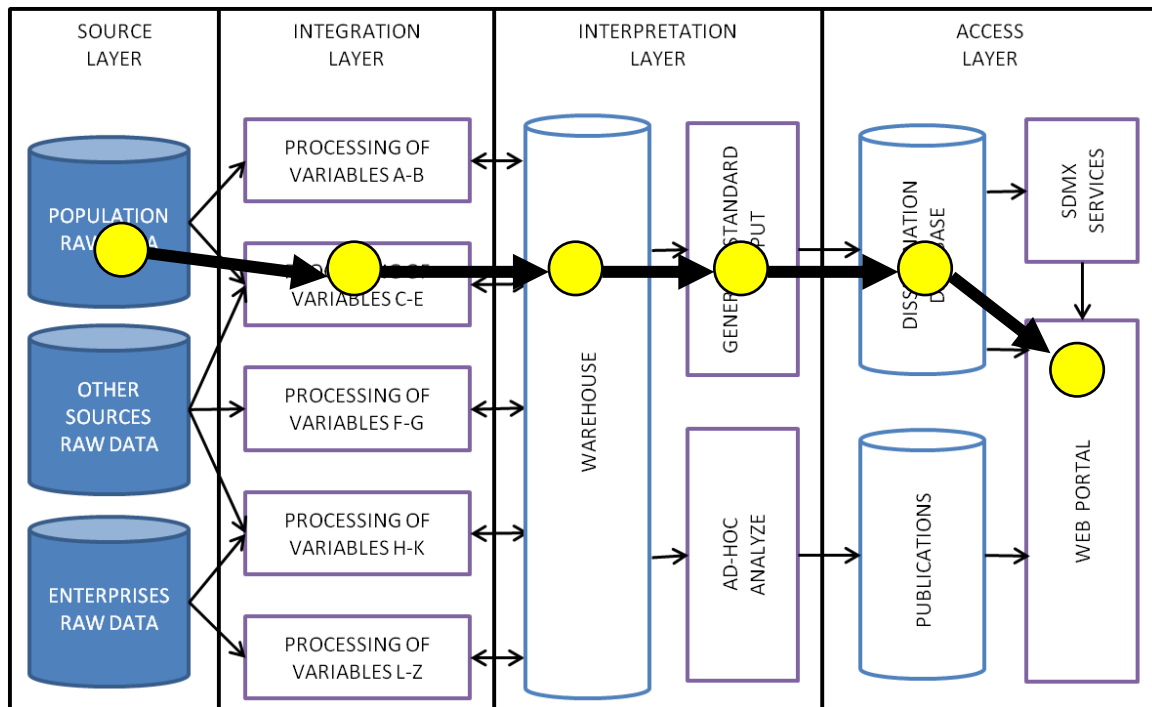


Figure 3 - Linear Workflow

3.4.3.2 Scenario 2: Monitoring collection

Sometime it is necessary to monitor collection process and analyse the raw data during the collection. Then the raw data is extracted from the collection raw database, processed in integration layer so that the data can be easily analysed with specific tools and be used for operational activities, or loaded to interpretation layer, where it can be freely analysed. This process is repeated as often as needed – for example, once a day, once a week or hourly.

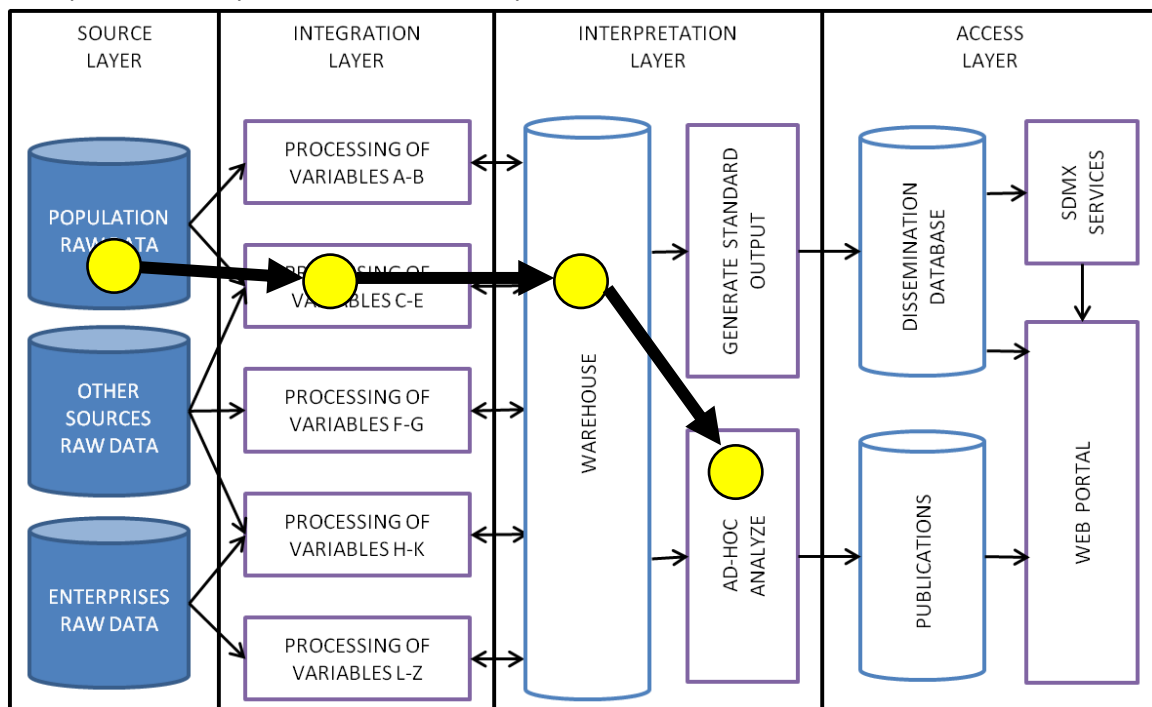


Figure 4 – Monitoring collection

3.4.3.3 Scenario 3: Evaluating new data source

When we receive a dataset from a new data source, it should be evaluated by statisticians. Dataset is loaded by the integration layer from the source to the interpretation layer, where statisticians can make their source-evaluation or, due to any changes on the administrative regulations, define new variables or new process-up-date for existents production process. From the technical perspective, this workflow is same as described in scenario 2. It is interesting to note that this update must be included in the coherent S-DWH by proper metadata.

3.4.3.4 Scenario 4: Re-using data for new standard output

Statisticians can analyse data already prepared in integration layer, compile new products and load them to access layer. If S-DWH is built correctly and correct metadata is provided, then compiling new products using already collected and prepared data should be easy and preferred way of working.

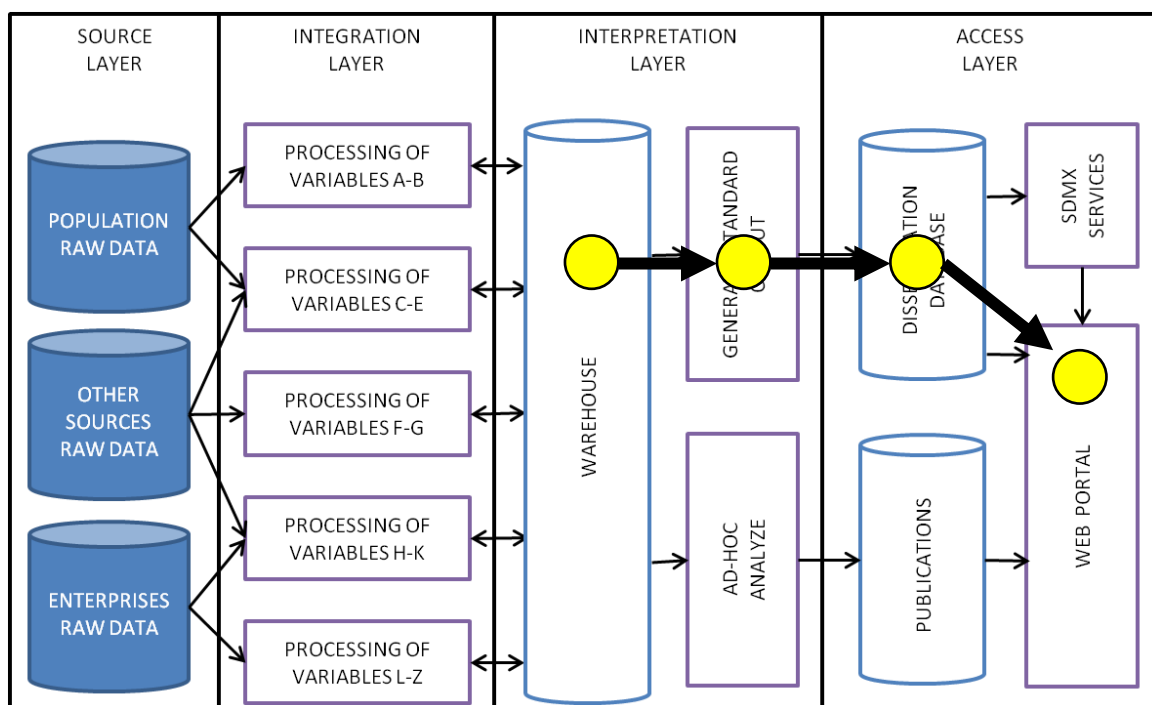


Figure 5 - Data Re-Use for New Output

3.4.3.5 Scenario 5: re-using data for complex custom query

This is variation from scenario 4, where instead of generating new standard output from data warehouse, statistician can make ad-hoc analysis by using data that is already collected and prepared in warehouse and prepare custom query for customer.

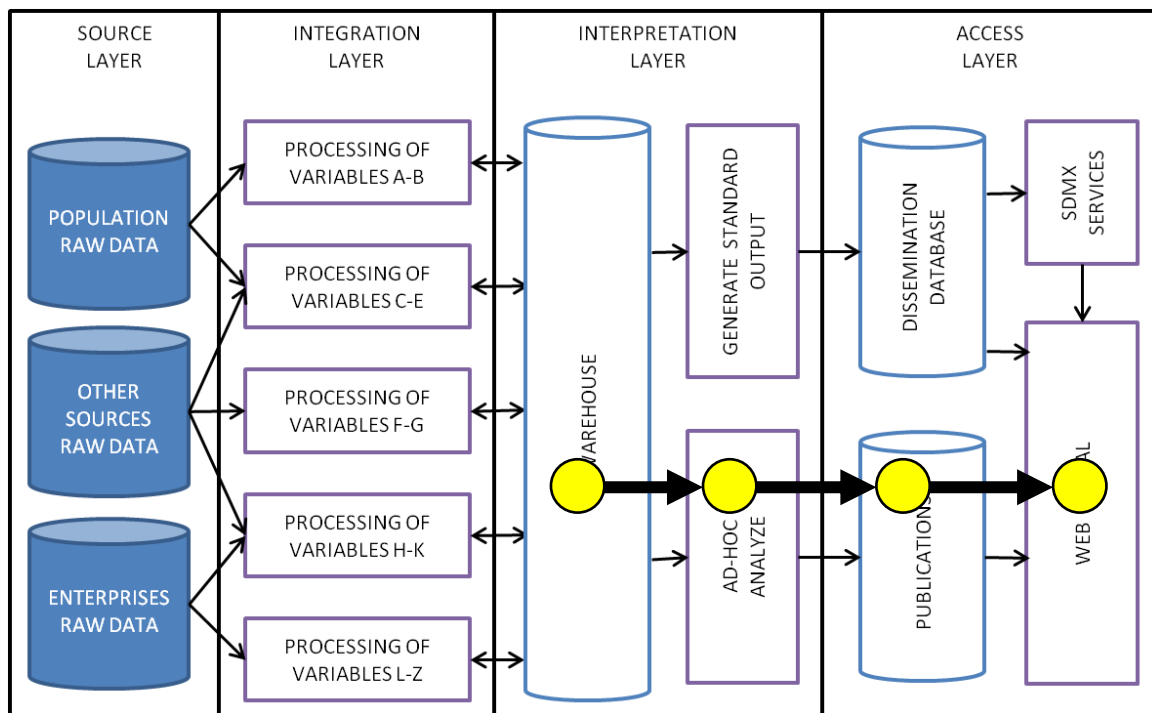


Figure 6 - Data Re-Use for Custom Query

3.5 Focus on SDMX in Statistical Data Warehouse

One of the goals of this chapter is to determine and describe relationships between SDMX and GSBPM and demonstrate how we can apply SDMX infrastructure to statistical processes.

The Statistical Data and Metadata Exchange (SDMX) is an initiative from a number of international organizations, which started in 2001 and aims to set technical standards and statistical guidelines to facilitate the exchange of statistical data and metadata using modern information technology. SDMX is confirmed as an ISO 17369:2013 “Statistical data and metadata exchange (SDMX)” standard designed to describe statistical data and metadata, standardise their exchange, and improve their efficient sharing across statistical and similar organisations. Main users of SDMX like the Bank for International Settlements, the European Central Bank, Eurostat, the International Monetary Fund, OECD, the United Nations Statistics Division, the United Nations Educational, Scientific and Cultural Organization and the World Bank recognized and supported the SDMX standards and guidelines as the preferred standard for the exchange and sharing of statistical data and metadata. A number of ESS member states and European organizations are also involved in developing these standards for various domains of official statistics.

From the practical point of view SDMX consists of:

- technical standards (including Information Model),
- statistical guidelines, and
- IT architecture and tools.

The SDMX usage aims at a reduction of development, maintenance and operation costs for an organisation through:

- logical unification of data stored inside and across national and international organisations through defining the common data model, harmonization of the statistical metadata (like code lists) and use of prescribed objects (like schemes, data structure definitions),
- application of common model and related standards effects in reduction of diversity among statistical data production processes and related business process,
- sharing of standard, generic software and IT infrastructures allowing automatic production, processing and exchange of data and metadata files among statistical organisations,
- use of standard software and standard data model allows machine to machine communication what in turn minimizes manual interventions and human errors,
- discovery and unification of distributed data shaped according to standard model.

An important component of SDMX standard is global SDMX Registry, which provides a platform for the automatic discovery of data products. In essence, the SDMX Registry services provide an online catalogue, listing all of the data available within a community. That community can be open or closed, depending on who is allowed access to the catalogue.

3.5.1 SDMX and the GSBPM

SDMX is more than a format for data exchange between separate organisations and information systems. Together, the technical standards, the statistical guidelines, and the IT architecture and tools can support improved business processes for any statistical organisation. Simultaneously, we are using the GSBPM as a description for statistical production processes from a business perspective. But there are some issues: how, where, and why is SDMX used here? Further we will demonstrate how SDMX fits into the work of a national-level NSI relating different phases of GSBPM, as well as determine relationships between SDMX and GSBPM and demonstrate how we can apply SDMX infrastructure to statistical processes.

3.5.1.1 SDMX and Analyse phase (Step 6)

It may not seem obvious that SDMX is relevant to the process of analysis of aggregates, but it can sometimes be very useful. This will depend on which tools are used at NSI to perform these various steps. Because most systems work well with XML generally – SDMX can provide some useful functions as the aggregates are analysed and further processed.



Figure 1 - GSBPM Step 6 Analyse

In the GSBPM sub-process **6.1** Prepare draft outputs, it may be helpful to use any of the various visualization tools based on SDMX when looking at the data. Especially if data is passed between several individuals while the draft outputs are prepared, it may be useful to exchange the SDMX-ML file, so that different individuals can use different visualizations of the same data while performing this work. Free tools exist for doing graphical visualizations of the SDMX data, using modern technology packages such as the Flex-CB.

The sub-process **6.2** Validate outputs requires more than just data visualization, and it is here that SDMX-ML can provide some solid benefit. Some of the validation rules exist within the data structure definitions, and these can be automatically checked using free SDMX data and metadata set tools, others exist within a SDMX Registry where cross references, versioning, and request for deletions are validated to ensure the integrity of the structural metadata.

Sub-process **6.3** Interpret and explain outputs is something which typically involves visualization of the data (as for sub-process 6.1) but may also include the creation of specific tabular views for inclusion in reports. The same tools which provide the ability to visualize SDMX data may also allow for the creation of tabular views for use in reports (Excel tables, etc.) but this will vary based on the systems within each NSI.

There is nothing in SDMX which directly addresses sub-processes **6.4** Apply disclosure control or **6.5** Finalise outputs, other than the use of visualization tools as described for earlier parts of Analyse phase. However, it should be noted that any corrections or edits to the data will need to be reflected in the SDMX-ML data to be reported. Depending on how the SDMX-ML is generated, this may involve going back to the tools and systems used to format the SDMX-ML in the first place, and making sure that the correct data is available in those tools for re-formatting as SDMX-ML.

3.5.1.2 SDMX and Disseminate phase (Step 7)

The most evident way of SDMX standards usage in S-DWH is its employment in the access layer which is intended for the final presentation, dissemination and delivery of information that end users need. The access layer is used by a wide range of users and computer instruments. In this layer the data organization must support automatic dissemination systems and free analysts but the statistical information is always macro data. Technical aspects should be thoroughly analyzed here and the data storage should be optimized to effectively present and compile data.



Figure 2 - GSBPM Step 7 Disseminate

According to the GSBPM this is covered by dissemination phase especially by its first two sub-processes. Step 7 of the GSBPM covers the process of dissemination in its broadest sense – that is, all users of the data are the target of this process step, including organizations which collect the aggregate data from NSIs. Thus, the GSBPM addresses dissemination as a single set of activities. There are several types of data dissemination, and when we consider dissemination using the internet and Web services this category looks very broad.

The first sub-process in dissemination phase is the **7.1 Update output systems**. This involves taking the aggregates as prepared in Analyse phase, and loading them into whatever systems are used to drive dissemination. Typically, this will involve database systems like Oracle and (if the same database is not used for Web dissemination) also loading data into whatever system drives the views of data on the Web site. SDMX can be used as a format for the exchange of data between systems, whether these systems are internal to an organization, or external, and thus it makes it a good format for loading databases used in all types of dissemination. Further, because it is an XML format, SDMX-ML can be used as input to systems for creating HTML, PDF, Excel, and other output formats. A SDMX Registry can make the reporting of such data more automated by using the data registration mechanism supported by a registry. The benefit of such a system is that once new data have been registered, the data user can simply query the service for the new data. This helps to ease the burden of data reporting.

Sub-process **7.2 Produce dissemination products** weakly bounds to SDMX as it includes preparing of explanatory text, tables, charts, quality statements and checking that dissemination products meet publication standards. However, SDMX visualizations may provide views of data for final outputs and outputs may be generated on-demand for dissemination on Web site.

The next sub-process in the GSBPM is **7.3 Manage release of dissemination products**. This covers a wide variety of potential products based on the data: textual or tabular reports (typically printed and disseminated as PDF, combining tabular views of the aggregate data with explanatory text and analysis), HTML pages displayed on a Web-site, data downloads in various formats (Excel, CSV, etc.), and Web-based interfaces for querying the data, and for doing graphic visualizations, which may even be interactive. At this place SDMX can be used as the single XML format for the creation of all other dissemination products, at least for providing the tabular views of the data.

SDMX is also directly useful in two more ways: as a format for reporting to data collectors and as a direct download format. The use of SDMX as a download format has become very popular and in some cases has proven to be the most accepted form of disseminated data available on Web sites. Many users prefer this format because it is easy to process and it is accompanied by rich metadata, including the structural metadata necessary for applications to process or visualize the data. Further, the format is predictable, allowing for easy use of the data coming from outside the organization. Eurostat is currently providing Census Hub Web service for collecting census data from ESS countries and then combining by the hub which use the same approach.

The last sub-process in the GSBPM which is related to IT is **7.4 Promote dissemination products**. SDMX is extremely useful in this regard, although not perhaps noticeable way. This process in the GSBPM is typically seen as the “advertising” of the statistical products, and SDMX can’t help here except that the use of high standards may offer some opportunities for promotion.

Far more interesting in increasing the visibility and use of data is the existence of the SDMX Registry services, which provide a platform for the automatic discovery of data products. It is a kind of Google machine for disseminating data for SDMX, and while the SDMX Registry services are not part of Google itself, they do provide an easy way of searching for all of the data produced within a domain, regardless of which site the data is published on.

An online catalogue provided by SDMX Registry services, list all of the data available within a community and enable the possibility to any Web site or application search for all of the data listed in that Registry, and then go to the site where that data is found. This is a very powerful feature firstly because this approach to locating data is being used more and more, and secondly, it leverages the latest generation of Web based technology making data more visible on the internet.

3.5.2 Brief description and classification of SDMX Tools

As already mentioned in the previous sections several SDMX-based IT tools exist today. Their purpose, availability and characteristics vary widely. Brief description of SDMX IT tools available on the market or developed on request in NSIs is presented in Annex 1 in sub-section **5 All layers**. Also their classification according to several important criteria is presented here.

4-Methodology

4.1 Data cleaning

Authors: Gary Brown

4.2 Data linkage

Authors: Gary Brown

4.3 Estimation

Authors: Gary Brown

4.4 Revisions

Authors: Gary Brown

4.5 Disclosure control

Authors: Gary Brown

References

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Pete Brodie; ESS-Net-DWH Deliverable 2.5: "Confidentiality aspects of combining data within a Statistical Data Warehouse"

Hannah Finselbach, Daniel Lewis, Orietta Luzi; ESS-Net-DWH Deliverable 2.6: "Selective editing options for a Statistical Data Warehouse"

Gary Brown; ESS-Net-DWH Deliverable 2.8: "Guidelines on detecting and treating outliers for the S-DWH"

Methodology: Data cleaning

1 Data cleaning

All data sources potentially include errors and missing values – data cleaning addresses these anomalies. Not cleaning data can lead to a range of problems, including linking errors, model mis-specification, errors in parameter estimation and incorrect analysis leading users to draw false conclusions.

The impact of these problems is magnified in the S-DWH environment¹ due to the planned re-use of data: if the data contain untreated anomalies, the problems will repeat. The other key data cleaning requirement in a S-DWH is storage of data before cleaning and after every stage of cleaning, and complete metadata on any data cleaning actions applied to the data.

The main data cleaning processes are editing, validation and imputation. Editing and validation are sometimes used synonymously – in this manual we distinguish them as editing describing the identification of errors, and validation their correction. The remaining process, imputation, is the replacement of missing values.

Different data types have distinct issues regards data cleaning, so data-specific processing needs to be built into a S-DWH.

- Census data – although census data do not usually contain a high percentage of anomalies, the sheer volume of responses, allied with the number of questions, so data cleaning needs to be automatic wherever possible
- Sample survey data – business surveys generally have less responses, more variables, and more anomalies than social surveys – and are more complex due to the continuous nature of the variables (compared to categorical variables for social surveys) – so data cleaning needs to be very differently defined for business and social surveys
- Administrative data – traditional data cleaning techniques do not work for administrative data due to the size of the datasets and the underlying data collection (which legally and/or practically precludes recontact to validate responses), so data cleaning needs to be automatic wherever possible

2 Editing

Data editing can take place at different levels, and use different methods – the choice is known as the data editing strategy. Different data editing strategies are needed for each data type – there is no “one size fits all” solution for a S-DWH.

Macro- and micro-editing

Editing can be at the micro level, editing individual records, or the macro level, editing (aggregate) outputs.

- Macro-editing is generally subjective – *eye-balling* the output, in isolation and/or relative to similar outputs/previous time periods, or calculating measures of growth and applying *rules of thumb* to decide whether they are realistic or not. This type of editing would not suit the S-DWH

¹ The option of cleaning the data outside the S-DWH, using legacy (or newly built systems), and then combining cleaned data in the S-DWH is not recommended here – due to additional costs and lack of consistency/coherence – but the basic theory is the same wherever data cleaning is performed.

environment, as outputs are separated by two layers from inputs, and given the philosophy of re-use of data it would be difficult to define a process where “the needs of the one (output) outweigh the needs of the many”. Hence nothing more is said about these methods.

- Micro-editing methods are numerous and well-established, and are appropriate for a S-DWH where editing should only take place in the sources layer. Hence these are the focus here.

Hard and soft edits

Editing methods – known as rules – detect errors, but once a response fails the treatment varies dependent on the rule type.

- Hard edits (*some* validity, consistency, logical and statistical) do not require validation and can be treated automatically – see below.
- Soft edits (all remaining) require external validation – see section 3.

Automatic editing

Automatic editing, mentioned in section 1 as a key option for census data, is also commonly used for business survey data as a cost- and burden-saving measure when responses fail hard edits. Given the high costs associated with development of a S-DWH, automatic editing should be implemented wherever possible – at least during initial development. However, another advantage of automatic editing applies both during development and beyond – it will lead to more timely data, as there will be less time spent validating failures, which will benefit all dependent outputs.

Selective editing

Selective (also significance) editing, like automatic editing, is a cost- and burden-saving measure. It reduces the amount of overall validation required by automatically treating the least important edit rule failures *as if they were not failures* – the remaining edit rule failures are sent for validation.

The decision whether to validate or not is driven by the selective editing score – all failures with scores above the threshold are sent for validation, all those with scores below are not validated.

The selective editing score is based on the actual return in period t (y_t), the expected return $E(y_t)$ (usually the return y_{t-1} in the previous period, but can also be based on administrative data), the weight in period t (w_t) – which is 1 for census and administrative data – and the estimated domain total in the previous period (Y_{t-1}):

$$\frac{w_t |y_t - E(y_t)|}{Y_{t-1}}$$

The selective editing threshold is set subjectively to balance cost versus quality: the higher the threshold, the better the savings but the worse the quality. In a S-DWH context, as responses can be used for multiple outputs, it is impossible to quantify the quality impact, so selective editing is of questionable utility. It should definitely be out of scope for all data in the backbone of the S-DWH.

3 Validation

Data validation takes place once responses fail edit rules, and are not treated automatically. The process involves human intervention to decide on the most appropriate treatment for each failure – based on three sources of information (in priority order):

- primary – answer given during a telephone call querying the response, or additional written information (*eg the respondent verified the response when recontacted*)
- secondary – previous responses from the same respondent (*eg if the current response, although a failure, follows the same pattern as previous responses then the response would be confirmed*)
- tertiary – current responses from similar respondents (*eg if there are more than one respondents in a household, their information could explain the response that failed the edit rule*)

In addition to these objective sources of information, there is also a valuable subjective source – the experience of the staff validating the data (*eg historical knowledge of the reasons for failures*).

In a S-DWH environment, the requirement for clean data needs to be balanced against the demand for timely data. This is a motivation for automatic editing, and is also a consideration for failures that cannot be automatically treated. The process would be more objective than outside a S-DWH, as the experience of staff working on a particular data source – the subjective information source for validation – would be lost given generic teams would validate all sources. This lack of experience could also mean that the secondary information source for validation – recognition of patterns over time – would also be less likely to be effective. This means that in a S-DWH, validation would be more likely to depend on the primary and tertiary sources of information – direct contact with respondents, and proxy information provided by similar respondents (or provided by the same respondent to another survey or administrative source).

4 Imputation

The final stage of data cleaning is imputation for partial missing response (item non-response) – the solution for total missing response (unit non-response) is estimation (see 1.3). To determine what imputation method to use requires understanding of the nature of the missing data.

Types of missingness

Missing data can be characterized as 3 types:

- MCAR (missing completely at random) – the missing responses are a random subsample of the overall sample
- MAR (missing at random) – the rate of missingness varies between identifiable groups, but within these groups the missing responses are MCAR
- NMAR (not missing at random) – the rate of missingness varies between identifiable groups, and within these groups the probability of being missing depends on the outcome variable

In a S-DWH environment, the ability to determine the type of missingness is in theory diminished due to the multiple groups and outcome variables the data could be used for, but in practice the type of missingness should be determined in terms of the primary purpose of the data source, as again it is impossible to predict all secondary uses.

Imputation methods

There is an intrinsic link between imputation and automatic editing: imputation methods define how to automatically replace a missing response based on an imputation rule; automatic editing defines how to automatically impute for a response failing an edit rule. Thus imputation methods are akin to automatic editing treatments, but the names are different.

There are a huge number of possible imputation methods – the choice is based on:

- the type of missingness – *generally* deterministic for MCAR, stochastic for MAR, deductive for NMAR
- testing each method against the truth – achieved by imputing existing responses, and measuring how close they imputed response is to the real response

In a S-DWH environment, the choice of imputation method should be determined based on the primary purpose of the data source – in concordance with the type of missingness. This chosen method, and its associated variance, must form part of the detailed metadata for each imputed response to ensure proper inference from all subsequent uses.

Methodology: Data linkage

1 Data linkage

Data linkage is a part of the process of data integration – linking combines the input sources (census, sample surveys and administrative data) into a single population, but integration also processes this population to remove duplicates/mis-matches. The first step in data linkage is to determine needs, the data availability, and whether a unique identifier exists:

- if a unique identifier exists – such as an identification code of a legal business entity, or a social security number – linking is a simple operation
- if a unique identifier does not exist, linking combines a range of identifiers – such as name, address, SIC code – to identify probable matches, but this approach can result in a considerable number of unmatched cases

2 Methods

Data linkage methods are usually deterministic or probabilistic, or a combination. The choice of method depends on the type and quality of linkage variables available on the data sets. In a S-DWH, good quality metadata is a crucial requirement for data linkage.

Deterministic linkage

Deterministic linkage ranges from simple joining of two or more datasets, by a single reliable and stable identifier, to sophisticated stepwise algorithmic linkage. The high degree of certainty required for deterministic linkage is achieved through the existence of a unique identifier for an individual or legal unit, such as company ID number or Social Security number. Combinations of linking variables (*eg first name, last name – for males, sex, dob*) can also be used as a “statistical linkage key” (SLK).

- Simple deterministic linkage – depends on exact matches, so linking variables (individually or as components of the SLK) need to be accurate, robust, stable over time and complete
- Rules-based linkage – pairs of records are determined to be links or non-links according to a set of rules, which can be more flexible than a SLK but are more labour intensive to develop as they are highly dependent on the data sets to be linked
- Stepwise deterministic linkage – uses auxiliary information to adjust SLKs for variation in component linking variables

Most SLKs for individuals are constructed (from last name, first name, sex and dob) as an alternative to a personal identifiers, hence protecting privacy and data confidentiality.

- A commonly used SLK is SLK 581 – comprising 5 characters for name (2nd/3rd/5th from the last name, 2nd/3rd from the first), 8 for dob (“ddmmyyyy”), and 1 for sex (“1” = male, “2” for female).

Data linkage using an SLK is usually deterministic, but this requires perfect linking variables. Two common imperfections leading to multiple SLKs for the same individual or multiple individuals with the same SLK are: incomplete or missing data, and variations/errors (*eg Smith/Smyth*). Probabilistic linkage is then applied as it requires less exacting standards of accuracy, stability and completeness.

Probabilistic linkage

Probabilistic linkage is applied where there are no unique entity identifiers or SLKs, or where linking variables are not as accurate, stable or complete as are required for deterministic linkage. Linkage depends on achieving a close approximation to unique identification through use of several linking

variables. Each of these variables only provides a partial link, but, in combination, the link is sufficiently accurate for the intended purpose.

Probabilistic linkage has a greater capacity to link when errors exist in linking variables, so can lead to much better linkage than simple deterministic methods. However, due to the implicit re-use of data, tolerance of errors in a S-DWH is lower than would be the case for ad hoc linkage projects, so deterministic methods are more likely to be applicable in a S-DWH than would usually be the case.

In deterministic linkage, pairs of records are classified as links if their linking variables predominantly agree, or as non-links if they predominantly disagree. There are 2^n possible link/non-link configurations of n fields, so probabilistic record linkage uses M and U probabilities for agreement and disagreement between a range of linking variables.

- M -probability – probability of a link given that the pair of records is a true link (constant for any given field), where a non-link occurs due to data errors, missing data, instability of values (*eg surname change, misspelling*)
- U -probability – probability of a link given that the pair of records is not a true link, or “the chance that two records will randomly link” (will often have multiple values for each field), typically estimated as the proportion of records with a specific value, based on the frequencies in the primary or more comprehensive and accurate data source

3 Processing

Data linkage can be project-based, ad hoc or systematic (systematic involves the maintenance of a permanent and continuously updated master linkage file and a master SLK). The data linkage process will vary according to the linkage model and method, but there are always 4 steps in common:

- data cleaning and data standardization – identifies and removes errors and inconsistencies in the data, much of which will be dealt with in a S-DWH by data cleaning but some of which is specific to linking (*eg Thomson/Thompson*), and analyzes the text fields so that the data items in each data file are comparable
- blocking and searching – when two large data sets (in a S-DWH the data sets will potentially be very large) are linked the number of possible comparisons equals the products of the number of records in the two data sets: blocking reduces the number of comparisons needed, by selecting sets of blocking attributes (*eg sex, dob, name*) and only comparing record pairs with the same attributes, where links are more likely
- record pair or record group comparisons – record pairs are compared on each linkage variable, with agreement scoring 1 and disagreement scoring 0, scores are weighted by field comparison weights, and the level of agreement is measured by summing weighted scores over linkage variables to form an overall record pair comparison weighted score
- a decision model – record pair comparison weights help decide whether a record pair belongs to the same entity, based on a single cut-off weight or on a set of lower and upper cut-off weights
 - under the single cut-off approach – all pairs with a weighted score equal to or above the cut-off weight are automatically links and all those below are automatic fails
 - under the lower and upper cut-off approach – all pairs with a weighted score below the upper cut-off are automatically links, all those below the lower cut-off are automatic fails, and pairs with weighted score between the upper and lower cut-offs are possible links, sent for clerical review – the optimum solution minimizes the proportion of pairs sent for clerical review, as it is costly, slow, repetitive and subjective

While data cleaning and data standardisation are common to both deterministic and probabilistic linkage, the other steps of the process are more relevant to the probabilistic method.

Quality – determinants and measurement

Key determinants of overall linkage quality are:

- the quality of the (blocking and linking) variables used to construct SLKs (*deterministic*)
- the quality of blocking and linking variables (*deterministic and probabilistic*)
- the blocking and linking strategy adopted (*probabilistic*)

Poor quality (eg if variables are missing, indecipherable, inaccurate, incomplete, inconstant, inconsistent) could lead to records not being linked – *missed links* – or being linked to wrong records – *false links*. The impact of these two types of errors may not be equal (eg a missed link may be more harmful than a false link), so this needs to be taken into account when designing a data linkage strategy, especially if the linking has legal or healthcare implications. The linkage strategy will be further complicated by the variety of sources in a S-DWH, and the recurrent nature of the process.

Linkage quality can be measured in terms of accuracy, sensitivity, specificity, precision and the false-positive rate (see Figure 1). However, not all these measures are easily calculated, because they depend on knowing the number of true non-matches or true negatives, which are unknown or difficult to calculate, hence the most widely used quality measures are:

- sensitivity or true positive rate – the proportion of matches that are correctly classified as matches – *true positive*, or the proportion of all records in a file with a match in another file that are correctly accepted as links – *true links*, calculated as $TP/(TP+FN)$
- precision or the positive predictor value – the proportion of all classified links that are true links or true positives, calculated as $TP/(TP+FP)$

		Match status (True)		
		Matches	Non-matches	
Link status (assigned by computer)	Links	True Links or True Positive (TP) (Matches that are linked)	False Links or False Negatives (FP) (Non-matches that are linked)	Total Links
	Non-links	Missed Links or False Non-links (FN) (Matches that are non-linked)	True Non-links or True Negatives (TN) (Non-matches that are not linked)	Total Non-links
		Total Matches	Total Non-matches	Total record pairs

Figure 1: Classification of matches and links

Linkage to the backbone of a S-DWH

The backbone of a S-DWH is the population (sampling frame) and auxiliary information about the population. The main characteristic of the backbone is that the integrated information is available at micro-data level. Both the backbone itself, and linking to the backbone, will be different for social (household, individual) and business data – linking for business data is explained in detail below.

The backbone for business data is based on statistical units, and contains information on activity, size, turnover and employment of (almost) every enterprise. Data linkage is not a problem when using surveys only, as these are generally based on statistical units from the statistical business register, but is a problem in a S-DWH which uses other data sources. The first step is to link sources to statistical units:

- ideally a unique identifier for enterprises based on the statistical unit would already exist – which would make data linkage simple
- in practice not all input data will link automatically to the statistical unit due to variation in the reporting level (the enterprise group, different parts of the enterprise group, the underlying legal units or tax units), which is driven by the enterprise size (one-to-one relationships can be assumed for small enterprises, but not for medium-sized or large) and national legislation – hence the relationship between the input and statistical units needs to be known before linking

Although most outputs are based on statistical units, some are produced for different units (*eg local units, LKAUs, KAUs, enterprises groups*). Therefore, relationships between the output and statistical units need to be known to generate flexible outputs – which are a fundamental element of a S-DWH.

Methodology: Estimation

1 Estimation

The three main data sources in a S-DWH – census, sample survey and administrative – have very different origins:

- census data are usually a result of a legislative act – a national census carried out at regular intervals – and are included in a S-DWH as they represent the fullest coverage and most definitive measurements of the population of interest, albeit only at limited points in time
- survey data are only collected when there is a requirement for information which cannot be met directly or indirectly from existing data sources – and are included in a S-DWH initially for the purpose of producing specific outputs
- administrative data are uniformly collected for an alternative purpose – for example, tax collection – and are included in a S-DWH as they are freely available (subject to data sharing agreements), even though they are not always directly relevant to statistical outputs

Estimation can involve all three sources of data – in isolation, or in combination. The implications for a S-DWH are very different in each case, and need to be explained at least at a high level of detail.

2 Single source estimation

The methods used in estimation of statistical outputs based on single sources are very different, by necessity, for the three data sources.

Census

In theory, estimation is unnecessary when using census data, but in practice there is nearly always a small amount of non-response that needs to be accounted for. If adjustment is not required, or the adjustment takes place in census-specific production systems outside the S-DWH, then within the S-DWH estimation can be based on census data as a single source – otherwise combined estimation is required. The common approach to adjusting for non-response is based on “capture-recapture” methodology, requiring an additional data source (*eg a census coverage survey*). In an S-DWH environment it is essential to include all the additional data required for non-response adjustment, and to ensure that appropriate metadata exists linking these to the census data.

Sample survey

Using sample survey data as a single source when estimating a statistical output will be decided:

- *a priori* when designing the survey – only when the data are used to estimate their primary output
- as a result of testing – either when the data are used to estimate their primary output, or secondary outputs

Single source estimation is based on survey design weights for both primary outputs and secondary outputs *eg derived variables or different geographical/socio-economic domains* – and hence is known as *design-based estimation*. In a S-DWH it is essential to have comprehensive metadata accompanying the survey data in order to estimate secondary outputs, and also to ensure methodological consistency when combining the survey data with other sources. The metadata should at least include:

- Variables (collected and derived)

- Definition of statistical units
- Classification systems used
- Mode of data collection
- Sample design – target population, sampling frame, sampling method, selection probability (or design weight, which is the inverse of the selection probability)

Administrative

Although administrative data generally represent a census, there are still many issues (in common to most administrative data) when using them for estimation of statistical outputs:

- coverage issues – the target population of the administrative data collection exercise is unlikely to correspond to the target population of the statistical output – if overcoverage is the problem, the administrative source could still be used in single source estimation, but if undercoverage is the problem then combined estimation would be required
- definitional issues – the variables in the administrative source are unlikely to have exactly the same definition as those required by the statistical output – if the variables can be redefined using other variables in the administrative source, or simply transformed, it can still be used in single source estimation, otherwise combined estimation is required
- timing issues – the timing of the collection of administrative data, or the timeframe they refer to, are based on non-statistical requirements, and so often do not align with the timing required by the statistical output – to align timing this requires time series analysis (interpolation or extrapolation commonly) using the same administrative data for other time periods, in which case the estimation is still single source, or using other data source(s), which is combined estimation
- quality issues – as with census data, administrative data generally suffer from some non-response, which needs to be adjusted for during estimation – if non-responses are recorded as null entries in the dataset, then estimation can still be single source, but if other data sources are needed to estimate for non-response, it becomes combined estimation

In a S-DWH, the impact of these issues is that for administrative data to be used in single source estimation, both additional data – the same administrative source in different time periods – and thorough metadata (*eg details of definitions, timing*) are essential.

3 Combined source estimation

Data sources are combined for estimation for a very wide range of purposes, but these can be categorized into 2 broad groups:

- calibration – to improve quality of estimates by enforcing consistency between different sources
- modelling – to improve quality of estimates by borrowing strength from complementary sources

Methodological consistency

Any sources can be combined at a micro-level if they share the same statistical unit definition, or at a macro-level if they share domains, but using combined sources in estimation requires further effort to determine whether methodology is consistent (via analysis of metadata) as this will have quality implications for resulting estimates (*eg even if variables share the same definition, differences in data collection modes and cleaning strategies could make results inconsistent and combining lead to biased estimates*).

When combined sources are consistent in terms of methodology, but results differ for the same domains and same variables, then the reliability of two sources needs to be investigated:

- if the sources are combined for calibration (see below) the more reliable source is pre-determined – by design – or the sources can be combined to form a new calibration total
- if the sources are combined for modelling, either the more reliable source needs to be determined via additional analysis – and identified as the priority source in processing rules – or the sources need to be combined as a composite estimator, acknowledging that neither is perfect, with weights reflecting their relative quality (*eg a classic composite estimator uses relative standard errors to weight the components*)

Calibration

The classic use of calibration is to use scale population estimates from a sample survey to published population estimates from a census, administrative source, or a larger equivalent sample survey. Known as *model-assisted estimation*, this adjusts the design weights to account for unrepresentative samples (*eg due to non-response*) based on the assumption that the survey variables are correlated with the variable that is used to calibrated to the population estimates (*eg business surveys are commonly calibrated to turnover on the statistical business register*). Hence this type of calibration is usually an intrinsic part of survey weighting. The assumption of correlated estimates and population variables is either made:

- *a priori* during design of the survey – when the data are used to estimate their primary output
- as a result of testing – either for the primary output, or secondary outputs

Calibration can also be an extrinsic process, such as contemporaneous benchmarking.

Modelling

Estimation based on modelling involving combined sources is rarely true *model-based estimation* – which assumes that a theoretical super-population model underpins observed sample data, allowing inference from small sample sizes – as the only practical application of model-based estimation is small area estimation (see below). More generally, modelling aims to replace poor quality or missing results – and is sometimes essentially *mass imputation*.

Modelling is generally used when a single source is unable to produce estimates of sufficient quality, or even at all, for domains (geographical or socio-economic) of interest. The additional source(s) either provide these estimates directly, or indirectly by specifying a model to predict them from existing data (or results) from the single source – this includes the mass imputation scenario.

A specific example of modelling is for census data, which require combined estimation to adjust for non-response. The common approach to census non-response is based on “capture-recapture” methodology, which requires a census sub-sample (*eg a census coverage survey*). In a S-DWH environment it is essential to ensure that appropriate metadata exists to link any additional source to the census data.

Small area estimation is a technique to provide survey-based estimates for small domains (often geographical), for which no survey data exist, and/or to improve the estimates for small domains where few survey data exist. The method involves a complex multilevel variance model, and borrowing strength from sources with full coverage of all domains – such as the census – selecting specific variables that explain the inter-area variance in the survey data. The chosen full coverage variables are used to estimate the domains directly, or in combination with the survey data as a

composite estimator. In a S-DWH environment, as long as the model is correctly specified in the analysis layer, the data requirements are still simply linked data sources – this time not at the micro- but the macro-level (aggregated for domains of interest) – and full and comprehensive metadata.

4 Outliers

Outliers are correct responses, as they are only identified once data cleaning is complete, which are either extreme in a distribution and/or have an undue influence on estimates. Outliers can cause distortion of estimates and/or models, so need to be identified and treated as part of estimation.

Common methods for identification and treatment are as follows:

- identification – visualisation, summary statistics, edit-type rules
- treatment – deletion, reweighting
- simultaneous identification and treatment – truncation, Winsorisation

In a S-DWH, identification and treatment both take place in the analysis layer.

Identification

Outliers can be identified qualitatively (*eg visual inspection of graphs*) or quantitatively (*eg values above a threshold*). Qualitative methods are more resource intensive, but are not necessarily of higher quality as the quantitative threshold is usually set subjectively, often to identify a desired number of outliers or a desired impact on estimates from treatment of the outliers.

Treatment

Outlier treatment fundamentally consists of weight adjustment:

- an adjustment to 0 percent (of original) equates to deleting the outlier (*eg truncation*)
- an adjustment to P percent (of original) equates to reducing the impact of the outlier (*eg reweighting and Winsorisation*)
- an adjustment to 100 percent (of original) equates to not treating the outlier (*eg ignoring it*)

All treatments reduce variance but introduce bias – so Winsorisation was developed to optimise this trade-off by minimising the mean squared error (the sum of the variance and the squared bias).

Outliers in a S-DWH

In a S-DWH environment there are three types of outliers – outliers in survey data, outliers in administrative data, and outliers in modelling:

- survey data – outliers are *unrepresentative* values, which means they only represent themselves in the population (population uniques) rather than representing (p-1) unsampled units in the population, as is assumed when weighting a unit sampled randomly with selection probability $1/p$ *eg footballers with extreme salaries randomly selected from a general population*
- administrative data – outliers are *atypical* values, which means they are simply extreme in the population as administrative data represent a census so do not require weighting and each unit

is treated as unique (*eg similar sources are prioritised for updating a statistical business register, but if the difference between them is above a certain limit, this identifies an outlier*)

- modelling¹ – outliers are *influential* values, which means they have an undue effect on the parameters of the model they are used to fit (*eg an extreme ratio when imputing & Fig Y below*)

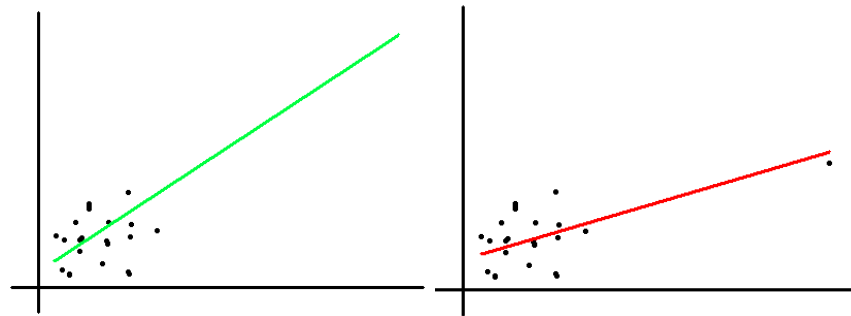


Figure Y: Modelling outlier – regression without extreme x-value (LHS, green) and with (RHS, red)

Identifying and treating outliers is complicated by the intended re-use of data in a S-DWH:

- survey data outliers are conditional on the target population (*eg if the target population was footballers only, a footballer's salary would no longer be an outlier*)
- administrative data outliers are conditional on their use (*eg if ratios of turnover to employment were consistent for two similar sources even though numerators and denominators were different – due to timing perhaps – the differences would no longer identify an outlier*)
- modelling outliers are conditional on the model fitted (*eg an outlying ratio for average of ratio's imputation, would no longer be an outlier for ratio of averages & in Fig Y above if the model is "average y-value" the extreme response with an x-value is no longer an outlier*)

In summary, any unit in a S-DWH can be an outlier (or not an outlier), conditional on the target population, the use in estimation, and the model being fitted. Hence, it is impossible to attach a meaningful outlier designation to any unit. The only statement that can be made with certainty is:

Every unit in a data warehouse is a potential outlier

It is not even possible to attach an outlier designation to any response by a unit – as it would have to record the use – ie the domain and period for estimation, and the fields combined and model used – and this will not be fixed given the intended re-use.

Given that neither the units in a S-DWH, nor the specific responses of units, can be identified as outliers *per se*, identification is domain- and context-dependent. This means that outliers are identified during processing of source data, and reported as a quality indicator of the output – if the output itself is stored in a SDWH, the outliers identified will become part of the metadata accompanying the output, but will not be identified as outliers at the micro-data level.

¹ Modelling includes setting processing rules (for example, editing/imputation), as well as statistical modelling

5 Further estimation

Often referred to as further analysis techniques, index numbers and time series analysis methods are often an integral part of the process leading to published estimates, but these methods will have no impact on metadata at the micro-level as they are applied to macro-data only. In a S-DWH, processing should be automatic, so these further steps are assumed to be part of estimation.

Index numbers

If users are more interested in temporal change than cross-sectional estimates (*eg growths not levels*), instead of releasing estimates as counts they are often indexed – by setting a base period as 100 and calculating indices as a percentage of that. Indices are also used, sometimes in combination with survey sources to provide weighting, to combine changes in prices or quantities across disparate products or categories into a single summary value. There are also many different index number formulae (*eg Paashche, Laspeyres*) that can be used, and different approaches to presenting time series of index numbers (*eg chained, unchained*).

Interpolation and extrapolation

If estimates are required for time periods that cannot be calculated directly from data sources, time series analysis techniques can provide estimates between existing time periods – *interpolation* – and before the earliest (*ie backcasting*) or after the latest (*ie forecasting*) existing time periods – *extrapolation*. Both interpolation and extrapolation can be used in single source or combined source estimation, and for both there are a huge variety of methods available (*eg ARIMA, splining*).

Benchmarking

Benchmarking is a time series analysis technique for calibrating different estimates of the same phenomenon. This usually requires physically bringing the data (micro- or macro-) together, aligning metadata, and then choosing a linking method – but in a S-DWH the data are already in the same environment, with consistent metadata, and pre-defined linking methods: so benchmarking will be easier in a S-DWH.

The most common use of benchmarking is *contemporaneous* – calibration at the same point in time – but *temporal* benchmarking – calibration over time – is also used, especially in the context of seasonal adjustment (see below) where the annual totals of the seasonally adjusted and unadjusted estimates are constrained to be consistent.

The aim of benchmarking is constant – to ensure consistency of estimates – but can be approached in two fundamentally different ways:

- binding benchmarking – defines one estimate as the benchmark, and calibrates the other estimates to it
- non-binding benchmarking – defines the benchmark as a composite of the different estimates, and calibrates all estimates to it

Non-binding benchmarking is theoretically appealing, as no estimate – by definition – is correct, and non-binding benchmarking combines all the estimates to form an improved estimate, but it is very rarely used in practice. The main reason for this is revisions – binding benchmarking means that the more reliable estimate, which is used as the benchmark, doesn't have to be revised. Given that the

benchmark estimate is usually a headline publication, it is understandable why producers do not want to change it – albeit possibly only by a small amount – based on a less public and less reliable estimate. Even if the headline estimate was released after non-binding benchmarking – which is feasible as being more reliable, is also likely to be less timely than the less public estimate(s) – any revisions to the less public estimate(s) would revise the non-binding benchmark, and hence cause revisions in the headline estimate.

Seasonal adjustment

Estimates have three (unobserved) components – long term change (*trend*), short-term sub-annual movements around the trend (*seasonal*) and random noise (*irregular*). As the seasonal component repeats annually (*eg increased retail sales at Christmas*) it can distort interpretation of short-term movements (*eg sales increases November to December do not imply an improving economy*). Hence the seasonal component is often removed from published estimates – they are *seasonally adjusted*. However, not all time series have a seasonal component (*eg sales of milk*) so seasonal adjustment is sometimes not required.

As the seasonal component is unobserved it has to be estimated – and as the nature of seasonality changes over time, the estimation parameters – and even the variables – also need to change to ensure the estimates are properly seasonally adjusted. The seasonal component can be estimated automatically, so this *moving seasonality* is not in itself a problem in a S-DWH. However, the nature of the seasonal component – a repeating annual effect – means that when the seasonal component is re-estimated, it is re-estimated for the entire time series. Hence any changes cause revisions throughout the time series. There are two common approaches to reducing these revisions – only revising the time series back to a certain point, and keeping the estimation variables for the seasonal component constant over a set time period (usually 1 year). The advantage of the first (*eg an up-to-date seasonal component for current estimates*) is the disadvantage of the second, but the advantage of the second (*eg a stable time series*) is not the disadvantage of the first, which is a discontinuity in the time series. Given that the chosen approach is usually applied to all outputs within an NSI, again this is not in itself a problem in a S-DWH.

However, a more problematic issue in a S-DWH is a *seasonal break*. This can be an abrupt change in the seasonal component (*eg in 1999 new car registrations in the UK changed from annual to biannual, and the seasonal component for new car sales immediately changed from having one annual peak to two*), or a series becoming seasonal (or non-seasonal). Although the treatment of seasonal breaks can be automated, their detection cannot be (with any degree of accuracy). As seasonal breaks can occur in **any** time series at **any** time, all seasonally adjusted estimates should be quality assured before release. Ideally, this quality assurance should be manual, but a compromise is to have an annual quality assurance supplemented by automatic checks to identify unexpected movements or differences (*eg between the unadjusted and seasonally adjusted estimates*).

Methodology: Revisions

1 Revisions

Revisions to estimates are a fact of life in statistical production – they reflect improvements in data or methods, and need to be incorporated in planning for systems – they should not be a surprise.

In contrast, revisions due to errors in production are a surprise, and can occur at any time due to manual mistakes or incorrect coding of software, but cannot be planned for – so are not discussed here. Suffice to say that in a S-DWH, clear response plans need to be in place in case of errors.

2 Micro-data

Revisions to micro-data are corrections – either due to updates from the data supplier or cleaning applied by the producer. The original micro-data, and all revised versions, need to be stored in a S-DWH along with comprehensive metadata to explain the reasons for the revisions.

Although the original dataset before cleaning is clearly the first vintage, not all data will be revised, and the timing of each datum being revised will vary, so later vintages of micro-data can only be defined at points in time. These “date” vintages are of overall value, as outputs are generally produced at certain dates from the latest available data, but to capture all the changes made to micro-data requires versions to be defined for each response for each unit: each change to the datum will define a new version, and each version needs to be accompanied by different metadata to explain the changes.

3 Macro-data (outputs)

Revisions to census and administrative macro-data are corrections, but revisions to sample survey macro-data outputs are also caused by methodological changes.

Some outputs are routinely revised due to estimation processes (*eg benchmarking and seasonal adjustment, above*), data cleaning and data updates, whereas others are never revised due to legal/financial implications (*eg HICP*). For the routinely revised outputs, a S-DWH needs to store all vintages (versions) of the estimates with appropriate metadata. These metadata (macrometadata?) are for outputs as distinct from the metadata for micro-data (micrometadata?) discussed above.

One-off revisions, due to fundamental methodological changes or revised classification systems (*eg NACE codes*), always have a major impact on outputs – and obviously need to be properly captured in macrometadata alongside the vintage – but have an even greater impact on production systems, so a S-DWH also needs to update processing to reflect these one-off events.

Methodology: Statistical Disclosure Control

1 Statistical disclosure control

When releasing any outputs, the confidentiality of personal and business information needs to be safeguarded, sometimes to comply with legal obligations but always to secure trust of respondents. However, the only way to guarantee zero risk of disclosure is not to release any outputs – so the risk is always balanced against the utility, ie how useful the outputs are to users. Given that micro-data in a S-DWH are designed for re-use, there may be multiple outputs and also multiple users, hence both risk and utility are more difficult to measure and confidentiality more difficult to guarantee.

Statistical disclosure control (SDC), sometimes referred to as the

art and science of protecting data

involves modifying data or outputs to reduce the risk to an *acceptable* level.

2 ESSnet on Statistical Disclosure Control

Substantial work was completed during the ESSnet on Statistical Disclosure Control and a comprehensive handbook¹ was produced in January 2010. The handbook aims to provide technical guidance on statistical disclosure control for NSIs on how to balance utility and risk:

- utility – preventing “information loss” providing users with the statistical outputs they need (*eg to determine policy, undertake research, write press articles, or find out about their environment*)
- risk – the “probability of disclosure” of confidential information and hence not protecting the confidentiality of survey respondents (*eg by releasing data at too fine a level of granularity*)

The main challenge for NSIs is to optimize SDC methods and solutions to maximize data usability whilst minimizing disclosure risks.

The handbook provides guidance for all types of statistical outputs. From a S-DWH perspective, there is reference within the handbook on dynamic databases whereby successive statistical queries to obtain aggregate information could possibly be combined with earlier data, leading to increased disclosure risk. There is also substantial discussion relating to the release of micro-data which is the newest sub-discipline of SDC. Chapters 3, 4 and 5 of the handbook examine the separate problems of micro-data, magnitude tabular data and finally frequency tables, and discuss available software. Chapter 6 focuses on remote access issues which is likely to have implications for any pan-European S-DWH and section 6.6 explains the confidentiality protection of analyses that are produced.

3 ESSnet on Data Integration

The handbook produced by the ESSnet on Statistical Disclosure Control is well supplemented by information produced by Work Package 1 of the ESSnet on Data Integration with a report outlining the “State of the art on Statistical Methodologies for Data Integration”², in which Chapter 4 is dedicated to a literature review update on data integration methods in SDC. The two main areas

¹ http://neon.vb.cbs.nl/casc/SDC_Handbook.pdf

² <http://www.cros-portal.eu/wp1-state-art>

covered are those of contingency tables and of micro-data dissemination, with section 4.2 focussing on SDC and data linkage. The main conclusion of the handbook is a strong recommendation that a system of disclosure risk measure be set up to monitor the data dissemination processes, in order to minimize the risk of compromising data confidentiality.

5-Metadata

5.1 Fundamental principles

Authors: Tauno Tamm

5.2 Business Architecture: metadata

Authors: Sónia Quaresma

5.3 Metadata System

Authors: Tauno.Tamm

5.4 Metadata and SDMX

Authors: Tauno.Tamm

References

Lars-Göran Lundell; ESS-Net-DWH Deliverable 1.1: "Framework of metadata requirements and roles in the S-DWH"

Colin Bowler, Michel Lindelauf, Jos Dressen; ESS-Net-DWH Deliverable 1.2: "Recommendations on the Impact of Metadata Quality in the Statistical Data Warehouse"

Maia Ennok, Kaia Kulla, Lars Goran Lundell, Colin Bowler, Viviana De Giorgi; ESS-Net-DWH Deliverable 1.4: "Definition of the functionalities of a metadata system to facilitate and support the operation of the S-DWH"

Viviana De Giorgi, Michel Lindelauf; ESS-Net-DWH Deliverable 1.5: "Recommendations and guidelines on the governance of metadata management in the S-DWH"

Maia Ennok; ESS-Net-DWH Deliverable 1.6: "Documentation of the mapping of the result of 1.4 on the 'ideal architecture' framework"

Antonio Laureti Palma, Björn Berglund, Allan Randlepp, Valerij Žavoronok; ESS-Net-DWH Deliverable 3.5: "Relate the 'ideal' architectural scheme into an actual development and implementation strategy"

1 Metadata

Metadata are data which describe other data.

When building and maintaining a S-DWH, the following types of metadata play significant roles:

- active metadata – the amount of objects (variables, value domains, etc.) stored makes it necessary to provide the users (persons and software) with active assistance finding and processing the data;
- formalized metadata – the amount of metadata items will be large and the requirement for metadata to be active makes it necessary to structure the metadata very well;
- structural metadata - active metadata must be structural, at least to some part;
- process metadata - since the data warehouse supports many concurrent users it is very important to keep track of usage, performance, etc. In a data warehouse that has been less than perfectly designed one user's choice of tool or operation could impair the performance for other users. An analysis of process metadata can be an input to correcting this anomaly.

The table below shows the possible combinations of metadata categories and subsets. In the cells are indicated which combinations are of general interest for statistics production ("gen") and which ones are of particular interest for a S-DWH ("sdw"). Most of the remaining combinations are possible, but less common or less likely to be useful.

Metadata subset	Metadata category							
	Formalized				Free-form			
	Reference		Structural		Reference		Structural	
	Act	Pas	Act	Pas	Act	Pas	Act	Pas
Statistical			sdw			gen		
Process	sdw		sdw	sdw	gen			gen
Quality		sdw				gen		
Technical			sdw					
Authorization			gen					
Data model				sdw				sdw

Metadata categories and subsets

Consistency within the metadata layer is an example of an attribute regarded as desirable in any statistics production environment, but that is considered essential in a S-DWH environment. In a S-DWH, all metadata items must be uniquely identified and there must be one-to-one relationships between identity and definition, and identity and name. The concept "statistical unit", for example, must be given an identity and a definition, and these must be consistently used in the S-DWH regardless of source, context, etc. If there will be a need for a slightly different definition, it must be given a new identity and a new name.

In the S-DWH it is desirable to be able to analyze data by time series at a low level of aggregation, or even to perform longitudinal analysis at unit level. To support these functions, metadata items should have validity information: “valid from 01-01-2001”, “valid until 31-12-2015”.

In order to be metadata driven the S-DWH has higher demands for process metadata, and it is more likely to have a built-in ability to produce process metadata.

The S-DWH is not only a data store, but it is also a system of processes to refine its data from input to output. These processes need active metadata: automated processes need formalized process metadata, such as programs, parameters, etc., and manual processes need process metadata such as instructions, scripts, etc.

1.1 Fundamental principles

In order to use metadata in a S-DWH, basic definitions and common terminology need to be agreed.

This section covers:

- basic definitions
- categories
- subsets
- architecture

1.1.1 Metadata and data basic definitions

General definitions of *metadata* can be found in many manuals. Most of them are very short and simple. The most commonly used generic definition states that “**Metadata** are data about data” but more precise definition states:

[Def 1.1] **Metadata** is data that defines and describes other data.¹

This definition will obviously cover all kinds of documentation which refer to any type of data in a data store. In context of S-DWH we use *statistical metadata* which is applicable to metadata that refer to data stored in a S-DWH.

[Def 1.2] **Statistical metadata** are data that define and describe statistical data

Since the definition of metadata shows that they are simply a special case of data, we need a reasonable definition of *data* as well. A derivative from a number of slightly varying definitions would be:

[Def 1.3] **Data** are characteristics or information, usually numerical, that are collected through observation²

In a statistical context:

[Def 1.4] **Statistical data** are data that are collected from statistical and/or non-statistical sources and/or generated by statistics in process of statistical observations or statistical data processing³

1.1.2 Categories

Metadata items can be described by three main metadata categories:

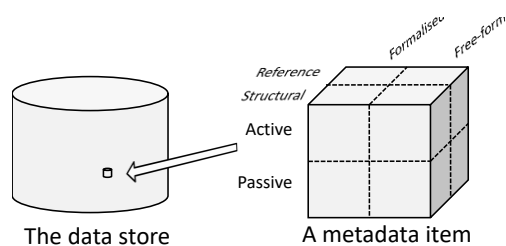
- Passive or Active;
- Formalized or Free-form;
- Reference or Structural.

Each metadata item can be then viewed as an element of a multi-dimensional metadata structure; this is shown in the figure below:

¹ ISO/IEC 11179-1:2004(E) ja Eurostat's Concepts and Definitions Database

² Eurostat's Concepts and Definitions Database

³ Eurostat's Concepts and Definitions Database



Multi-dimensional metadata structure

1.1.2.1 Passive or Active category dimension

Traditionally, metadata have been seen as the documentation of an existing object or a process, such as a statistical production process that is running or has already finished. Metadata will become more active if they are used as input for planning, for example a new survey period or a new statistical product.

[Def 2.1] **Passive metadata** are all metadata used for documentation of an existing object or a process.

This indicates a passive, recording role, which is useful for documenting.

Examples: Quality report for a survey/census/register; documentation of methods that were used during a survey; most log lists; definitions of variables

[Def 2.2] **Active metadata** are metadata stored and organized in a way that enables operational use, manual or automated, for one or more processes.

The term active metadata should, however, be reserved for metadata that are operational. Active metadata may be regarded as an intermediate layer between the user and the data, which can be used by humans or computer programs to search, link, retrieve or perform other operations on data. Thus active metadata may be expressed as parameters, and may contain rules or code (*algorithmic* metadata).

Examples: Instruction; parameter; script (SQL, XML).

1.1.2.2 Formalized or Free-form category dimension

All metadata could be structured, or could be created and stored in completely free form. In practice all metadata probably follow some kind of structure, which may be more or less strict.

[Def 2.3] **Formalized metadata** are metadata stored and organized according to standardized codes, lists and hierarchies

This means that only pre-determined codes or numerical information from a pre-determined domain may be used. Formalized metadata can be easily used actively.

Examples: Classification codes; parameter lists; most log lists.

Formalized metadata are obviously well suited for use in an active role and since active metadata are vital to building an efficient S-DWH, it follows that its metadata should also be formalized whenever possible.

[Def 2.4] **Free-form metadata** are metadata that contain descriptive information using formats ranging from completely free-form to partly formalized (semi-structured)

Free-form metadata mainly refers to documentation that is not organized in a pre-defined manner. Free-form metadata is typically text-heavy, but may contain data such as dates, numbers, and facts as well.

Unstructured metadata, for example a set of chapters, subdivisions, headings, etc., may be mandatory or optional and their contents may adhere to some rules or may be entered in a completely free form (text, diagrams, etc.).

Examples: Quality report for a survey, a census or register; methodological description; process documentation; background information.

1.1.2.3 **Reference or Structural category dimension**

Generally reference metadata (also known as business, conceptual, logical, quality, methodological) help the user understand, interpret and evaluate the contents, the subject matter, the quality, etc, of the corresponding data, whilst structural metadata (also known as technical) help the user, who in this case may be human or machine, find, access and utilize the data operationally.

[Def 2.5] **Reference metadata** are metadata that describe used concepts, used methods and quality measures for the statistical data

Preferably, reference metadata should describe the concepts used and their practical implementation, allowing users to understand what the statistics are measuring and, thus, their fitness for use; the methods used for the generation of the data ; the different quality dimensions of the resulting statistics.

Reference metadata are typically passive and stored in a free format, but with more effort they can be active and formalized by storing them in a structured way. *Examples:* Quality information on survey, register and variable levels; variable definitions; reference dates; confidentiality information; contact information; relations between metadata items.

[Def 2.6] **Structural metadata** are metadata that help the user find, identify, access and utilize the data

Particularly in a S-DWH, structural metadata can be defined as any metadata that can be used actively or operationally. The user may in this case be a human or a machine (a program, a process, a system). Structural metadata describe the physical locations of the corresponding data, such as names or other identities of servers, databases, tables, columns, files, positions, etc. *Examples:* Classification codes; parameter lists.

1.1.3 **Metadata subsets**

In a S-DWH, each metadata item should belong to one of the following metadata subset:

- Statistical
- Process
- Quality
- Technical
- Authorization

- Data models

Several more types may be identified to serve special purposes, but are not further described here. The indicated subsets are described below.

1.1.3.1 Statistical metadata

Statistical metadata directly refer to central concepts in the statistics. This still means that the statistical metadata subset may – at least partly – overlap some other subsets, but will exclude some more administrative and technical ones. Statistical metadata may use any of the metadata formats.

Examples: Variable definition; register description; code list

1.1.3.2 Process metadata

Information on an operation, such as when it started and ended, the resulting status, the number of records processed, which resources were used is known as process metadata (also process data, process metrics, or paradata). These data may contain either expected values or actual outcomes. In both cases, they are primarily intended for planning – in the latter case by evaluating finished processes in order to improve recurring or similar ones. If process metadata are formalized, this will obviously facilitate computer-aided evaluation. Process metadata are less likely to be categorized as free-form, but may be active or passive, and reference or structural.

[Def 3.1] **Process metadata** are metadata that describe the expected or actual outcome of one or more processes using evaluable and operational metrics

Examples: Operator's manual (passive, formalized, reference); parameter list (active, formalized, structural); log file (passive, formalized, reference/structural).

1.1.3.3 Quality metadata

Keeping track of, maintaining and perhaps raising the quality of the data in the S-DWH is an important governance task that requires support from metadata. Quality information should be available in different forms and serve several purposes: to describe the quality achieved, to serve the end users of the data, or to measure the outcome to support governance and future improvements. Most quality metadata can be categorized as passive, free-form and reference metadata.

[Def 3.2] **Quality metadata** are any kind of metadata that contributes to the description or interpretation of the quality of data.

Examples: Quality declarations for a survey, a census or a register; documentation of methods that were used during a; most log lists.

1.1.3.4 Technical metadata

Technical metadata are usually categorized as formalized, active and structural.

[Def 3.3] **Technical metadata** are metadata that describe or define the physical storage or location of data.

Examples: Server, database, table and column names and/or identifiers; server, directory and file names and/or identifiers

1.1.3.5 Authorization metadata

Every computerized system needs some way of handling user privileges, access rights, etc. Users need to be classified or assigned a role as, or to be given an explicit privilege to “read”, “write”, or “update” a certain item, etc. In a S-DWH, having a large amount of data and many users performing various tasks, there is a need for a comprehensive authorization subsystem. This system will need to store and use its own administrative data, which may be defined as authorization metadata. Authorization metadata are categorized as active, formalized and structural.

[Def 3.4] **Authorization metadata** are administrative data that are used by programs, systems or subsystems to manage users' access to data.

Examples: User lists with privileges; cross references between resources and users.

1.1.3.6 Data models

The various types of data models are an often overlooked type of metadata. The reason is probably that these metadata are usually only seen as useful to the technical staff (IT personnel).

[Def 3.5] A **data model** is an abstract documentation of the structure of data needed and created by business processes.

Important types of data models for the S-DWH include the *conceptual model* that usually gives a high-level overview, and the *physical model* that describes the details of databases, files, etc. The *metadata model* (see 2.3) can also be described conceptually as well as physically.

[Def 3.5.1] A **metadata model** is a special case of a data model: an abstract documentation of the structure of metadata used by business processes.

1.1.4 Metadata architecture

In order to find, retrieve and use metadata efficiently their locations must be known to users on some level. A S-DWH is often described as consisting of several layers that serve separate functions⁴. Since metadata is a vital part of the S-DWH, the term metadata layer is sometimes used to refer to both the metadata store and metadata functions in the S-DWH.

[Def 4.1] A **metadata layer** is a conceptual term that refers to all metadata in a data warehouse, regardless of logical or physical organization.

Metadata need to be organized in some kind of structured, logical way in order to make it possible to find and use them. A logical structure may be physically stored in several distributed, coordinated structures. A distinction can be found in the level of formal organization of the metadata store, the restrictions and approval rules required to perform changes, and the coordination of the contents. The term *registry* often refers to a more strictly administered, regulated and coordinated environment than the more general term *repository*.

[Def 4.2] A **metadata registry** is a central point where logical metadata definitions are stored and maintained using controlled methods.

⁴ Palma, S-DWH Business Architecture, 2013

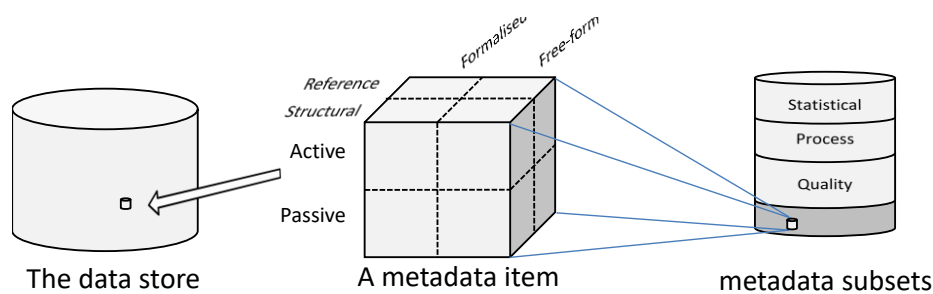
In order to load a metadata item into the registry it must fulfil requirements regards structure, contents and relations to other metadata items. Normally the registry does not define any links between metadata and the data they describe. Usually the definition of a metadata repository does not require the metadata to adhere to strict rules in order to be loaded. However, the repository usually implies storing metadata for operational use, so it is expected to contain a link to the corresponding data and it is operationally used to locate and retrieve data.

[Def 4.3] A **metadata repository** is a physical location where metadata and their links to data are stored.

In a repository we consider active, formalized and structural metadata for all kind of subsets.

- active metadata - the amount of objects (variables, value domains, etc.) stored makes it necessary to provide the users with active assistance finding and processing the data
- formalized metadata - the amount of metadata items will be large and the requirement for metadata to be active makes it necessary to structure the metadata very well
- structural metadata - especially technical metadata; active metadata must be structural, at least to some degree

The metadata layer is used to locate and retrieve data, as shown below:



Using the metadata to locate and retrieve data

Since a S-DWH supports many concurrent users, it is very important to keep track of usage, performance, etc. In a S-DWH that has been less than perfectly designed, one users' choice of tool or operation could impair the performance for other users. An analysis of process metadata can be an input to correcting this anomaly.

1.2 Business Architecture: metadata

In general discussions on metadata for the statistical production lifecycle, several attempts have been made to link metadata to the generic processes: what metadata are produced during a process, what metadata are needed to perform a process, and what metadata are forwarded from one process to the next one.

The GSBPM is applicable to any statistics production, including a S-DWH. There are, however, alternative or complementing models that may be used to describe the specific metadata needs for a S-DWH.

This section covers:

- Preparatory work
- Metadata of the S-DWH layers
- Summary of S-DWH layers and metadata categories

1.2.1 Preparatory work - Specify needs (Phase I of GSBPM)

In this phase the organization determines the need for the statistics, identifies the concepts and variables etc.

The result of this step is the description of every sub-process. The deep analysis helps to avoid errors when similar information is already available. Therefore financial and human resources can be saved significantly. The description created in this phase is needed for the second phase (GSBPM 2, design), where the definitions of the variables are created.

The problems of this phase are as follows:

- Methodological consistency: The organization could determine the needs for similar information, but different in methodological terms. There is a problem of the methodological consistency.
- The integration problem of different data sources could arise: The organization could determine the data sources: survey sampling, administrative sources or statistical register, or integrated e.g. survey and administrative.

In this phase the metadata should be identified and defined for:

- User needs
- Survey objectives
- Legal framework
- Statistical outputs
- Quality measures for statistical outputs
- Concepts
- Data sources

1.2.2 Preparatory work – Design (Phase 2 of GSBPM)

The result of this phase is the defined variables, the described methodology of the data collection, the design of the frame and the sample, the statistical processing, and the design of the production systems and the workflow.

The methodological side of this step is important to the rest of steps of GSBPM. The main tasks (when there is more than one data source) are as follows.

- To compare descriptions of variables from different sources
- To compare methodologies of design data collection, design frame and sample, design statistical processing
- To compare the design of the production systems and the workflows

During the phase of identify needs (1 of GSBPM), the integration problem of data from different sources was indicated. If there is more than one variable (with similar characteristics), in this phase must be clearly defined similarities and differences of the variables. Explanations of methodologie aspects must be presented. The integration procedure should be documented and manuals for the stuff should be prepared.

In the frame of S-DWH the priorities or rules for different data sources when we integrate similar variables should be defined in this phase. We can correct the priorities (rules) if there is a necessity and document them. The main set of metadata is defined in this phase:

- Indicators
- Indicators (derived)
- Statistical unit
- Classification/Code list
- Data collection mode
- Questionnaire
- Target population
- Register
- Frame design
- Sampling method
- Processing methods (description of the methods that cover all the GSBPM phases)
- Operational methods (methods that are mainly related to the specification of IT)

Example 1. We provide an example of two possible scenarios when similar variables from different data sources are integrated. (Figures 1, 2).

- 1) **We can integrate** similar variables (A1, A2, A3..) from different data sources (respectively S1, S2, S3,...) and obtain only one variable A. (Integration priorities or rules should be defined)

Input from datasources (S1, S2, S3, ...)

Output

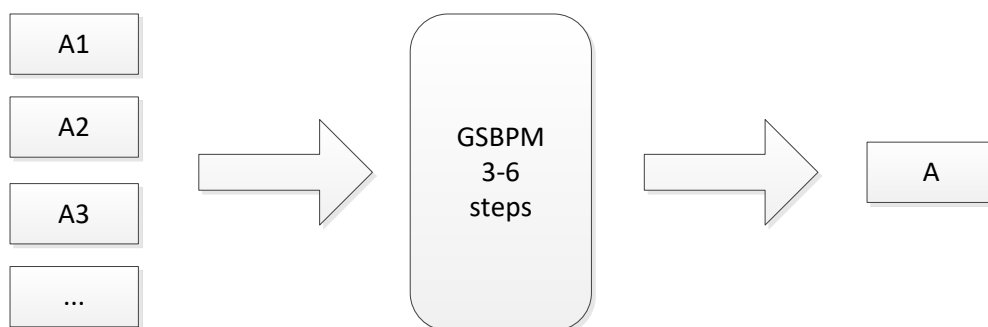


Figure 1.Data integration from different sources

- 2) **We cannot integrate** (there are objective reasons like different definitions of the variable, different methodologies, and other) all variables (A1, A2, A3...) from different data sources (respectively S1, S2, S3,...). The output is variables A1*, A2*, A3*,...

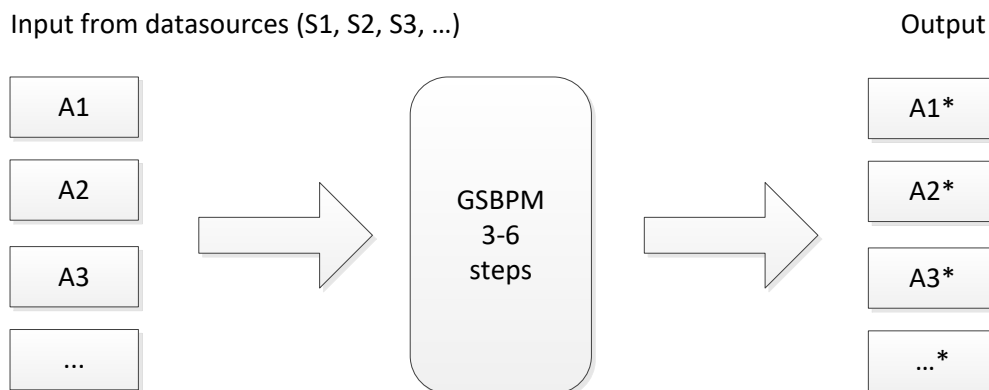


Figure 2. Data integration from different sources

Fursova (2013) summarised the problems that can arise during the data integration process in S-DWH. When we are linking data from different sources like sample surveys, combined data and administrative data we can meet such problems as data missing, data overlapping, “unlinked data” etc. Errors might be detected in statistical units and target population when linking other data to this information. If these errors are influential they need to be corrected in S-DWH.

One of the problems is the conflict between the sources. Data conflict is when two or more sources contain data for the same variable (with different values). In many cases, when two (or more) reliable sources conflict, one (or more) of those sources can be demonstrated to be unreliable. The main goal is to define data source priority for each indicator and rules determining the quality of priority data source.

It is needed to be defined what data source for what indicator is more reliable and it is needed to be determined by different criteria. In some other cases it could be needed additional analysis, and could be used more sophisticated methods, or even manual techniques. To determine priority source need to be defined priority rules: like quality of data source, completeness, update time, and the consultation with experts.

When there is no unique identifier, we use more sophisticated methods for matching and linking several identifiers. It could cause that some data could be “unlinked”. Poor quality of selected linkage variables or of probabilistic methods can lead to some record not being linked or linked to the wrong records, some records are unable to be linked because of missing, incomplete or inaccurate variables.

1.2.3 Preparatory work - Build (Phase 3 of GSBPM)

The objective of this phase is building and testing the production system. Processing and operational methods are tested and completely defined.

The result of this phase is the tested production system. The components of the process and technical aspects should be documented; the user manuals should be prepared. Concerning the S-DWH the additional metadata should be described. The additional metadata will identify the similarities or differences between different cases at the level of separate sub-process. There are two possible ways to compare the cases:

- To analyze the specific (critical) metadata in every sub-process and fix the similarities or difference.
- To analyze the specific (critical) metadata in the phase 5 where the data integration process is performed. E.g. if the statistical data from two data sources are integrated (sub-process 5.1), the specific metadata of the priorities of data sources should be defined on the phases 1-3 of GSBPM.

1.2.4 Critical area

More than a half of metadata is defined in phases 1-3 of GSBPM. In phases 4-6 of GSBPM the metadata

- Could be used as defined in phases 1-3,
- Could be replaced/supplemented (according to the additional information of the sub-process. E.g. we defined some metadata in phases 1-3 but sometimes we need to make corrections in separate sub-process.),
- Could be updated (when the metadata is used in a particular sub-process).

Metadata of the phases 4-6 of the GSBPM is discussed in more detailed level in metadata chapter (reference).

In S-DWH different statistical processes are integrated. In order to link the information from different sources at the level of separate sub-process we need to have additional meta information. We define this group of information “critical area” with the main objectives of comparing different processes at separate sub-process level.

Critical area could help to analyse the differences between different processes. It is useful to define the metadata of critical area for all or selected sub-processes. Possible examples of metadata of critical area at the level of sub-process of GSBPM are provided in Table 1. Also the description and examples of comparison is given in that table.

Using metadata of critical area there is possibility to compare what similarities and differences are between these processes. E.g. in the 4.1 Select sample there are several metadata for critical area defined (The same classification / Not the same; Frozen frame /Not Frozen frame,...). We could check if all processes use the same or not the same classification, e.g. Nace2; if all surveys uses frozen or not frozen frame.

Metadata of critical area	Description of the objectives
Select sample:	
same classification / not the same	to check if all processes use the same or not the same classification, e.g. Nace 2.
frozen frame / not frozen frame	to check if frozen or not frozen frame is used for the selection of enterprises.
survey sampling / census survey	to check if survey sampling or census survey is used.
same/different criteria for the selection of enterprises	to check if the same criteria for selection of enterprises are used, e.g. selection 80 per cent of enterprises with biggest annual income.

Integrate data: unique ID/ not unique ID the same priorities/ not the same to make correction / no corrections	to analyse if the enterprise has a unique identification code. to check if the same (similar) priorities are used for the integration of statistical data from different data sources. to check if some corrections for the statistical data are provided during the integration sub-process.
Review, validate and edit: the same editing rules / not the same	to check if the same (similar) or different editing rules used for different surveys.
Calculate weights: weights calculation/ no weights	to check if weights are calculated or no weights are used.
Prepare draft outputs: the same / not the same quality rules	to check if the same (similar) quality rules for the statistical output are used.
Validate outputs: the same validation rules / not the same	to check if the same (similar) validation rules for the statistical output are used.
Apply disclosure control: the same disclosure control rules / not the same	to check if the same (similar) disclosure control rules for the statistical output is used.
Finalise output: the same procedure of validation / not the same	to check if the same (similar) procedure for the validation of statistical output is used.

Table 1. Metadata of critical area

1.2.5 Metadata of the S-DWH layers

The metadata layer, at the left-hand side of the S-DWH schema in Figure 1, indicates the necessity of metadata support to each layer. In practice, metadata are used and produced in every sub-process of the statistical production lifecycle as an input, to perform each sub-process and as an output, to predispose metadata for the next sub-process.

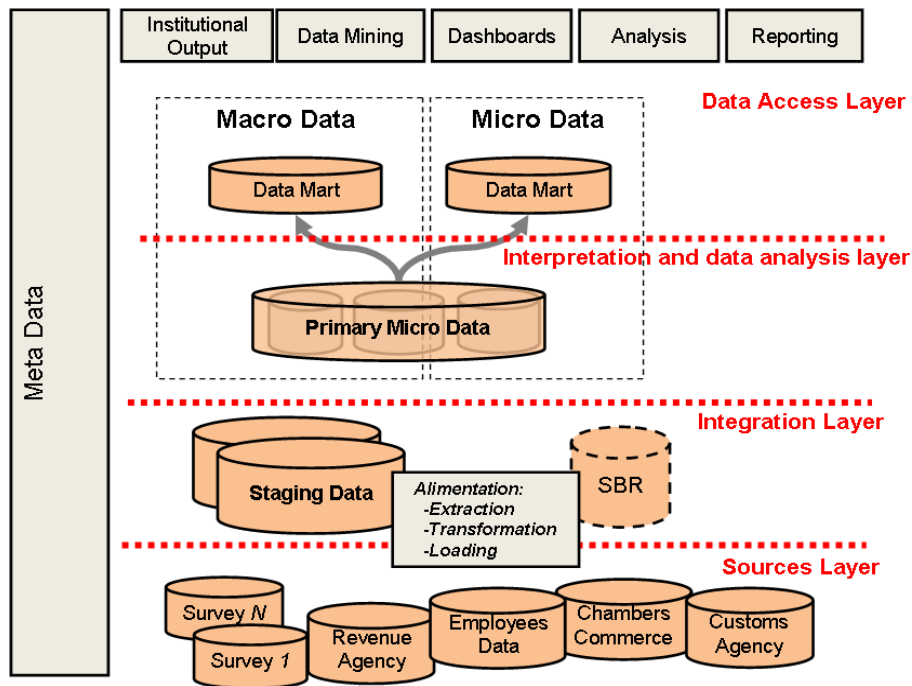


Figure 1 The S-DWH layers

1.2.5.1 Source layer metadata

The source layer is the entry point to the S-DWH regarding data as well as metadata. Data are collected from various sources outside of the control of the S-DWH, spanning from surveys and censuses conducted within the organization to administrative registers kept by other organizations. Hence, the original metadata that accompany the data will vary in content and quality, and the potential to influence the metadata will vary as well.

The source layer, being the entry point, has the important role of gatekeeper, making sure that data entered into the S-DWH and forwarded to the integration layer always have matching metadata of at least the agreed minimum extent and quality. The metadata may be either already available, for example loaded earlier with a previous periodic delivery, or supplied with the current data delivery.

The main responsibilities for this layer include:

- to make sure that all relevant data are collected from the sources, including their metadata,
- to add or complete missing or bad metadata,
- to deliver data and metadata in the best possible formats to the integration layer.

The source layer is the foundation for metadata to be used in the other layers. Consistency in definitions and standardization of code lists are examples of areas where efforts should be made to influence the sources in order to build the strongest possible metadata foundation.

1.2.5.2 Integration layer metadata

The efficiency of data linking and other tasks carried out in the integration layer will depend on the quality of the metadata carried forward from the source layer.

In the integration layer, data are *extracted* from the sources, *transformed* as necessary, and *loaded* into their places in the data warehouse (ETL operations). These tasks need to use active metadata,

such as descriptions and operator manuals as well as derivation rules being used, for example scripts, parameters and program code for the tools used.

The ETL operations will also create several types of metadata:

- Structural process metadata
 - Automatically generated formalized information, log data on performance, errors, etc.
 - Manually added, more or less formalized information
- Structural statistical metadata
 - Automatically generated additions to, or new versions of, code lists, linkage keys, etc.
 - Manually added additions, corrections and updates to the new versions
- Reference metadata
 - Manually added information (quality, process etc.), regarding a dataset, or a new version

1.2.5.3 Interpretation and data analysis layer metadata

The interpretation and data analysis layer stores cleaned, versioned and well-structured final micro-data. Once a new dataset or a new version has been loaded, few updates are made to the data in this layer. Consequently, metadata are normally only added to, with few or no changes being made.

On loading data to this layer, the following additions should be made to metadata:

- Structural process metadata
 - Automatically generated log data
- Structural statistical metadata
 - New versions of code lists, etc.
- Reference metadata
 - Optional additions to quality information, process information, etc.

Relatively few users will access this layer, but those who do will need metadata to perform their tasks:

- Structural process metadata
 - Estimation rules, descriptions, code, etc.
 - Confidentiality rules
- Structural statistical metadata
 - Variable definitions
 - Derivation rules
- Reference metadata
 - Quality information, process information, etc.

1.2.5.4 Data access layer metadata

Loading data into the access layer means reorganizing data from the analysis layer by derivation or aggregation into relevant stores, or data marts. This will require metadata that describe and support

the process itself (derivation and aggregation rules), but also new metadata that describe the reorganized data.

Necessary metadata to load the data access layer include:

- Structural process metadata
 - Derivation and aggregation rules
- Structural technical metadata
 - New physical references, etc.

Using the data access layer will require:

- Structural statistical metadata
 - Optional additional definitions of derived entities or attributes, aggregates, etc.
- Structural technical metadata
 - Physical references, etc.
- Reference metadata
 - Information on sources, links to source quality information

1.2.5.5 Summary of S-DWH layers and metadata categories

The table below gives a rough overview of where in the S-DWH layers the three important metadata categories are created (indicated by *c*) and used (*u*).

Layer	Statistical metadata	Process metadata	Quality metadata
Data access	u	cu	u
Interpretation	cu	cu	cu
Integration	cu	cu	c
Source	c	c	c

Metadata creation and use

The table shows that the lower layers mainly create metadata, but can't make much use of them, while in the higher layers metadata are used, but relatively little is created. This very much agrees with the rule that metadata should be defined as close to the source, or as early in the process as possible.

The S-DWH architecture should make it possible to trace any changes made to data as well as to metadata by using process metadata and versioning both data and metadata. Thus, a metadata item is normally never changed, updated or replaced. Instead, a new version is created when necessary, which means that there will always be a possibility to identify which metadata were considered correct at a certain point in time, even if they have later been revised.

A more detailed analysis of the metadata subsets and their use in the S-DWH layers can be found in *Definition of the functionalities of a metadata system to facilitate and support the operation of the S-DWH*⁵

⁵ Ennok, Lundell, Bowler, de Giorgi, Kulla (2013)

1.3 Metadata System⁶

The S-DWH is a logically coherent data store, but is not necessarily a single physical unit.

The logical coherence means that it must be possible to uniquely identify a data item throughout the S-DWH to trace its path over time, or cross-sectionally at a point in time, and to track all changes, for example the ETL processes through the S-DWH logical layers. This means that all data in the S-DWH must have corresponding metadata, all metadata items must be uniquely identifiable, metadata must be versioned to enable longitudinal use and metadata must provide “live” links to the physical data.

According to the Common Metadata Framework⁷, statistical metadata system should be tool that enables effectively perform the following functions:

- Planning, designing, implementing and evaluating statistical production processes.
- Managing, unifying and standardizing workflows and processes.
- Documenting data collection, storage, evaluation and dissemination.
- Managing methodological activities, standardizing and documenting concept definitions and classifications.
- Managing communication with end-users of statistical outputs and gathering of user feedback.
- Improving the quality of statistical data and transparency of methodologies. It should offer a relevant set of metadata for all criteria of statistical data quality.
- Managing statistical data sources and cooperation with respondents.
- Improving discovery and exchange of data between the statistical organization and its users.
- Improving integration of statistical information systems with other national information systems.
- Disseminating statistical information to end users. End users need reliable metadata for searching, navigation, and interpretation of data.
- Improving integration between national and international organizations. International organizations are increasingly requiring integration of their own metadata with metadata of national statistical organizations in order to make statistical information more comparable and compatible, and to monitor the use of agreed standards.
- Developing a knowledge base on the processes of statistical information systems, to share knowledge among staff and to minimize the risks related to knowledge loss when staff leave or change functions. Improving administration of statistical information systems, including administration of responsibilities, compliance with legislation, performance and user satisfaction.

The main functions of a metadata system are to gather and store metadata in one place, provide an overview of metadata (queries, searches etc.), create and maintain metadata, evaluate metadata, manage access by role-based security.

This section covers:

- Metadata model

⁶ Workpackage 1.4

⁷ <http://www1.unece.org/stat/platform/display/metis/The+Common+Metadata+Framework>

- Metadata functionality groups
- Metadata functionalities by layers:
 - Source layer
 - Integration layer
 - Interpretation and data analysis layer
 - Data access layer

1.3.1 Metadata model⁸

Metadata system requires metadata layer to have comprehensive *registry* functionality as well as *repository* functions. The registry functions are needed to control data consistency, so data contents are searchable. The repository functions are needed to enable operations on the data. Whether one or more repositories are needed will depend on local circumstances. The recommendation from a functional and governance point of view is a solution with one single installation that covers both registry and repository functions. However, in a decentralized or geographically dispersed organization, building one single metadata repository may be technically difficult, or at least less attractive.

1.3.1.1 Metadata model, general references

General references for a metadata model can be seen in “Guidelines for the Modelling of Statistical Data and Metadata” produced from the Conference of European Statisticians Steering Group on Statistical Metadata (usually abbreviated to "METIS Steering Group").

The most important standards in relationship to the use of metadata models are:

ISO/IEC 11179-3 ⁹	ISO/IEC 11179 is a international standard for representing metadata in a metadata registry. It has two main purposes: definition and exchange of concepts. Thus it describes the semantics and concepts, but does not handle physical representation of the data. It aims to be a standard for metadata-driven exchange of data in heterogeneous environments, based on exact definitions of data.
Neuchâtel Model for Classifications and Variables ¹⁰	The main purpose of this model is to provide a common language and a common perception of the structure of classifications and the links between them. The original model was extended with variables and related concepts. The discussion includes concepts like object types, statistical unit types, statistical characteristics, value domains, populations etc.
CMR ¹¹	Corporate Metadata Repository Model, CMR, statistical metadata model integrates a developmental version of edition 2 of ISO/IEC 11179 and a business data model derivable from the Generic Statistical Business Process Model. It includes the constructs necessary for a registry.

⁸ Workpackage 1.1 and 1.3

⁹ <http://metadata-stds.org/11179/#A3>

¹⁰ <http://www1.unece.org/stat/platform/pages/viewpage.action?pageId=14319930>

¹¹ <http://www.unece.org/stats/documents/1998/02/metis/11.e.pdf>

Nordic Metamodel ¹²	The Nordic Metamodel was developed by Statistics Sweden, and has become increasingly linked with their popular "PC-Axis" suite of dissemination software. It provides a basis for organizing and managing metadata for data cubes in a relational database environment.
CWM ¹³	Common Warehouse Metamodel, CWM, enables exchange of metadata between different tools.
SDMX ¹⁴	Statistical Data and Metadata eXchange, SDMX, a standards for the exchange of statistical information. SDMX has its focus on macro data, even though the model also supports micro data. It is an adopted standard for delivering and sharing data between NSIs and Eurostat. Recently, SDMX more and more has evolved to a framework with several sub frameworks for specific use (ESMS, SDMX-IM, ESQRS, MCV, MSD).
DDI ¹⁵	Data Documentation Initiative, DDI, an XML based standard has its roots in the data archive environment, but with its latest development, DDI 3 or DDI Lifecycle, it has become an increasingly interesting option for NSIs. DDI is an effort to create an international standard for describing data from the social, behavioral, and economic sciences.
GSIM ¹⁶	Generic Statistical Information Model, GSIM, a reference framework of information objects, which enables generic descriptions of data and metadata definition, management, and use throughout the statistical production process. GSIM will facilitate the modernization of statistical production by improving communication at different levels: ▪ Between the different roles in statistical production (statisticians, methodologists and information technology experts); ▪ Between the statistical subject matter domains; ▪ Between statistical organizations at the national and international levels. The GSIM is designed to be complementary to other international standards, particularly the Generic Statistical Business Process Model (GSBPM). It should not be seen in isolation, and should be used in combination with other standards.
MMX metadata framework ¹⁷	The MMX metadata framework is not an international standard, it is a specific adaptation of several standards by a commercial company. The MMX Metamodel provides a storage mechanism for various knowledge models. The data model underlying the metadata framework is more abstract in nature than metadata models in general.

From the metadata perspective, the ultimate goal is to use one single model for statistical metadata, covering the total life-cycle of statistical production. But considering the great variety in statistical production processes (for example surveys, micro data analysis or aggregated outputs), all with their

¹² http://www.scb.se/Pages/List_314010.aspx

¹³ <http://www.omg.org/spec/CWM/1.1/>

¹⁴ http://sdmx.org/?page_id=10

¹⁵ <http://www.ddialliance.org/>

¹⁶ <http://www1.unece.org/stat/platform/display/gsim/Generic+Statistical+Information+Model>

¹⁷ <http://www.mmxframework.org/>

own requirements for handling metadata, it is very difficult to agree upon one single model. The biggest risk is duplication of metadata, which should be avoided - this can best be achieved by the use of standards for describing and handling statistical metadata.

1.3.1.2 Metadata models guidelines

The guidelines below recommend how to establish a uniform policy and governance:

1. Do not strive for 100% perfection but keep everything as simple as possible;
2. Determine the subset(s) of metadata to describe and for what purpose;
3. Select per subset a model or standard that covers most of the needs determined in step 2;
4. Use this model or standard as a starting point to define your final solution. It is very important that the selected model or standard applies to most of the attributes in the subset to be described. But only use a single model or standard for each subset to be described within the S-DWH.
5. Only make adjustments to a model or standard when it is really necessary;
6. When it is necessary to make adjustments in the starting model or standard it is mandatory that you do describe these adjustments per subset;
7. Publish the final model or standard and make sure that users know about it and will use it the same way;
8. Make sure that there is a change management board where users can report errors and shortcomings. Then let the board decide whether the model or standard should be adjusted and how that is being done.

Always document the adjustments approved by the board and make sure all users are aware of them on time and act in accordance with these adjustments.

1.3.2 Metadata functionality groups

Core requirements of metadata systems are record creation/modification/deletion, multi-value attributes, select-list menu, simple and advanced search, simple display, import and export using XML or CSV documents, links to other databases, cataloguing history, and authorization management.

A metadata system has to:

- provide different levels of information granularity,
- convert legacy systems and records into new ones,
- equip customized options for generating reports,
- incorporate miscellaneous tools, in terms of metadata creation, retrieval, display,
- implement structured relations for existing metadata standards,
- enable multi-lingual processing (inc. Unicode character sets),
- include a built-in process for managing the workflow evaluation of metadata,
- support a role-based security system controlling access to all features of the system.

In the Common Metadata Framework¹⁸, a model for managing the development phases of an statistical metadata system (SMS) life cycle is presented. SMS management has following phases: design, implementation, maintenance, use and evaluation.

Considering all of the above, the following metadata functionality groups can be specified for the management of a metadata system for a S-DWH:

- metadata creation;
- metadata usage;
- metadata maintenance;
- metadata evaluation.

Metadata management also includes user training and composing a user guide for the metadata system.

1.3.2.1 Metadata creation

Metadata in the metadata system are either created or collected.

List of functionalities related to metadata:

- manual creation;
- automated creation;
- harvesting from other systems:
 - automated extraction (a regular process of collecting descriptions from different sources to create useful aggregations of metadata and related services);
 - converting;
 - manual import from files (XML, CSV);
- creating data access authorization metadata;
- implementing a metadata repository;
- creating links between metadata objects and processes;
- defining metadata objects.

1.3.2.2 Metadata usage

Users of S-DWH metadata can be both humans (statisticians, IT specialists, end-users etc.) and machines (other systems). The metadata must be available to users in the right form and with the right tools. The metadata system must be integrated with other systems and S-DWH components;

List of functionalities related to metadata usage:

- search;
- navigation;
- metadata export;
- international use.

¹⁸ Common Metadata Framework Part A, page 26

1.3.2.3 Metadata maintenance

All metadata stored in the metadata repository need to be up-to-date for ongoing use.

List of functionalities related to metadata maintenance:

- maintenance of metadata history (versioning, input, update, delete);
- updating meta models in metadata repository;
- updating links between metadata objects;
- users and rights (of metadata) management.

1.3.2.4 Metadata evaluation

To ensure metadata are of high quality, the metadata system should have the functionality to evaluate metadata according to the quality indicators/requirements chosen locally.

List of functionalities related to metadata evaluation:

- metadata validation (for example, check value domains and links between metadata objects);
- collection of standards used.

1.3.3 Metadata functionalities by layers: Source layer

Layers are defined in the S-DWH Business Architecture¹⁹ document and metadata subsets by layers are defined in the Metadata framework²⁰.

The source layer is data's entry point to the S-DWH. It is responsible for receiving and storing the original data from internal or external sources and making data available to the ETL functions that bring data to the integration layer.

In an ideal situation all metadata necessary to forward data from the source layer to the integration layer have either already been created by the external data suppliers and are delivered to the S-DWH, or can be created automatically either in the source layer or in the integration layer. In any case, a minimum requirement is that the technical metadata that describe the incoming data are provided by the data suppliers.

If the metadata created by the external sources are delivered in standardized formats, such as DDI, SDMX, etc., the source layer should be able to create the metadata needed in the S-DWH by extracting them and, if necessary, converting them to the required formats automatically.

Creating metadata by manually adding them to the S-DWH metadata repository should be a last resort, but will probably often be necessary to some degree. For example, metadata that documents a questionnaire may be created automatically or may need manual creation depending on what software has been used for the questionnaire design.

The source layer in itself uses relatively few metadata. It needs information on the sources, such as:

- responsibilities for data deliveries (who makes source data available to the S-DWH, which access rights are needed, etc.), the methods to be used (are data going to be delivered to the

¹⁹ Laureti Palma A. (2012) S-DWH Business Architecture. Deliverable 3.1

²⁰ Lundell L.G. (2012) Metadata Framework for Statistical Data Warehousing, ver. 1.0. Deliverable 1.1

S-DWH – “pushed”, physically collected by the S-DWH from some agreed location – “pulled”, or directly accessed from the original location – “virtual storage”),

- if relevant and possible the expected frequencies (when will new source data be available),
- source data formats (record layout, storage type, location).

One of the main tasks of the source layer is to act as the warehouse’s gatekeeper, the function that makes sure that all data entered into the S-DWH adhere to an agreed set of rules (recommendations on metadata quality are described in Recommendations on the Impact of Metadata Quality in the Statistical Data Warehouse²¹). These rules are expressed as technical and process metadata. This means that in order to accept a delivery of source data (“raw data”) and allow them to be forwarded to the next layer, relevant and correct metadata must be available, i.e. they must already exist or they be created.

Regardless of whether metadata are entered manually or created automatically they must always be validated. New metadata should be compared with and checked against already existing metadata and, if relevant, data to ascertain consistency within the metadata repository, and between data and metadata.

The source layer’s gatekeeper responsibility requires that all codes that appear in the data must appear in the metadata as enumerated value domains. Since many of these codes will be used as dimensions in the following layers it is vital that no values are missing. A check that no mismatches exist must be carried out in the source layer, and any found errors must be corrected by editing the metadata or the data. In case metadata contain minimum and maximum values (e.g., a percentage value must be within the range of 0-100) the corresponding data values should be checked, and corrected when needed.

1.3.4 Metadata functionalities by layers: Integration layer

According to the S-DWH Business Architecture²² in the integration layer all clerical operational activities typical of a statistical production process are carried out. This means operations carried out, automatically or manually, by users to produce statistical information in an IT infrastructure.

All classical ETL processes are covered in the integration layer of S-DWH.

Most of statistical metadata is created manually, process metadata is created manually and automatically, technical metadata is created mostly automatically, same for quality metadata.

As much as possible standards are used for creating metadata of integration layer for example for statistical metadata Neuchâtel is used, for quality metadata ESQRS is used.

Metadata harvesting depends on how S-DWH is developed, for example in integration layer process and technical metadata are usually created in S-DWH and harvested by metadata system.

If metadata of integration layer is in other format, there should convert metadata to suitable format for example transformation rules in collection systems are often different format than needed in data processing.

²¹ Bowler. C (2013) Recommendations on the Impact of Metadata Quality in the Statistical Data Warehouse.Deliverable 1.2

²² Laureti Palma A. (2012) S-DWH Business Architecture. Deliverable 3.1

Data access metadata (authorization metadata) is created for data warehouse (data marts) and data staging areas.

Metadata users in integration layer are both humans and machines. In every process of integration layer metadata should be navigable and searchable (example browsing metadata of variables by statistical activities and domains). All metadata objects in metadata system are related (example variable is related to statistical activity and classifier).

Metadata is multilingual (English, local language), possible to share internationally via unified services with standard format (like XML, SDMX).

S-DWH shares its metadata with other systems via metadata system. In S-DWH data object has reference to metadata object (example by metadata object id) in metadata system. Metadata of integration layer can be exported from metadata system. S-DWH uses metadata from metadata system that retrieves metadata also from other systems.

By creating integration layer metadata (data processing algorithms), this metadata is validated: controlling existing required values, data type controls, linking only existing objects, data models are with comments. Metadata is validated according to usable standards.

Some evaluation controls are built-in a SMS for metadata fill-in processes, some are systematic a built-in processes for managing the workflow evaluation of metadata that control following (validation queries), some are organizational processes.

1.3.5 Metadata functionalities by layers: Interpretation and data analysis layer

This layer is mainly aimed at 'expert' (i.e. statisticians/domain experts, and data scientists) users for carrying out advanced analysis, and data manipulation and interpretation functions, and access would be mainly interactive. The work in generating the analysis is effectively a design of potential statistical outputs. This layer might produce such things as data-marts, which would contain the results of the analysis. In many cases, however, the results of an investigation into the data required for a particular analysis may identify a shortfall in the availability of the required information. This may trigger the identification of requirements for whole new sets of variables, and methodologies around the processing of them.

1.3.6 Metadata functionalities by layers: Data access layer

The access layer is the fourth and last layer identified in a generic S-DWH; it is the layer at the end of the process of an S-DWH that together with the interpretation layer represents the operational IT infrastructure. The access layer is the layer for the final presentation, dissemination and delivery of the information sought²³.

Actually, metadata creation about data of the S-DWH at the data access level is merely an operation of converting/harvesting metadata already created in the other layers, in order to be used for dissemination. What is needed at this level is the procedure of harvesting metadata already provided.

²³ Laureti Palma A. (2012) S-DWH Business Architecture. Deliverable 1.3.

In access level will be created metadata about data access, for example statistics about users access on data and metadata, which data is the most requested, which year, which disaggregation, etc. and users evaluation metadata, e.g. assessment of easiness of finding information.

Metadata about users and uses are created in an automated way. Users' evaluation metadata should be generated automatically.

At the access level the main users of data/metadata are final users (researcher, students, organizations, etc.), who want to know in general the meaning of data and also the accuracy, the availability, and other important aspects of the quality of data. This is in order to be able to correctly identify and retrieve potentially relevant statistical data for a certain study/research/purpose, as well as for correctly interpret and (re)use statistical data. Metadata concerning quality, contents and availability aspects of data and processes is an important part of a feedback system, as well as the users' evaluation and users' data access.

	CREATION	USAGE	MAINTENANCE	EVALUATION
SOURCE	technical metadata that describe the incoming data are provided by the data suppliers (standardized formats, such as DDI, SDMX, etc) and will be created automatically or created in advance	- responsibilities for data deliveries, the methods to be used - if relevant and possible the expected frequencies (when will new source data be available) - source data formats (record layout, storage type, location)	Metadata may be closely linked to one particular data delivery (metadata may be part of the data delivery and entered automatically) or might be valid for several deliveries (metadata should be entered in advance)	- Metadata must always be validated - All codes that appear in the data must appear in the metadata - A check that no mismatches exist must be carried out and errors must be corrected by editing the (meta)data
INTEGRATION	- Metadata of statistical activity - Classifier, variable and validation metadata (algorithms) - Frame, sample and stratum metadata - Data model metadata - Imputation and pre-fill metadata - Dissemination metadata - Algorithms of statistical confidentiality - Data processing algorithms (incl. aggregation, weights calculation) and scheduling metadata - Questionnaires design - Data collection structure technical metadata - Quality metadata - Data finalizing metadata	- Checking data availability by using metadata - Design production systems and workflow by using variable, classifiers and coding tables metadata - Configuring workflows scheduling by using statistical and process metadata - Integrate data by using variable, pre-filling, collection and sample metadata, data model of raw data - For classify & code coding algorithms, classifiers, coding-tables metadata is used. - Imputation metadata is used. - Calculate weights by using stratum and frame metadata - For finalize data files finalization data warehouse data model metadata is used.	- Maintain (create, update, delete, versioning) integration metadata (data processing algorithms, data warehouse data models etc.). - User rights are according to the S-DWH system operations of all S-DWH processes in all layers - Integration metadata can be stored in different meta models (maintaining meta models) - All users of S-DWH can access to metadata (only for viewing metadata), but for changing metadata there should be granted privileges by S-DWH operations and statistical activities.	- By creating integration layer metadata (data processing algorithms), this metadata is validated: controlling existing required values, data type controls, linking only existing objects, data models are with comments - Some evaluation controls are built-in a SMS for metadata fill-in processes, some are systematic a built-in processes for managing the workflow evaluation of metadata that control following (validation queries), some are organizational processes

INTERPRETATION & ANALYSIS	▪ Design for a new analysis or output definition ▪ Variable definitions for the new analysis or output ▪ Methodology design for the statistical processing ▪ Scripts encompassing the data selection rules required to carry out the identification of the data to be used ▪ Quality report (reference metadata) ▪ Interpretation Document metadata in text form to accompany any data sets	▪ To carry out examination of the metadata in order to evaluate suitability for a new analysis or identified output ▪ Run the scripts to extract and integrate the data from different sources in the DWH ▪ Utilize disclosure rule metadata in the disclosure checking process for the intended output datasets being created by the run of the analysis ▪ Utilize quality metadata as input to any interpretation documentation accompanying output data sets	▪ Check that appropriate rights exist in the S-DWH for the user who is attempting to create a new analysis design ▪ Delete old or defunct analysis descriptions and their associated scripts, as part of a maintenance/archive function	▪ Recording quality characteristics from the different elements of the analysis during the preparation of a draft output. This might take the form of quality indicator attributes attached to variables ▪ Following evaluation of the output as a whole, the statistical content would need to have some approval status metadata
ACCESS	▪ metadata about data access, for example statistics about users access on data and metadata ▪ users evaluation metadata	▪ locating, searching and filtering metadata ▪ obtaining information on metadata availability ▪ obtaining feedback/evaluation from users, by working out statistics and tracking usage of data/metadata ▪ foreseeing accessibility and availability other systems	▪ creating, updating, deleting, reviewing metadata ▪ harmonizing and exploiting (meta)data ▪ exporting, converting metadata ▪ managing (meta)data libraries through the catalogues and descriptors ▪ managing authentications ▪ managing metadata about users of data	▪ ensuring valid (default)values and structures ▪ standardized and harmonized metadata formats for official statistics ▪ updating metadata as soon as it is available ▪ dissemination of related metadata with data ▪ multilingual aids for users

Figure 3: mapping S-DWH layers and metadata functions

1.4 Metadata and SDMX

1.4.1 The SDMX standard

The Statistical Data and Metadata eXchange (SDMX) standard utilizes the terms ‘Structural’ and ‘Reference’ as defined in 2.1.2.3 (Metadata subsets) to distinguish between the types of metadata which can be represented in an SDMX data exchange message, or even more generically, within a data/metadata repository which might conform to the SDMX information model.

1.4.2 Structural metadata

Structural metadata in SDMX is (as indicated by the name) used to define the structure of a dataset. SDMX is mostly associated with aggregated, or time-series multi-dimensional data sets (although it can also be used to define unit-level datasets).

The structure of a dataset in SDMX is described using a Data Structure Definition (DSD), in which the metadata elements are (1) **dimensions**, which form the identifiers for the statistical data, and (2) **attributes**, which provide additional descriptive information about the data.

Both dimensions and attributes are manifested by statistical concepts which may be underpinned by code lists or classifications, to provide some sort of value domain, such as:

- FREQUENCY – which could take values in a range such as ‘A -Annual’, Q-‘Quarterly’, M-‘Monthly’;
- TIME – point in time or period to which the data refers (example value could be ‘March 2011’)
- SOC – standard occupational classification – e.g. ‘2121 – Civil Engineer’
- COUNTRY – e.g. NL, EE, FI, IT, PT, LT, UK
- TOPIC - subject matter domain, e.g. ‘Labour market ‘
- UNIT – e.g. population might be measures in ‘000’s of people, or steel foundry output might be measured in TONNES

A combination of dimensions (e.g. TIME, SOC, COUNTRY using the examples above) would be the **dimensions** which would uniquely identify a cell of data, or a single measure, which would refer to the number of people employed in a particular industry at a particular time in a particular country. The UNIT (‘000s of people) would be an **attribute**, because it gives additional information to the reader about the data item, aiding understanding.

Within an SDMX-ML message, any code lists associated with the concepts defining the data form part of the message.

From the overall ESS perspective, the standardization and harmonization of these code lists and classifications will greatly help in terms of comparability and efficiency when collating, aggregating, and comparing data at the European level (see 2.5.4 Content Oriented Guidelines below).

1.4.3 Reference Metadata

Reference metadata relates to descriptive/narrative information often associated with datasets. It can be particular to any level of the dataset to which the reference metadata refers/is linked to, and is usually sent as an independent message from the dataset.

Reference metadata is usually in textual form, and would cover such information as:

- Methodological statements/reports
- Quality reports
- Concept descriptions

Reference metadata would normally be transmitted in a separate XML message from that of the dataset.

The ESS has a set of standard concepts relating to the reference metadata:

The revised version of Euro-SDMX Metadata Structure(ESMS 2.0)

	Concept Name		Concept Name		Concept Name
1	Contact	7	Confidentiality	15	Accessibility and clarity
1.1	Contact organisation	7.1	Confidentiality - policy	15.1	News release
1.2	Contact organisation unit	7.2	Confidentiality - data treatment	15.2	Publications
1.3	Contact name	8	Release policy	15.3	On-line database
1.4	Contact person function	8.1	Release calendar	15.4	Micro-data access
1.5	Contact mail address	8.2	Release calendar access	15.5	Other
1.6	Contact email address	8.3	User access	15.6	Documentation on methodology
1.7	Contact phone number	9	Frequency of dissemination	15.7	Quality documentation
1.8	Contact fax number	10	Quality management	16	Cost and burden
2	Metadata update	10.1	Quality assurance	17	Data revision
2.1	Metadata last certified	10.2	Quality assessment	17.1	Data revision - policy
2.2	Metadata last posted	11	Relevance	17.2	Data revision - practice
2.3	Metadata last update	11.1	User needs	18	Statistical processing
3	Statistical presentation	11.2	User satisfaction	18.1	Source data
3.1	Data description	11.3	Completeness	18.2	Frequency of data collection
3.2	Classification system	12	Accuracy and reliability	18.3	Data collection
3.3	Sector coverage	12.1	Overall accuracy	18.4	Data validation
3.4	Statistical concepts and definitions	12.2	Sampling error	18.5	Data compilation
3.5	Statistical unit	12.3	Non-sampling error	18.6	Adjustment
3.6	Statistical population	13	Timeliness and punctuality	19	Comment
3.7	Reference area	13.1	Timeliness		
3.8	Time coverage	13.2	Punctuality		
3.9	Base period	14	Coherence and comparability		
4	Unit of measure	14.1	Comparability - geographical		
5	Reference period	14.2	Comparability - over time		
6	Institutional mandate	14.3	Coherence - cross domain		
6.1	Legal acts and other agreements	14.4	Coherence - internal		

There is also a set of concepts which specific the ESS Standard for the Quality Report Structure (ESQRS):

Item	Concept Name	Item	Concept Name	Item	Concept Name
1	Contact	6	Accuracy and reliability	9	Accessibility and clarity
1.1	Contact organisation	6.1	Accuracy - overall	9.1	News release
1.2	Contact organisation unit	6.2	Sampling error	9.2	Publications

1.3	Contact name	6.2.1	Sampling error - indicators	9.3	Online database
1.4	Contact person function	6.3	Non-sampling error	9.3.1	Data tables - consultations
1.5	Contact mail address	6.3.1	Coverage error	9.4	Microdata access
1.6	Contact email address	6.3.1.1	Over-coverage - rate	9.5	Other
1.7	Contact phone number	6.3.1.2	Common units - proportion	9.6	Documentation on methodology
1.8	Contact fax number	6.3.2	Measurement error	9.7	Quality documentation
2	Statistical presentation	6.3.3	Non response error	9.7.1	Metadata completeness - rate
2.1	Data description	6.3.3.1	Unit non-response - rate	9.7.2	Metadata - consultations
2.2	Classification system	6.3.3.2	Item non-response - rate	10	Cost and Burden
2.3	Sector coverage	6.3.4	Processing error	11	Confidentiality
2.4	Statistical concepts and definitions	6.3.4.1	Imputation - rate	11.1	Confidentiality - policy
2.5	Statistical unit	6.3.5	Model assumption error	11.2	Confidentiality - data treatment
2.6	Statistical population	6.4	Seasonal adjustment	12	Comment
2.7	Reference area	6.5	Data revision - policy		
2.8	Time coverage	6.6	Data revision - practice		
2.9	Base period	6.6.1	Data revision - average size		
3	Statistical processing	7	Timeliness and punctuality		
3.1	Source data	7.1	Timeliness		
3.2	Frequency of data collection	7.1.1	Time lag - first result		
3.3	Data collection	7.1.2	Time lag - final result		
3.4	Data validation	7.2	Punctuality		
3.5	Data compilation	7.2.1	Punctuality - delivery and publication		
3.6	Adjustment	8	Coherence and comparability		
4	Quality management - assessment	8.1	Comparability - geographical		
4.1	Quality assurance	8.1.1	Asymmetry for mirror flow statistics - coefficient		
4.2	Quality assessment	8.2	Comparability - over time		
	Relevance	8.2.1	Length of comparable time series		
5.1	User Needs	8.3	Coherence - cross domain		
5.2	User Satisfaction	8.4	Coherence - sub annual and annual statistics		
5.3	Completeness	8.5	Coherence - National Accounts		
5.3.1	Data completeness - rate	8.6	Coherence - internal		

These concepts would form the basis of the structure of the message containing the reference metadata of interest. Use of harmonized and common concepts obviously aids the collation of information across the ESS.

These lists of concepts above are to be brought together into the Single Integrated Metadata Structure [which also contains *Process metadata* concepts, but which are not discussed in the context of SDMX here].

A1-Annex: Technology Architecture

I.1 Technology Architecture

Authors: Sónia Quaresma

I.2 Classification of SDMX Tools

Authors: : Valerij Zavoronok, Sónia Quaresma

References

Valerij Žavoronok, Maksim Lata, Lina Amšiejūtė; ESS-Net-DWH Deliverable 3.4: "Overview of various technical aspects in SDWH"

Annex 1

Technology Architecture

The **Technology Architecture** is the combined set of software, hardware and networks able to develop and support IT services. This is a high-level map or plan of the information assets in an organization, including the physical design of the building that holds the hardware.

This annex is intended to be an overview of software packages existing on the market or developed on request in NSIs in order to describe the solutions that would meet NSI needs, implement S-DWH concept and provide the necessary functionality for each S-DWH level.

1 Source layer

The Source layer is the level in which we locate all the activities related to storing and managing internal or external data sources. Internal data are from direct data capturing carried out by CAWI, CAPI or CATI while external data are from administrative archives, for example from Customs Agencies, Revenue Agencies, Chambers of Commerce, Social Security Institutes.

Generally, data from direct surveys are well-structured so they can flow directly into the integration layer. This is because NSIs have full control of their own applications. Differently, data from others institution's archives must come into the S-DWH with their metadata in order to be read correctly.

In the early days extracting data from source systems, transforming and loading the data to the target data warehouse was done by writing complex codes which with the advent of efficient tools was an inefficient way to process large volumes of complex data in a timely manner. Nowadays ETL (Extract, Transform and Load) is essential component used to load data into data warehouses from the external sources. ETL processes are also widely used in data integration and data migration. The objective of an ETL process is to facilitate the data movement and transformation. ETL is the technology that performs three distinct functions of data movement:

- The extraction of data from one or more sources.
- The transformations of the data e.g. cleansing, reformatting, standardisation, aggregation.
- The loading of resulting data set into specified target systems or file formats.

ETL processes are reusable components that can be scheduled to perform data movement jobs on a regular basis. ETL supports massive parallel processing for large data volumes. The ETL tools were created to improve and facilitate data warehousing.

Depending on the needs of customers, there are several types of tools. One of them performs and supervises only selected stages of the ETL process like data migration tools (EtL Tools, "small t" tools), data transformation tools (eTL Tools, "capital T" tools). Another are complete (ETL Tools) and have many functions that are intended for processing large amounts of data or more complicated ETL projects. Some of them (like server engine tools) execute many ETL steps at the same time from more than one developer, while other like client engine tools are simpler and execute ETL routines on the same machine as they are developed. There are two more types. The first one is called Code base tools and is a family of programming tools which allow you to work with many operating systems and programming languages. The second one called GUI base tools removes the coding layer and allows you to work without any knowledge (in theory) about coding languages.

The first task is data **extraction** from internal or external sources. After sending queries to the source, system data may go indirectly to the database. However, usually there is a need to monitor or gather more

information and then go to staging area. Some tools extract only new or changed information automatically so we don't have to update it by our own.

The second task is **transformation** which is a broad category:

- transforming data into a structure which is required to continue the operation (extracted data has usually a structure typical to the source);
- sorting data;
- connecting or separating;
- cleansing;
- checking quality.

The third task is **loading** into a data warehouse. ETL Tools have many other capabilities (next to the main three: extraction, transformation and loading) like for instance sorting, filtering, data profiling, quality control, cleansing, monitoring, synchronization and consolidation.

The most popular **commercial ETL Tools** are:

- IBM InfoSphere DataStage

IBM InfoSphere DataStage integrates data on demand with a high performance parallel framework, extended metadata management, and enterprise connectivity. It supports the collection, integration and transformation of large volumes of data, with data structures ranging from simple to highly complex. It also provides support for big data and Hadoop, enabling customers to directly access big data on a distributed file system, thereby helping customers address the most challenging data volumes in the systems. It offers in addition a scalable platform that enables customers to solve large-scale business problems through high-performance processing of massive data volumes, as well as supports real-time data integration and completes connectivity between any data source and any application.

- Informatica PowerCenter

Informatica PowerCenter is a widely used extraction, transformation and loading (ETL) tool used in building enterprise data warehouses. PowerCenter empowers its customers to implement a single approach to accessing, transforming, and delivering data without having to resort to hand coding. The software scales to support large data volumes and meets customers' demands for security and performance. PowerCenter serves as the data integration foundation for all enterprise integration initiatives, including data warehousing, data governance, data migration, service-oriented architecture (SOA), B2B data exchange, and master data management (MDM). Informatica PowerCenter also empowers teams of developers, analysts, and administrators to work faster and better together, sharing and reusing work, to accelerate project delivery.

- Oracle Warehouse Builder (OWB)

Oracle Warehouse Builder (OWB) is a tool that enables designing a custom Business Intelligence application. It provides dimensional ETL process design, extraction from heterogeneous source systems, and metadata reporting functions. Oracle Warehouse Builder allows creation of both dimensional and relational models, and also star schema data warehouse architectures. Except of being an ETL (Extract, Transform, Load) tool, Oracle Warehouse Builder also enables users to design and build ETL processes, target data warehouses, intermediate data storages and user access layers. It allows metadata reading in a wizard-driven form from a data dictionary or Oracle Designer but also supports over 40 metadata files from other vendors.

- SAS Data Integration Studio

SAS Data Integration Studio is a powerful visual design tool for building, implementing and managing data integration processes regardless of data sources, applications, or platforms. An easy-to-manage, multiple-user environment enables collaboration on large enterprise projects with repeatable processes that are

easily shared. The creation and management of data and metadata are improved with extensive impact analysis of potential changes made across all data integration processes. SAS Data Integration Studio enables users to quickly build and edit data integration, to automatically capture and manage standardized metadata from any source, and to easily display, visualize, and understand enterprise metadata and your data integration processes. SAS Data Integration Studio is part of the SAS software offering, SAS Enterprise Data Integration Server.

- SAP Business Objects Data Services (SAP BODS)

SAP Business Objects Data Services (SAP BODS) is one of the fundamental capabilities of Data Services. It is used for extracting, transforming, and loading (ETL) data from heterogeneous sources into a target database or data warehouse. Customers can create applications (jobs) that specify data mappings and transformations by using the Designer. Also it empowers users to use any type of data, including structured or unstructured data from databases or flat files to process, cleanse and remove duplicate entries. Data Services RealTime interfaces provide additional support for real-time data movement and access. Data Services RealTime reacts immediately to messages as they are sent, performing predefined operations with message content. Data Services RealTime components provide services to web applications and other client applications. The Data Services product consists of several components including: Designer, Job server, Engine and Repository.

- Microsoft SQL Server Integration Services (SSIS)

Microsoft SQL Server Integration Services (SSIS) is a platform for building enterprise-level data integration and data transformations solutions. Integration Services are used to solve complex business problems by copying or downloading files, sending e-mail messages in response to events, updating data warehouses, cleaning and mining data, and managing SQL Server objects and data. The packages can work alone or together with other packages to address complex business needs. Integration Services can extract and transform data from a wide variety of sources such as XML data files, flat files, and relational data sources, and then load the data into one or more destinations. Integration Services includes a rich set of built-in tasks and transformations, tools for constructing packages, and the Integration Services service for running and managing packages. You can use the graphical Integration Services tools to create solutions without writing a single line of code, or you can program the extensive Integration Services object model to create packages programmatically and code custom tasks and other package objects.

The most popular freeware (open-sources) ETL Tools are:

- Pentaho Data Integration (Kettle)

Pentaho Data Integration (Kettle) is a part of the Pentaho Open Source Business intelligence suite. It includes software for all areas of supporting business decisions making - the data warehouse managing utilities, data integration and analysis tools, software for managers and data mining tools. Pentaho data integration is one of the most important components of this business intelligence platform and seems to be the most stable and reliable. Pentaho Data Integration is well known for its ease of use and quick learning curve. PDI implements a metadata-driven approach which means that the development is based on specifying WHAT to do, not HOW to do it. Pentaho lets administrators and ETL developers create their own data manipulation jobs with a user friendly graphical creator, and without entering a single line of code. Advanced users know, that not every user friendly solution is as effective as it could be, so skilled and experienced users can use advanced scripting and create custom components. Pentaho Data Integration uses a common, shared repository which enables remote ETL execution, facilitates team work and simplifies the development process. There are a few development tools for implementing ETL processes in Pentaho:

- Spoon – data modelling and development tool for ETL developers. It allows creation of transformations (elementary data flows) and jobs (execution sequences of transformations and other jobs);
- Pan – executes transformations modelled in Spoon;
- Kitchen – is an application which executes jobs designed in Spoon;
- Carte – a simple webserver used for running and monitoring data integration tasks.

- CloverETL

CloverETL is a data transformation and data integration tool (ETL) distributed as a Commercial Open Source software. As the Clover ETL framework is Java based, it is independent and resource- efficient. CloverETL is used to cleanse, standardize, transform and distribute data to applications, databases and warehouses. It is a Java based program and thanks to its component based structure customization and embedded ability are possible. It can be used as standalone, as well as a command-line application or server application or can be even embedded in other applications as Java library. Clover ETL has been used not only on the most wide spread Windows platform but also on Linux, HP-UX, AIX, AS/400, Solaris and OSX. It can be both used on low-cost PC as on high-end multi processors servers. Clover ETL pack includes Clover ETL Engine, Clover ETL Designer and CloverETL Server.

- JasperETL

JasperETL – JasperETL is considered to be one of the easiest solutions for data integration, cleansing, transformation and movement on the market. It is a data integration platform-ready-to-run and high performing, that can be used by any organization. JasperETL is not a sole data integration tool, but it is a part of the Jaspersoft Business Intelligence Suite. Its capabilities can be used when there is a need for:

- aggregation of large volumes of data from various data sources;
- scaling a BI solution to include data warehouses and data marts;
- boosting of performance by off-loading query and analysis form systems.

JasperETL provides an impressive set of capabilities to perform any data integration task. It extracts and transforms data from multiple systems with both consistency and accuracy, and loads it into optimized store. Thanks to the technology of JasperETL, it is possible for database architects and data store administrators to:

- use the modeler of the business to get access to a non-technical view of the workflow of information;
- display and edit the ETL process using a graphical editing tool - Job Designer;
- define complex mapping and transformation using Transformation Mapper and other components;
- be able to generate portable Java or Perl code which can be executed on any machine;
- track ETL statistics from start to finish using real-time debugging;
- allow simultaneous input and output to and from various sources using flat files, XML files, web services, databases and servers with a multitude of connectors;
- make configurations of heterogeneous data sources and complex data formats (incl. positional, delimited, XML and LIDF with metadata wizards);
- use the AMC (Activity Monitoring Console) to monitor data volumes, execution time and job events.

2 Integration layer

The integration layer is where all operational activities needed for all statistical elaboration processes are carried out. This means operations are carried out automatically or manually by operators to produce

statistical information in an IT infrastructure. With this aim, different sub processes are predefined and preconfigured by statisticians as a consequence of the statistical survey design in order to support the operational activities.

In general, for the Integration layer dedicated software applications are mostly available and are usually defined as Data Integration tools. This kind of software is used for metadata management and usually is developed and implemented on NSI request. This is because of specific needs and requirements from customer. It has a user friendly graphic interface to help the integration of different input sources and their manipulation.

In next chapters we will provide some solutions from several NSIs on what are main features of their custom software.

➤ Italy

Italy (Istat) has self-implemented system SIQual as metadata system. This is an information system for quality assessment. It contains information on the execution of Istat, primary surveys and secondary studies, and activities developed to guarantee quality of the produced statistical information metadata managing system developed solution. This is also a tool to generate quality reports.

To manage this system Istat has a dedicated developed solution, named SIDI, in which is possible to update all information. SIDI main feature is common management of metadata documentation standards:

- Thesaura: lists of standard items to be used to document process activities and quality control actions.
- Content: topics of the survey, analysis units, questionnaire.
- Process: Reporting unit (sources of the secondary study), survey design, data collection, data transformation, data processing.
- Quality: Activities carried out to prevent, monitor and evaluate survey errors.
- Metadata qualitative descriptions: free notes supporting standard metadata items.

Istat doesn't have a metadata managing system for operational activities yet.

➤ Lithuania

Statistics Lithuania don't use a single, centralized metadata management system yet. Most of the systems have been developed independently of each other. Any kind of metadata can be found in most of the systems. This is the reason why some metadata are stored as different copies in different systems.

Metadata related to quality of statistical data (such as relevance, accuracy, timeliness, punctuality, accessibility, clarity, coherence and comparability), as well as statistical method descriptions are stored as free text using MS Office tools. Currently, Official statistics portal is functional, all metadata are to be stored in it and any user is to be able to access it. Official statistics portal is running on MS SQL server.

Statistical metadata such as indicators and related data (definitions, measurement units, periodicities of indicators, links to the questionnaires in which indicators are used), classifications, and code lists are managed in e. statistics (an electronic statistical business data preparation and transmission system). This system has the ability to export metadata (which is stored in this system) to defined XML format. The statistical data submission from business management systems standard has been developed. It is possible to submit the statistical data described according to the said standard from the business management or accounting systems used in respondent's enterprises. E. statistics is run on MS SQL server.

Metadata which are relevant to the dissemination of data were previously stored in PC-Axis. Now they are moved to Official statistics portal.

Almost all of metadata used to analyse and process statistical data of business surveys is stored in Oracle DB with the much of the results processing being carried out in SAS, only one business survey is carried out in FoxPro, while all the statistical data and metadata of social surveys is stored in MS SQL server.

Statistics Lithuania also uses several other software systems, which have some basic metadata storage and management capability, in order to fulfil basic everyday needs.

➤ Portugal

Statistics Portugal (INE) has implemented the SMI (Integrated Metadata System) which is in production since June, 2012.

The Integrated Metadata System integrates and provides concepts, classifications, variables, data collection instruments and methodological documentation in the scope of the National Statistical System (NSS). The various components of the system are interrelated, aim to support statistical production and document the dissemination of Official Statistics. As in other NSI's, it is a solution developed by request and until now it is only used internally.

The main goals of this system are:

- Support survey design.
- Support data dissemination, documenting indicators disseminated through the dissemination database.

It is intended that this system constitutes an instrument for coordination and harmonization within the NSS.

➤ United Kingdom

United Kingdom's Office for National Statistics (ONS) doesn't have a single, centralised metadata management system. The operational metadata systems are developed and supported on a variety of technology platforms:

- Most business survey systems (including the business register) are run on Ingres DBMS with the much of the results processing being carried out in SAS.
- Most new developments (including the Census and Web Data Access redevelopment) are carried out in Oracle/Java/SAS.
- Older systems supporting Life Events applications (births, marriages, deaths etc.) are still maintained on Model 204 database which is old fashioned preSQL and prerelational database product.

As a result, each system or process supported by each of these technology implementations have their own metadata, which are managed by using the specific applications developed for the statistical system storage, along with the data itself.

➤ Estonia

Statistics Estonia (SE) has implemented centralised metadata repository based on MMX metadata framework. MMX metadata framework is a lightweight implementation of OMG Metadata Object Facility built on relational database technology.

Statistical metadata such as classifications, variables, code lists, questionnaires etc. is managed in iMeta application. The main goal of iMeta is to support survey design.

Operational metadata is managed in VAIS application – extendable metadata-driven data processing tool to carry out all data manipulations needed in statistical activities. VAIS was first used in production for Population and Housing Census 2011 data processing.

3 Interpretation and Data Analysis layer

The interpretation and data analysis layer is specifically for statisticians and would enable any data manipulation or unstructured activities. In this layer expert users can carry out data mining or design new statistical strategies.

➤ Statistical Data Mining Tools

The overall goal of the data mining tools is to extract information from a data set and transform it into an understandable structure for further use. Aside from the main goal of the data mining tools they should also be capable to visualise data/information, which was extracted in data mining process. Because of this feature, a lot of tools from this category have been already covered in Graphics and Publishing tools section, such as:

- IBM SPSS Modeler (data mining software provided by IBM)
- SAS Enterprise Miner (data mining software provided by the SAS Institute)
- STATISTICA Data Miner (data mining software provided by StatSoft)

This list of statistical data mining tools can be increased by adding some other very popular and powerful commercial data mining tools, such as:

- Angoss Knowledge Studio (data mining tool provided by Angoss)
- Clarabridge (enterprise class text analytics solution)
- E-NI (e-mining, e-monitor) (data mining tool based on temporal pattern)
- KXEN Modeler (data mining tool provided by KXEN)
- LIONsolver (an integrated software application for data mining, business intelligence, and modelling that implements the Learning and Intelligent Optimization (LION) approach)
- Microsoft Analysis Services (data mining software provided by Microsoft)
- Oracle Data Mining (data mining software by Oracle)

One of data mining tools widely used among statisticians and data miners is open source software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and Mac OS.

- R (programming language and environment for statistical computing and graphics)

R is an implementation of the S programming language combined with lexical scoping semantics inspired by Scheme. R is a GNU project. The source code for the R software environment is written primarily in C, Fortran, and R. R is freely available under the GNU General Public License, and pre-compiled binary versions are provided for various operating systems. R uses a command line interface. However, several graphical user interfaces are available for use with R. R provides a wide variety of statistical and graphical techniques, including linear and nonlinear modelling, classical statistical tests, time-series analysis, classification, clustering, and others. R is easily extensible through functions and extensions, and the R community is noted for its active contributions in terms of packages. There are some important differences, but much code written for S runs unaltered. Many of R's standard functions are written in R itself, which makes it easy for users to follow the algorithmic choices made. For computationally intensive tasks, C, C++, and Fortran code can be linked and called at run time. Advanced users can write C or Java code to manipulate R objects directly. R is highly extensible through the use of user-submitted packages for specific functions or specific areas of study. Due to its S heritage, R has stronger object-oriented programming facilities than most statistical computing languages. Extending R is also eased by its permissive lexical scoping rules. Another strength of R is static graphics, which can produce publication-quality graphs, including mathematical symbols. Dynamic and interactive graphics are available through additional packages. R has its own LaTeX-like documentation format, which is used to supply comprehensive documentation, both on-line in a number of formats and in hard copy. R functionality has been made accessible from several scripting languages such as Python (by the RPy interface package), Perl (by the Statistics: R module), and Ruby (with the rruby rubygem). PL/R can be used alongside, or instead of, the PL/pgSQL scripting language in the PostgreSQL and Greenplum database management system. Scripting in R itself is possible via `littler` as well as via `Rscript`. Other major commercial software systems supporting connections to or integration with R include: SPSS, STATISTICA and SAS.

- Business Intelligence Tools for data analysis in a direct connection with data base.

Business Intelligence tools which allow users to create visual reports/'dashboards' and other summaries of specific sets of data for trending and other data analysis needs are Reporting Tools. Reporting tools often come as packages that include tools for extracting, transforming and loading (ETL) transactional data from multiple operational repositories/database tables, and for creating specialised reporting cubes (OLAP to speed response/add insight, etc.), and finally presentational tools for displaying flat file/tabular data read from specialised reporting views in a database for end users. All reporting tools can be categorized into two categories:

Open source software such as:

- Eclipse BIRT Project

Eclipse BIRT Project is a project that provides reporting and business intelligence capabilities for rich client and web applications, especially those based on Java and Java EE. BIRT is a top-level software project within the Eclipse Foundation, an independent not-for-profit consortium of software industry vendors and an open source community. BIRT has two main components: a visual report designer within the Eclipse IDE for creating BIRT Reports, and a runtime component for generating reports that can be deployed to any Java environment. The BIRT project also includes a charting engine that is both fully integrated into the report designer and can be used standalone to integrate charts into an application. BIRT Report designs are persisted as XML and can access a number of different data sources including JDO datastores, JFire Scripting Objects, POJOs, SQL databases, Web Services and XML.

- JasperReports

JasperReports as an open source Java reporting tool that can write to a variety of targets, such as screen, a printer, into PDF, HTML, Microsoft Excel, RTF, ODT, Comma-separated values or XML files. It can be used in Java-enabled applications, including Java EE or web applications, in order to generate dynamic content. It reads its instructions from an XML or .jasper file. JasperReports is part of the Lisog open source stack initiative.

- OpenOffice Base

OpenOffice Base is a database module roughly comparable to desktop databases such as Microsoft Access and Corel Paradox. They can connect to external full-featured SQL databases such as MySQL, PostgreSQL and Oracle through ODBC or JDBC drivers. OpenOffice Base can hence act as a GUI front-end for SQL views, table-design and query. In addition, OpenOffice.org has its own Form wizard to create dialog windows for form filling and updates. Starting with version 2.3, Base offers generation of reports based on Pentaho software.

Some commercial software for reporting is:

- Oracle Reports

Oracle Reports is a tool for developing reports against data stored in an Oracle database. Oracle Reports consists of Oracle Reports Developer (a component of the Oracle Developer Suite) and Oracle Application Server Reports Services (a component of the Oracle Application Server). The report output can be delivered directly to a printer or saved in the following formats: HTML, RTF, PDF, XML, Microsoft Excel.

- SAS Web Report Studio

SAS Web Report Studio is an art of the SAS Enterprise Business Intelligence Server, which provides access to query and reporting capabilities on the Web. It is aimed at non-technical users.

- SQL Server Reporting Services (SSRS)

SQL Server Reporting Services (SSRS) is a server-based report generation software system from Microsoft. Administered via a web interface, it can be used to prepare and deliver a variety of interactive and printed reports. Reports are defined in Report Definition Language (RDL), an XML markup language. Reports can be

designed using recent versions of Microsoft Visual Studio, with the included Business Intelligence Projects plug-in installed or with the included Report Builder, a simplified tool that does not offer all the functionality of Visual Studio. Reports defined by RDL can be generated in a variety of formats including Excel, PDF, CSV, XML, TIFF (and other image formats), and HTML Web Archive. SQL Server 2008 SSRS can also prepare reports in Microsoft Word (DOC) format.

- Crystal Reports

Crystal Reports is a business intelligence application used to design and generate reports from a wide range of data sources. Crystal Reports allows users to graphically design data connection(s) and report layout. In the Database Expert, users can select and link tables from a wide variety of data sources, including Microsoft Excel spreadsheets, Oracle databases, Business Objects Enterprise business views, and local file system information. Fields from these tables can be placed on the report design surface, and can also be used in custom formulas, using either BASIC or Crystal's own syntax, which are then placed on the design surface. Formulas can be evaluated at several phases during report generation as specified by the developer. Both fields and formulas have a wide array of formatting options available, which can be applied absolutely or conditionally. The data can be grouped into bands, each of which can be split further and conditionally suppressed as needed. Crystal Reports also supports subreports, graphing, and a limited amount of GIS functionality.

- Zoho Reports

Zoho Reports is online business intelligence and reporting application in the Zoho Office Suite. It can create charts, pivots, summary and other wide-range of reports through a powerful drag & drop interface.

- Tools for designing OLAP cubes

- SAS OLAP Cube Studio

SAS OLAP Cube Studio provides an easy-to-use graphical user interface to create and manage SAS OLAP cubes. You can use it to build and edit SAS OLAP cubes, to incrementally update cubes, to tune aggregations, and to make various other modifications to existing cubes. SAS OLAP Cube Studio is part of the SAS software offerings, SAS OLAP Server and SAS Enterprise BI Server.

- SQL Server Analysis Services (SSAS)

SQL Server Analysis Services (SSAS) delivers online analytical processing (OLAP) and data mining functionality for business intelligence applications. Analysis Services supports OLAP by letting you design, create, and manage multidimensional structures that contain data aggregated from other data sources, such as relational databases. For data mining applications, Analysis Services lets you design, create, and visualize data mining models that are constructed from other data sources by using a wide variety of industry-standard data mining algorithms.

- Analytic Workspace Manager 11g (AWM 11g)

Analytic Workspace Manager 11g (AWM 11g) is a tool for creating, developing, and managing multidimensional data in an Oracle 11g data warehouse. With this easy-to-use GUI tool, you create the container for OLAP data, an analytic workspace (AW), and then add OLAP dimensions and cubes. In Oracle OLAP, a Cube provides a convenient way of collecting stored and calculated measures with similar characteristics, including dimensionality, aggregation rules, and so on. A particular AW may contain more than one cube, and each cube may describe a different dimensional shape. Multiple cubes in the same AW may share one or more dimensions. Therefore, a cube is simply a logical object that helps an administrator to build and maintain data in an AW. After creating cubes, measures, and dimensions, you map the dimensions and stored measures to existing star, snowflake, and normalized relational sources and then load the data. OLAP data can then be queried with simple SQL.

- Pentaho Schema Workbench (PSW)

Pentaho Schema Workbench (PSW) provides a graphical interface for designing OLAP cubes for Pentaho Analysis (Mondrian). The schema created is stored as a regular XML file on disk.

4 Access layer

The principal purpose of the data warehouse is to provide information to its users for strategic decision-making. These users interact with the warehouse through the Access layer using end user access tools. The examples of some of the end user access tools can be:

➤ Specialised Business Intelligence Tools for data access

Business intelligence tools are a type of software that is designed to retrieve, analyse and report data. This broad definition includes everything from Reporting and Query Tools, Application Development Tools to Visual Analytics Software, as well as Navigational Tools (OLAP viewers). The main makers of business intelligence tools are:

- Oracle
- Microsoft
- SAS Institute
- SAP
- Tableau
- IBM Cognos
- QlikView

➤ Office Automation Tools (used for regular productivity and collaboration instruments)

By Office automation tools we mean all software programs which make it possible to meet office needs. In particular, an office suite usually contains the following software programs: word processing, a spreadsheet, a presentation tool, a database, and a scheduler. Among the most common office automation tools around are:

- Microsoft Office
- Corel WordPerfect
- iWork
- IBM's Lotus SmartSuite
- OpenOffice (open source/freeware).

➤ Graphic and Publishing tools

Graphic and publishing tools provide the ability to create one or more infographics from a provided data set or to visualize information. There are a vast variety of tools and software to create any kind of information graphics, depending on the organization's needs:

- PSPP

PSPP is a free software application for analysis of sampled data, intended as a free alternative for IBM SPSS Statistics. It has a graphical user interface and conventional command-line interface. It is written in C, uses GNU Scientific Library for its mathematical routines, and plot utils for generating graphs. This software provides a basic set of capabilities: frequencies, cross-tabs comparison of means (T-tests and one-way ANOVA); linear regression, reliability (Cronbach's Alpha, not failure or Weibull), and re-ordering data, non-parametric tests, factor analysis and more. At the user's choice, statistical output and graphics are done in ASCII, PDF, PostScript or HTML formats. A limited range of statistical graphs can be produced, such as histograms, pie-charts and np-charts. PSPP can import Gnumeric, OpenDocument and Excel spreadsheets, PostgreSQL databases, comma-separated values- and ASCII-files. It can export files in the SPSS 'portable', 'system' file formats and to ASCII files. Some of the libraries used by PSPP can be accessed programmatically. PSPP-Perl provides an interface to the libraries used by PSPP.

- SAS

SAS is a well-known integrated system of software products provided by SAS Institute Inc., which enables programmers to perform information retrieval and data management, report writing and graphics, statistical analysis and data mining, forecasting, operations research and project management, quality improvement, applications development and data warehousing (extract, transform, load). SAS is driven by SAS programs, which define a sequence of operations to be performed on data stored as tables. Although non-programmer graphical user interfaces to SAS exist (such as the SAS Enterprise Guide), these GUIs are most often merely a front-end that automates or facilitates the generation of SAS programs. The functionalities of SAS components are intended to be accessed via application programming interfaces, in the form of statements and procedures. SAS has an extensive SQL procedure, allowing SQL programmers to use the system with little additional knowledge. SAS runs on IBM mainframes, Unix, Linux, OpenVMS Alpha, and Microsoft Windows. SAS consists of a number of components which organizations can separately license and install as required.

- SPSS

SPSS Statistics is a software package used for statistical analysis, officially named "IBM SPSS Statistics". Companion products in the same family are used for survey authoring and deployment (IBM SPSS Data Collection), data mining (IBM SPSS Modeler), text analytics, and collaboration and deployment (batch and automated scoring services). SPSS is among the most widely used programs for statistical analysis in social science. The many features of SPSS are accessible via pull-down menus or can be programmed with a proprietary 4GL command syntax language. Command syntax programming has the benefits of reproducibility, simplifying repetitive tasks, and handling complex data manipulations and analyses. Additionally, some complex applications can only be programmed in syntax and are not accessible through the menu structure. Programs can be run interactively or unattended, using the supplied Production Job Facility. Additionally, a "macro" language can be used to write command language subroutines and a Python programmability extension can access the information in the data dictionary as well as data and dynamically build command syntax programs. In addition, the Python extension allows SPSS to run any of the statistics in the free software package R. From version 14 onwards SPSS can be driven externally by a Python or a VB.NET program using supplied "plug-ins". SPSS can read and write data from ASCII text files (including hierarchical files), other statistics packages, spreadsheets and databases. SPSS can read and write to external relational database tables via ODBC and SQL. Statistical output is set to a proprietary file format (*.spv file, supporting pivot tables) for which, in addition to the in-package viewer, a stand-alone reader can be downloaded. The proprietary output can be exported to text or Microsoft Word, PDF, Excel, and other formats. Alternatively, output can be captured as data (using the OMS command), as text, tab-delimited text, PDF, XLS, HTML, XML, SPSS dataset or a variety of graphic image formats (JPEG, PNG, BMP and EMF).

- Stata

Stata is a general-purpose statistical software package created by StataCorp. It is used by many businesses and academic institutions around the world. Stata's capabilities include data management, statistical analysis, graphics, simulations, and custom programming. Stata has always emphasized a command-line interface, which facilitates replicable analyses. Starting with version 8.0, however, Stata has included a graphical user interface which uses menus and dialog boxes to give access to nearly all built-in commands. This generates code which is always displayed, easing the transition to the command line interface and more flexible scripting language. The dataset can be viewed or edited in spreadsheet format. From version 11 on, other commands can be executed while the data browser or editor is opened. Stata can import data in a variety of formats. This includes ASCII data formats (such as CSV or databank formats) and spreadsheet

formats (including various Excel formats). Stata's proprietary file formats are platform independent, so users of different operating systems can easily exchange datasets and programs.

- Statistical Lab

The computer program Statistical Lab (Statistiklabor) is an explorative and interactive toolbox for statistical analysis and visualization of data. It supports educational applications of statistics in business sciences, economics, social sciences and humanities. The program is developed and constantly advanced by the Center for Digital Systems of the Free University of Berlin. Their website states that the source code is available to private users under the GPL. Simple or complex statistical problems can be simulated, edited and solved individually with the Statistical Lab. It can be extended using external libraries. Via these libraries, it can also be adapted to individual and local demands like specific target groups. The versatile graphical diagrams allow demonstrative visualization of underlying data. Statistical Lab is didactically driven. It is focused on providing facilities for users with little statistical experience. It combines data frames, contingency tables, random numbers, and matrices in a user friendly virtual worksheet. This worksheet allows users to explore the possibilities of calculations, analysis, simulations and manipulation of data. For mathematical calculations, the Statistical Lab uses the Engine R, which is a free implementation of the language S Plus.

- STATISTICA

STATISTICA is a suite of analytics software products and solutions provided by StatSoft. The software includes an array of data analysis, data management, data visualization, and data mining procedures, as well as a variety of predictive modelling, clustering, classification, and exploratory techniques. Additional techniques are available through integration with the free, open source R programming environment. Different packages of analytical techniques are available in six product lines: Desktop, Data Mining, Enterprise, Web-Based, Connectivity and Data Integration Solutions, and Power Solutions.

STATISTICA includes analytic and exploratory graphs in addition to standard 2- and 3-dimensional graphs. Brushing actions (interactive labelling, marking, and data exclusion) allow for investigation of outliers and exploratory data analysis. Operation of the software typically involves loading a table of data and applying statistical functions from pull-down menus or (in versions starting from 9.0) from the ribbon bar. The menus then prompt for the variables to be included and the type of analysis required. It is not necessary to type command prompts. Each analysis may include graphical or tabular output and is stored in a separate workbook.

- Web services tools (machine oriented)

- Stylus Studio

Stylus Studio has many different components like a powerful Web Service Call Composer that enables you to locate and invoke Web service methods directly from within Stylus Studio XML IDE. Stylus Studio's Web Service Call composer supports all of the core Web service technologies like Web Service Description Language (WSDL), Simple Object Access Protocol (SOAP), Universal Description Discovery and Integration (UDDI). It is an ideal Web services tool for testing Web services, inspecting WSDL files, generating SOAP envelopes, and automating or accelerating many other common XML development tasks encountered when developing Web service enabled applications. It also has a powerful schema-aware WSDL editor, which can greatly simplify your work with Web Services and the Web Service Description Language (WSDL) – an XML format for describing network services as a set of endpoints operating on messages containing either document-oriented or procedure-oriented information. Stylus Studio's WSDL editor supports working with WSDL files, making editing WSDL files and validating them a breeze.

- Microsoft Visual Studio

Microsoft Visual Studio contains a bunch of dedicated tools for creating and supporting web services, such as: **Web Services Description Language Tool** which generates code for XML Web services and XML Web services clients from Web Services Description Language (WSDL) contract files, XML Schema Definition (XSD) schema files, and .discomap discovery documents; **Web Services Discovery Tool** – discovers the URLs of XML Web services located on a Web server, and saves documents related to each XML Web service on a local disk; **Soapsuds Tool** - helps you compile client applications that communicate with XML Web services using a technique called remoting.

- Apache Axis

Apache Axis is an open source, XML based Web service framework. It consists of a Java and a C++ implementation of the SOAP server, and various utilities (**WSIF, SOAP UDDI, Ivory, Cauchy Hessian, Cauchy Burlap, Metro, Xfire, Gomba, Crispy** and etc.) and APIs for generating and deploying Web service applications. Using Apache Axis, developers can create interoperable, distributed computing applications. Axis is developed under the auspices of the Apache Software Foundation.

5 All layers

Some kind of software is using not in one particular layer but passes through several layers. It could be core components of S-DWH like database management systems. Choosing of specific DBMS for use in NSI depends on more factors like institutional policy, experience in administering, compatibility issues. Commonly used and known are these database servers:

- Microsoft SQL Server
- Oracle Database
- IBM DB2
- Informix
- Sybase Adaptive Server
- PostgreSQL
- MySQL

➤ SDMX tools

Another set of tools that can be used in S-DWH are SDMX tools. SDMX tools in S-DWH are intended for data dissemination and transferring from one S-DWH component to another. Their purpose, availability and characteristics vary widely and so we present in this section a brief inventory of the currently available SDMX-based tools and classify them according to several important criteria.

- SDMX Connectors for Statistical Software

This framework is developed by Bank of Italy. It represents a set of plug-ins which enable to the end user easier manipulation with data that come from different sources by using their standard statistical tools. Connectors are available for the following software: R, SAS, STATA and MATLAB. The connector for EXCEL is about to be published.

The framework can be downloaded for free in the following link: <https://github.com/amattioc/SDMX>.

- ECB SDMX Java Suite

This framework represents a set of libraries developed by European Central Bank. It is used for reading and checking SDMX-EDI and SDMX-ML data files. The Suite consists of two parts:

- ECB Checker - checks the syntax of incoming data files and convert the files between defined set of formats
- ECB visualization framework – represents set of libraries which can be used to optimize visualization tools for statistical data and metadata expressed in SDMX-ML. Some of the

visualization tools based on SDMX are [ECB inflation dashboard](#), [Euro area yield curve](#), [Euro foreign exchange reference rates](#);

The framework can be downloaded for free in the following link (upper right corner “visit site for download” button): http://www.sdmxtools.org/tool_info.php?id=26

- Flex-CB Visualization

This framework represents a set of libraries which enable us develop Flash visualization tools for statistical data and metadata, under the condition that data is provided by one of the SDMX-ML data formats. The framework can also be used as optimization of the tools already developed in a way of improving some parts of it like interoperability with different data sets, information support for an expanded user base, and improved presentation layer. The Flex-CB libraries are written in ActionScript 3 and are therefore meant to be included into Adobe Flex projects. The deliverable is a SWF (Flash) file.

The framework can be downloaded for free in the following link: <https://code.google.com/p/flex-cb/downloads/list>.

- DSW (Data Structure Wizard)

DSW is a desktop application that is able to convert/edit commonly used metadata formats into SDMX-ML formats. It is a Java standalone application that supports version 2 of the SDMX standard, and it can be used both in offline and on-line mode. The off-line mode is intended to be used for the maintenance of Data Structure Definitions, Code Lists, Concept Schemes, Data Flows, Hierarchical Code lists, Category Schemes and Organization Schemes. In on-line mode, users can perform the same operations as in off-line mode, except they have the possibility to interact with an SDMX compliant Registry, such as the Eurostat SDMX Registry.

- SDMX Converter

SDMX converter is an open source application which enables conversion between all the existing formats of the SDMX 2.0 standard, GESMES (SDMX-EDI 2.0), FLR and CSV formats. It also support conversions from DSPL (Google's Dataset Publishing Language) messages to SDMX-ML and backwards.

The user can setup the converter as web service or standalone application. In the last case it can be setup with a platform independent installer or a windows installer.

Interaction with the Converter is possible using a Graphic User Interface (GUI), command line interface (CLI) (via its programming API) as well as a Web Service interface. GUI will likely be used by human users, CLI by other applications (CLI can be utilized to perform conversions in a batch-processing mode without user interaction), and Web Service will be used to offer SDMX converter functionalities over the Internet, with nevertheless some overhead in the time processing compared to GUI or CLI due to the nature of the Internet communication paths.

SDMX converter can be downloaded for free in the following link: <https://circabc.europa.eu/faces/jsp/extension/wai/navigation/container.jsp>.

- SDMX-RI (Reference Infrastructure)

SDMX-RI is an infrastructure that can be used partially or completely by any company which intends to start SDMX projects related to data exchange. It consists of many modules, which can be used together or separately, depends on the needs of the company. Most commonly used modules are the following:

- SDMX Query Parser - XML parsing API implementation for incoming SDMX-ML Query messages
- Data Retriever - retrieves respective data from dissemination databases
- Structure Retriever - translates SDMX Structure query to an SQL statement and takes the SDMX Structural Metadata from the Mapping Store, delivering at the end an SDMX-ML structure message

- SDMX Data Generator - translates the Data Message to an SDMX-ML Dataset in the requested data format
- Mapping Assistant - developed in order to make the mapping of the data easier: structural metadata provided by an SDMX-ML Data Structure Definition (DSD) and those that are stored in a dissemination database.

The Census HUB project uses the intermediate version of this product.

SDMX-RI can be downloaded for free in the following link:

<https://circabc.europa.eu/faces/jsp/extension/wai/navigation/container.jsp>.

- SDMX Registry

SDMX Registry is metadata registry which provides a web-based user interface and web services to use within Eurostat and its statistical partners. It provides structure, organization, maintenance and query interfaces for most of the SDMX components required to support sharing the data. The "data sharing" model has a task to discover easily where data and metadata are available and how to access them. The SDMX Registry is one of the main modules in the whole system and can be seen as a central application which is accessible to other programs over the Internet (or an Intranet or Extranet) to provide information needed for reporting, collection and dissemination of statistics. In its broad terms, the SDMX Registry – as understood in web services terminology – is an application which stores metadata for querying, and which can be used by any other application in the network with sufficient access privileges.

Web user interface of SDMX Registry can be accessed in the following link:

<https://webgate.ec.europa.eu/sdmxregistry/>

The application can be downloaded for free in the following link:

<https://circabc.europa.eu/faces/jsp/extension/wai/navigation/container.jsp>

- XSD Generator

XSD Generator is a tool which produces XML Schema Definitions (XSD) based on a DSD received. Current version of XSD Generator can be used in the following ways:

- As a re-usable building block, through its API
- Through a standalone Graphic User Interface (GUI)
- Through Command Line Interface (CLI)
- Through a web GUI

All versions of the tool can be accessed in the following link:

<https://circabc.europa.eu/faces/jsp/extension/wai/navigation/container.jsp>

- OpenSDMX

OpenSDMX provides components which can be used in various ways where SDMX is implemented. The OpenSDMX web-service component produces SDMX data and metadata in a RESTful way. OpenSDMX offers as well a web-application with a SDMX 2.1 REST interface. There are also libraries available which can be used in any context where SDMX is needed. If we want to integrate our own data, we have to write our own adapter. OpenSDMX enables you to exclude one part of it, if you don't want to use it, or to adapt it to your needs. That is why it is considered to be very flexible. OpenSDMX web-applications can be deployed in an application server like Tomcat. The UI currently consists of a few simple html pages which can be replaced by the ones we make.

Source code can be accessed in the following link:

<https://svn.code.sf.net/p/opensdmx/svn/>

- SDMX Framework Project

SDMX Framework Project is developed by National Statistical Institute of Italy. It represents a set of tools which are used for managing data and metadata in SDMX format. This general set is divided into three branches which contain a set of tools which user can choose from:

- SDMX Data Project
- SDMX Metadata Reference Project
- SDMX Data Project for EGR

The framework could be used entirely from the reporting phase to the dissemination phase, or alternatively using the modules separately.

- Fusion Family of Products
 - Fusion Audit

Fusion Audit is a standalone web application that can receive audit and log events from any Fusion Application that has been configured to audit itself. It has been built specifically for SDMX specific information to be captured and organized into categories. It enables the user to have an aggregated view of audit events which can be filtered upon defined criteria.

- Fusion Matrix

Fusion Matrix is used for data and metadata storage retrieval for any statistical domain. The heart of Fusion Matrix is the SDMX Information model which describes the data. It supports all SDMX versions for both consumption and dissemination, and it has been designed in a way that it will support all versions of the SDMX standard that are about to appear with doing minor changes to current application. It also can be easily adopted to support other standards and formats as required. Web services provided by Fusion Matrix make data accessible by human and machines as well. Fusion Matrix has as well a web interface which offers a user to view and navigate through datasets. In addition, it provides an API which can optimize development of dissemination systems. MatrixJS library is also available, if we want to develop a web application with some of the functionalities it provides.

- Fusion Registry

Fusion Registry is the registry used in the SDMX Global Registry. It is built on a syntax-independent set of SDMX components, and structures which are uploaded in one version of the software can be retrieved in a different version without any problem. It can be used as a web service REST interface for queries, but it also has a GUI in addition to the web services interfaces.

- Fusion Security

Fusion Security is a web application that manages user accounts for the Fusion Product range. It provides an authentication service that is used by the Fusion Products to authenticate user login requests.

- Fusion Transformer

Fusion Transformer is a command-line application, which allows the conversion of SDMX data or structure files from one format to another. The application uses data streaming, and that means no restriction on the size of input or output files exists. One of the interesting options the application has, is the ability to split a single file which contains multiple datasets into multiple files.

- Fusion Transformer Pro

Fusion Transformer Pro has all of the facilities of Fusion Transformer plus many additional ones. It is a web application which enables the users to load, validate, transform, map, and export data through web browsers. Pro version also provides a lightweight command line client. It supports most formats and versions of SDMX, including SDMX-EDI. In addition, it supports both reading and writing to and from CSV and Microsoft Excel. For users who have to report data in SDMX format, the Fusion Transformer Pro offers a file storage area where the datasets can be retrieved in any SDMX format via its web service.

- Fusion Weaver

Fusion Weaver is a GUI-desktop tool developed for SDMX structural metadata, as well as for data set validation, transformation, and schema generation for SDMX data sets which have specific structure.

The Fusion tools can be downloaded following the link: <http://metadatatechnology.com/downloads.php>

- SAE SDMX Editor

SDMX SAE editor offers a simple way for managing and accessing statistical metadata. It is developed by a private company called NextSoft GmbH, and it can be used for the following: metadata entry, navigation through metadata, storage and organization of SDMX files, access the metadata, and management of statistical metadata.

The tool can be downloaded for free in the following link: <http://www.sdmx.ch/Download.aspx>

- SDMX.NET

SDMX.NET is framework for the .NET platform. It is developed by UNESCO Institute for Statistics. It represents an accurate implementation of SDMX standard and enables developers to easily create SDMX applications. Below are some basic properties of it:

- Completely open source
- Uses as input as well output SDMX from any data source
- Accurately implements SDMX
- Written in C#, and therefore is compatible with any .NET application
- Easy for API usage
- Optimized for scalability and performance

A framework can be downloaded for free in the following link: <https://code.google.com/p/sdmxdotnet/downloads/list>

- PandaSDMX

PandaSDMX is an extensible SDMX library which is written in Python. It is platform-independent and it can be run wherever the Python runs. It has an option to export data to the data analysis toolkit "pandas".

A library can be downloaded for free in the following link:

<http://pandasdmx.readthedocs.org/en/v0.2.2dev/>

6 Classification of SDMX Tools

As already mentioned before several SDMX-based IT tools exist today. Their purpose, availability and characteristics vary widely. This chapter gives a basic overview on how the tools can be classified in terms of the various features they provide.

- License Type

The following table displays classification of currently available tools, according to their license type:

Free License	Permissive Free License	Proprietary License
SDMX Connectors for Statistical Software DSW (Data Structure Wizard) MA (Mapping Assistant) SDMX Converter SDMX-RI (Reference Infrastructure) SDMX Registry XSD Generator SDMX Framework Project Fusion Registry Fusion Security Fusion Transformer Fusion Weaver SAE SDMX Editor SDMX.NET	ECB SDMX Java Suite Flex-CB Visualisation Framework OpenSDMX	Fusion Audit Fusion Matrix Fusion Transformer Pro

Table 1: Classification of the tools according to their license type

- **Platform developed**

SDMX-based IT tools are developed in different platforms: JAVA, Adobe Flex and .NET platform. The following table displays a classification of the tools, according to their platform:

JAVA	Adobe Flex	.NET
SDMX Connectors for Statistical Software ECB SDMX Java Suite DSW (Data Structure Wizard) MA (Mapping Assistant) SDMX Converter SDMX-RI (Reference Infrastructure) SDMX Registry XSD Generator OpenSDMX Fusion Audit Fusion Matrix Fusion Registry Fusion Security Fusion Transformer Fusion Transformer Pro	Flex-CB Visualisation Framework Fusion Registry Fusion Weaver	SDMX-RI (Reference Infrastructure) SDMX Framework Project SAE SDMX Editor SDMX.NET

Table 2: Classification of the tools according to platform developed

- **Main features type**

The following table displays the classification of the tools according to their main features.

IT tool	Feature type											
	SM	A	SV	ST	RA	RI	WS	DV	AT	MO	SG	SD
SDMX Connectors for Statistical Software					X		X		X			
ECB SDMX Java Suite			X	X					X			
Flex-CB Visualisation								X				
DSW (Data Structure Wizard)	X					X			X	X		
MA (Mapping Assistant)	X								X			
SDMX Converter				X			X		X			
SDMX-RI (Reference Infrastructure)	X						X		X	X		
SDMX Registry	X				X	X	X		X	X	X	
XSD Generator							X		X	X		
OpenSDMX					X		X			X	X	
SDMX Framework Project		X					X			X		X
Fusion Matrix		X	X				X	X				X
Fusion Registry	X				X	X	X			X	X	
Fusion Transformer			X	X								
Fusion Transformer Pro				X			X				X	
Fusion Weaver			X	X							X	
SAE SDMX Editor		X				X	X		X	X		
SDMX.NET	X	X	X	X						X		
PandaSDMX									X			

Table 3: Review of main features of the tools

Explanations of the abbreviations of feature types which have been used are presented in the table below:

Abbreviation	Meaning of abbreviation
SM	Structure Maintenance
A	Authoring
SV	Structure Visualization – HTML Transformation
ST	Structure Transformation
RA	Registry API
RI	Registry User Interfaces
WS	Web Services
DV	Data Visualization – HTML Transformation
AT	Analytical Tools
MO	SDMX maintenance objects
SG	Schema generation
SD	SDMX database

Table 4: Abbreviations and their explanations

- **Type of the tool provided**

The following table displays a review of tools according to the type of tool.

IT tool	Type of the tool			
	Application		Web Service	Library
	Web	Desktop		
SDMX Connectors for Statistical Software			x	x
ECB SDMX Java Suite				x
Flex-CB Visualisation				x
DSW (Data Structure Wizard)		x		
MA (Mapping Assistant)		x		
SDMX Converter		x	x	
SDMX-RI (Reference Infrastructure)	x	x	x	
SDMX Registry	x		x	
XSD Generator	x	x	x	x
OpenSDMX	x		x	x
SDMX Framework Project	x		x	
Fusion Audit	x			
Fusion Matrix	x		x	x
Fusion Registry	x		x	
Fusion Security	x			
Fusion Transformer		x		
Fusion Transformer Pro	x	x	x	
Fusion Weaver		x		
SAE SDMX Editor		x	x	
SDMX.NET				x
PandaSDMX				x

Table 5: Classification of the tools according to their type

Reference List

- https://webgate.ec.europa.eu/fpfis/mwikis/sdmx/index.php/Main_Page
- <http://www.sdmxtools.org/index.php>
- <http://metadatatechnology.com/sdmx.php#whatis>
- <https://code.google.com/p/sdmxdotnet/>
- <http://www.metadatatechnology.com/products/audit/product.php>
- <http://sdmx.org/>
- <https://github.com/amattioc/SDMX/wiki/SDMX-Connector-for-STAT>
- <https://code.google.com/p/flex-cb/>
- <https://en.wikipedia.org/wiki/SDMX>
- <http://ec.europa.eu/eurostat/data/sdmx-data-metadata-exchange>
- <http://www.sdmxsource.org/>
- <http://sourceforge.net/p/opensdmx/home/Home/>
- <http://www.oecd.org/std/SDMX%202013%20Session%207.4%20-%20How%20to%20implement%20an%20SDMX%20infrastructure%20for%20dissemination%20and%20reporting.pdf>
- <http://www.unecce.org/fileadmin/DAM/stats/documents/ece/ces/2011/49.e.pdf>
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- <http://www.oecd.org/std/47610077.pdf>

